A collection of results in quantum information and computation

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Entanglement cost of antisymmetric Werner states

Abstract

For the antisymmetric Werner state α_d, we prove $E_F(\alpha_d^{\otimes n}) = n$ for every $d \geq 2$ and $n \geq 1$, and hence entanglement cost $E_C(\alpha_d) = 1$ ebit. The proof establishes a sharp companion-channel purity inequality for the antisymmetric-subspace channel, together with its equality cases.

We prove that the antisymmetric state α_d has entanglement cost $E_C(\alpha_d) = 1$ ebit for every $d \geq 2$. Yura established the case $d = 3$ [2]; the case $d = 2$ is a pure singlet. Christandl, Schuch, and Winter proved the dimension-independent lower bound $\log_2(4/3)$ and left open whether the cost equals one [3, 4]. The result below establishes that value also for $d \geq 4$, through the exact identity $E_F(\alpha_d^{\otimes n}) = n$. The associated antisymmetric-subspace channel also occurs in the work of Grudka, Horodecki, and Pankowski on minimum-output Rényi entropy [5]. Here the channel estimate is a sharp purity inequality, including its equality cases, from which the entanglement-cost statement follows.

1 Definitions and statements

All Hilbert spaces are finite-dimensional complex Hilbert spaces. Traces and Schatten norms are unnormalized. Let $H_d = \mathbb{C}^d$, $d \geq 2$, and let $X_d = \Lambda^2 H_d$, with $D = \binom{d}{2}$. Its orthonormal coordinate basis is e_{ab}, $a < b$. The canonical isometry and its partial-trace channel are

$$V_d e_{ab} = \frac{|a,b\rangle - |b,a\rangle}{\sqrt{2}}, \qquad \Phi_d(T) = \mathrm{Tr}_B(V_d T V_d^*). \tag{1.1}$$

For a quantum channel $N : \mathcal{L}(Y) \to \mathcal{L}(B)$, with nonzero input and output spaces, define

$$P_{\max}(N) = \max_{\|y\|=1} \mathrm{Tr}\, N(|y\rangle\langle y|)^2. \tag{1.2}$$

Compactness and continuity give an attained maximum. If $m = \dim B$, every output density has purity at least $1/m$, so $P_{\max}(N) > 0$. A normalized Slater is an element $u = x \wedge y \in X_d$ with $x, y \in H_d$ orthonormal. Thus its edge coordinates are $u_{ab} = x_a y_b - x_b y_a$, and its physical vector is $(x \otimes y - y \otimes x)/\sqrt{2}$. An overall unit phase can be absorbed into either factor.

Theorem 1.1 (Sharp companion bound and equality cases)**.** *Let $d \geq 2$, let N be any channel on its entire input space Y, and set $\beta = P_{\max}(N)$. For every unit $\eta \in X_d \otimes Y$, put $\sigma = \mathrm{Tr}_{X_d}|\eta\rangle\langle\eta|$ and $s = \mathrm{Tr}\,\sigma^2$. Then*

$$\mathrm{Tr}[(\Phi_d \otimes N)(|\eta\rangle\langle\eta|)]^2 \leq \frac{\beta}{2} - \frac{\beta}{d+1}(1 - s). \tag{1.3}$$

Moreover,

$$P_{\max}(\Phi_d \otimes N) = \frac{1}{2} P_{\max}(N). \tag{1.4}$$

A unit input attains this maximum if and only if it is $u \otimes y$, where u is a normalized Slater and y is a unit pure-input maximizer of N. For each $d \geq 3$, the coefficient $\beta/(d+1)$ in (1.3) is the largest possible coefficient valid for all channels and all unit inputs with that normalization.

Let F_d exchange the two physical factors and let

$$\alpha_d = \frac{V_d V_d^*}{D} = \frac{I - F_d}{d(d-1)}. \tag{1.5}$$

The entanglement of formation is measured in bits:

$$E_F(\rho) = \inf_{\rho=\sum_j p_j|\psi_j\rangle\langle\psi_j|} \sum_j p_j S_2(\mathrm{Tr}_B|\psi_j\rangle\langle\psi_j|), \qquad S_2(\tau) = -\operatorname{Tr}\tau\log_2\tau. \tag{1.6}$$

The infimum ranges over every finite normalized complex pure ensemble; zero eigenvalues contribute zero to entropy. Tensor factors are regrouped across the full Alice–Bob cut.

Theorem 1.2 (Tensor powers and formation)**.** *For every $d \geq 2$ and integer $n \geq 0$,*

$$P_{\max}(\Phi_d^{\otimes n}) = 2^{-n}, \qquad E_F(\alpha_d^{\otimes n}) = n. \tag{1.7}$$

For $n \geq 1$, maximum-purity inputs are exactly tensor products of normalized Slaters. Equivalently, every unit physical vector supported on $(V_d X_d)^{\otimes n}$ has Alice purity at most 2^{-n}, with equality exactly for physical tensor products of normalized Slaters. At $n = 0$ the space is $\mathbb{C}$, and the maximizing inputs are unit scalars. Every supported physical pure vector has Alice entropy at least n bits.

Corollary 1.3 (Ordinary entanglement cost)**.** *The ordinary vanishing-error LOCC entanglement cost of α_d is one singlet per copy. Every achievable singlet rate is at least one, and rate one is achievable.*

2 The physical replica operator

Write F for the exchange on $X_d \otimes X_d$. The four-particle alternating subspace embeds in $\mathrm{Sym}^2 X_d$; write W for its orthogonal projector, and put

$$Z = \frac{I+F}{2} - W, \qquad J = \frac{I-F}{2}. \tag{2.1}$$

These are mutually orthogonal projectors with sum I. For each four-subset $Q = \{a < b < c < f\}$, define the real symmetric edge matrix B_Q by the complementary-edge entries

$$(ab, cf) : 1, \qquad (ac, bf) : -1, \qquad (af, bc) : 1,$$

including their transposes, and with every other entry zero. If P_Q projects onto the six edges of Q, then $B_Q^2 = P_Q$ and $\operatorname{rank} B_Q = 6$. Set $\Omega_Q = \operatorname{vec} B_Q$, where vectorization uses the edge basis. The vectors $\Omega_Q/\sqrt{6}$ are an orthonormal basis of the embedded four-particle alternating subspace. In particular,

$$\|\Omega_Q\|^2 = 6, \qquad \sum_Q |\Omega_Q\rangle\langle\Omega_Q| = 6W. \tag{2.2}$$

Lemma 2.1 (Replica identity)**.** *The purity kernel of Φ_d is*

$$T_d = \frac{I + F - 6W}{4} = \frac{Z}{2} - W. \tag{2.3}$$

Proof. On four physical factors let $P = P_{\mathrm{a},12}P_{\mathrm{a},34}$ be the projection onto the antisymmetric subspace of each pair, and let A_4 be the full antisymmetrizer. Decomposing its permutation sum into the six cosets of the within-pair permutation subgroup gives

$$6A_4 = P(I + F - F_{13} - F_{14} - F_{23} - F_{24})P.$$

Here F exchanges the two pairs. The two within-pair swaps act as -1 on the range of P. Conjugating by them equates the four compressed cross swaps. The compression of F_{13} is exactly the swap-trick purity kernel T_d. Consequently $6W = I + F - 4T_d$ after transport by $V_d \otimes V_d$. Expanding a normalized four-particle wedge gives $\Omega_Q/\sqrt{6}$ and proves (2.2). This proves the identity on the whole replica space, including the sector J. □

Choose a Stinespring isometry for N and let N^c be its complementary channel. Its Hermitian swap kernel K_N is characterized by

$$\langle u \otimes v | K_N | x \otimes y \rangle = \operatorname{Tr} N(|x\rangle\langle u|) N(|y\rangle\langle v|). \tag{2.4}$$

This kernel need not be positive. The transition identity for a Stinespring isometry gives $K_{N^c} = K_N F_Y$, where F_Y is the companion replica exchange. For example, if the isometry sends x to a coefficient matrix C_x on $B \otimes E$, then

$$N(|x\rangle\langle u|) = C_x C_u^*, \qquad N^c(|x\rangle\langle u|) = (C_u^* C_x)^T;$$

cyclicity of trace proves the kernel identity.

Write the given input as $\eta = \sum_e e \otimes a_e$ and let A be the matrix with columns a_e. Thus $\sigma = AA^*$ and $\|A\|_{\mathrm{HS}}^2 = 1$. Regroup $\Xi = \eta \otimes \eta$ as $(X_d \otimes X_d) \otimes (Y \otimes Y)$. Define

$$\begin{aligned} q &= \operatorname{Tr} N(\sigma)^2, & q^c &= \operatorname{Tr} N^c(\sigma)^2, \\ z &= \langle \Xi, (Z \otimes K_N)\Xi \rangle, & w &= \langle \Xi, (W \otimes K_N)\Xi \rangle, \\ z_0 &= \langle \Xi, (Z \otimes I)\Xi \rangle, & w_0 &= \langle \Xi, (W \otimes I)\Xi \rangle, \\ M_Q &= AB_Q A^T, & C &= \sum_Q \langle \operatorname{vec} M_Q, K_N \operatorname{vec} M_Q \rangle. \end{aligned} \tag{2.5}$$

Every one of these contractions uses the same matrix A.

Lemma 2.2 (Exact scalar identities)**.** *With the definitions above, the actual output purity P satisfies*

$$z_0 + w_0 = \frac{1+s}{2}, \qquad q + q^c = 2(z+w), \qquad C = 6w, \qquad P = \frac{q + q^c - C}{4} = \frac{z}{2} - w. \tag{2.6}$$

Proof. The expectation of the first-site exchange on Ξ is s, by the swap trick and equality of the two marginal spectra of a pure state. This proves the first identity. The identity part of $I + F$ gives q. Since $(F \otimes F_Y)\Xi = \Xi$, its first-site exchange term gives q^c by $K_{N^c} = K_N F_Y$. Thus the symmetric sector gives $q + q^c = 2(z + w)$. Contraction with Ω_Q gives exactly $\operatorname{vec}(AB_Q A^T)$, proving $C = 6w$ from (2.2). Finally (2.3) and the swap trick prove the purity formula. □

3 A rank-six channel contraction estimate

Lemma 3.1 (Coherence bound)**.** *For orthogonal unit companion inputs ψ_j, ψ_l, set*

$$P_j = \operatorname{Tr} N(|\psi_j\rangle\langle\psi_j|)^2, \qquad B_{jl} = \operatorname{Tr} N(|\psi_l\rangle\langle\psi_j|)^2.$$

Then

$$|B_{jl}| \le \min\left\{ \frac{\beta}{2}, \sqrt{P_j P_l} \right\}. \tag{3.1}$$

Proof. Put $O = \operatorname{Tr} N(|\psi_j\rangle\langle\psi_j|)N(|\psi_l\rangle\langle\psi_l|)$ and $X = \operatorname{Tr} N^c(|\psi_j\rangle\langle\psi_j|)N^c(|\psi_l\rangle\langle\psi_l|)$. The two transition Hilbert–Schmidt norms are X and O, respectively, and their squared traces agree. Therefore $|B_{jl}| \le \min(O, X)$. Hilbert–Schmidt Cauchy–Schwarz gives $O \le \sqrt{P_j P_l}$. Average the output purities of $(\psi_j + e^{i\theta}\psi_l)/\sqrt{2}$ and $(\psi_j - e^{i\theta}\psi_l)/\sqrt{2}$. Choosing θ so that the squared transition contributes positively gives

$$4\beta \ge P_j + P_l + 2O + 2X + 2|B_{jl}| \ge 8|B_{jl}|.$$

For the last inequality use $P_j + P_l \ge 2\sqrt{P_j P_l} \ge 2|B_{jl}|$ and $O, X \ge |B_{jl}|$. This proves both bounds. □

Lemma 3.2 (Rank-six estimate)**.** *For every complex symmetric matrix M of rank at most six, define $C_N(M) = \langle \operatorname{vec} M, K_N \operatorname{vec} M\rangle$. Then*

$$C_N(M) \ge -\frac{4\beta}{5}\|\Lambda^2 M\|_1, \qquad C_N(M) \ge -2\beta\|M\|_{\mathrm{HS}}^2. \tag{3.2}$$

Proof. Take a Takagi decomposition $M = \sum_{j=1}^{r} \lambda_j \psi_j \psi_j^T$, where $r \le 6$, $\lambda_j \ge 0$, and the ψ_j are orthonormal. By (2.4),

$$C_N(M) = \sum_j \lambda_j^2 P_j + 2\sum_{j<l} \lambda_j \lambda_l \operatorname{Re} B_{jl}.$$

The weighted average of the two bounds in (3.1) gives $|B_{jl}| \le (4/5)(\beta/2) + (1/5)\sqrt{P_j P_l}$. Set $x_j = \lambda_j \sqrt{P_j}$. Retaining the diagonal terms yields

$$\begin{aligned} C_N(M) &\ge \frac{6\sum_j x_j^2 - (\sum_j x_j)^2}{5} - \frac{4\beta}{5}\sum_{j<l}\lambda_j\lambda_l \\ &\ge -\frac{4\beta}{5}\sum_{j<l}\lambda_j\lambda_l. \end{aligned}$$

The omitted bracket is nonnegative by $r \le 6$. The singular values of $\Lambda^2 M$ are the pair products $\lambda_j\lambda_l$, proving the first inequality. Also $\sum_{j<l}\lambda_j\lambda_l \le (5/2)\sum_j \lambda_j^2$, proving the second. Zero Takagi values require no separate assumption. □

Since $\operatorname{rank}(AB_Q A^T) \le 6$, applying the second estimate to the actual correction matrices gives

$$w \ge -2\beta w_0. \tag{3.3}$$

Indeed, $\sum_Q \|AB_Q A^T\|_{\mathrm{HS}}^2 = 6w_0$ by (2.2).

4 The Slater moment and the positive sector

Lemma 4.1 (Normalized Slater-square moment)**.** *Let $R(U) = \Lambda^2 U$ for $U \in U(d)$ and let $u(U) = R(U)e_{12}$. With normalized Haar measure,*

$$\int_{U(d)} |u(U)\otimes u(U)\rangle\langle u(U)\otimes u(U)|\, dU = \frac{Z}{z_d}, \qquad z_d = \operatorname{rank} Z = \frac{d^2(d^2-1)}{12}. \tag{4.1}$$

Proof. Every Slater square is symmetric and has zero four-form component, because $(x\wedge y)\wedge(x\wedge y) = 0$. The integral is therefore supported on Z and commutes with $R(U)\otimes R(U)$. We verify that a commuting operator on Z is scalar.

Diagonal physical phases separate coordinate weights. The weight of $e_{ab} \otimes e_{ab}$ is one-dimensional, and physical permutations equate the scalars on all these squares. Rotating coordinate b into c in such a square and comparing the middle coefficient forces the same scalar on every common-index symmetric pair $e_{ab} \odot e_{ac}$. Here $p \odot q = (p \otimes q + q \otimes p)/\sqrt{2}$ for orthogonal p, q.

For four distinct $a < b < c < f$, let $x = e_{ab} \odot e_{cf}$, $y = e_{ac} \odot e_{bf}$, $t = e_{af} \odot e_{bc}$. The four-form line is $x - y + t$, and its orthogonal complement is the two-dimensional Z weight space. Rotate a into c in $e_{ab} \odot e_{af}$: after removing the common-index terms the middle coefficient is $x - t$. Rotating a into f in $e_{ab} \odot e_{ac}$ similarly produces $-(x + y)$. Commutation gives the same scalar on these two independent vectors. Coordinate squares, common-index pairs, and these weight planes span Z, proving the assertion. The argument also covers $d = 3$, where the four-distinct weight spaces are absent, and $d = 2$ is one-dimensional.

The integral has trace one. Finally

$$z_d = \frac{D(D+1)}{2} - \binom{d}{4} = \frac{d^2(d^2-1)}{12},$$

which proves the stated normalization. □

For a normalized Slater u, contracting the first-site square of the fixed Ξ gives $v_u \otimes v_u$, with $v_u = A\overline{u}$. The definition of β on the whole companion input space gives

$$\langle v_u \otimes v_u, K_N(v_u \otimes v_u)\rangle = \operatorname{Tr} N(|v_u\rangle\langle v_u|)^2 \le \beta \|v_u\|^4.$$

This is homogeneous and also holds when $v_u = 0$. Integrating and using (4.1) on both sides yields

$$z \le \beta z_0. \tag{4.2}$$

Define $\Delta = \beta(1+s) - (q + q^c)$. The exact identities (2.6) now give

$$C + 2\Delta = 4(\beta z_0 - z) + 2(w + 2\beta w_0) \ge 0. \tag{4.3}$$

Thus $C + \Delta \ge C/2$. This conclusion uses contractions on the given $\eta \otimes \eta$; it does not require K_N to be positive.

5 The invariant exterior correction bound

Lemma 5.1 (Common-matrix exterior bound)**.** *For $d \ge 4$, the same correction C satisfies*

$$C \ge -2\beta \frac{d-3}{d+1}(1-s). \tag{5.1}$$

Proof. For a physical unitary U, write $R = \Lambda^2 U$. The vectors $(R \otimes R)\Omega_Q/\sqrt{6}$ form another orthonormal basis of the same four-form subspace. Their actual bra contractions are

$$M'_Q = A\overline{RB_QR^T}A^T.$$

Accordingly $C = \sum_Q C_N(M'_Q)$ for every U. Each matrix still has rank at most six. Put $L = \Lambda^2 A$. The symmetric matrix $\Lambda^2\overline{RB_QR^T}$ is a partial isometry of rank fifteen; its left support is

$$Q'_Q = \Lambda^2(\overline{R}P_QR^T).$$

All its nonzero Takagi values are one, since the six nonzero singular values of B_Q are one. Hence it has a factorization VV^T with $V^*V = I_{15}$ and $VV^* = Q'_Q$. Multiplicativity of exterior powers and Schatten Cauchy–Schwarz imply

$$\|\Lambda^2 M'_Q\|_1 = \|(LV)(LV)^T\|_1 \le \|LV\|_{\mathrm{HS}}^2 = \operatorname{Tr} L^*LQ'_Q. \tag{5.2}$$

The first estimate of Lemma 3.2 therefore gives, for every U,

$$C \ge -\frac{4\beta}{5} \operatorname{Tr} L^*L \sum_Q Q'_Q. \tag{5.3}$$

The unitary action on $\Lambda^2(\Lambda^2 H_d)$ has scalar commutant. Here is a coordinate verification. Its one-dimensional repeated-index weight spaces are spanned by $e_{ab} \wedge e_{ac}$. Permutations equate their scalars. For four distinct $a < b < c < f$, set $x = e_{ab} \wedge e_{cf}$, $y = e_{ac} \wedge e_{bf}$, $t = e_{af} \wedge e_{bc}$. Rotating the common index a into f, c, and b in, respectively, $e_{ab} \wedge e_{ac}$, $e_{ab} \wedge e_{af}$, and $e_{ac} \wedge e_{af}$ produces the middle coefficients

$$-x + y, \qquad x + t, \qquad y - t.$$

The remaining coefficients are repeated-index vectors already controlled. These three middle coefficients are independent and span the four-distinct weight space. Commutation forces the same scalar on it. All coordinate outer wedges belong to one of these types, proving the assertion. Conjugating the representation preserves this property.

Consequently the Haar average of $\sum_Q Q'_Q$ is scalar. Its trace is $15\binom{d}{4}$, and the ambient dimension is $\binom{D}{2}$, so

$$\int \sum_Q Q'_Q \, dU = R_d I, \qquad R_d = \frac{15\binom{d}{4}}{\binom{D}{2}} = \frac{5(d-3)}{d+1}. \tag{5.4}$$

The matrix L stays fixed during this integration. If a_j are the singular values of A, then $\sum_j a_j^2 = 1$ and

$$\|L\|_{\mathrm{HS}}^2 = \sum_{j<l} a_j^2 a_l^2 = \frac{1 - \operatorname{Tr}(AA^*)^2}{2} = \frac{1-s}{2}.$$

Integrating (5.3) proves (5.1). □

6 Sharpness and rigidity

Proof of Theorem 1.1. For $d \ge 4$, combine (2.6), (4.3), and (5.1):

$$\begin{aligned} P &= \frac{\beta(1+s) - (C+\Delta)}{4} \\ &\le \frac{\beta(1+s) + \beta\frac{d-3}{d+1}(1-s)}{4} = \frac{\beta}{2} - \frac{\beta}{d+1}(1-s). \end{aligned}$$

For $d = 3$, $W = 0$ and (4.2) gives $P \le (\beta/4)(1+s)$, the same formula. For $d = 2$, X_d is one-dimensional, so every input is a product and $s = 1$; its first output has purity $1/2$.

For a unit $u \in X_d$, Lemma 2.1 gives directly

$$\operatorname{Tr} \Phi_d(|u\rangle\langle u|)^2 = \frac{1}{2} - \frac{3}{2}\|W(u \otimes u)\|^2 \le \frac{1}{2}. \tag{6.1}$$

A coordinate Slater attains $1/2$. Thus a product with any pure-input maximizer of N attains $\beta/2$, proving (1.4).

For the complete equality case of (6.1), let u_{ab} be extended as a skew matrix to all ordered coordinates. The condition $W(u \otimes u) = 0$ is exactly the family of Plücker relations

$$u_{ab}u_{cf} - u_{ac}u_{bf} + u_{af}u_{bc} = 0 \quad \text{for all } a, b, c, f. \tag{6.2}$$

Indeed these are the coefficients of $u \wedge u$; the normalized four-form basis above has nonzero constant normalization. Choose a pivot $u_{ab} \neq 0$. Applying (6.2) to arbitrary i, j yields

$$u_{ij} = \frac{u_{ai}u_{bj} - u_{aj}u_{bi}}{u_{ab}}.$$

Thus $u = x \wedge y$ with $x_i = u_{ai}$ and $y_i = u_{bi}/u_{ab}$. Gram–Schmidt, together with $\|x \wedge y\|^2 = \|x\|^2\|y\|^2 - |\langle x, y\rangle|^2$, writes the same unit u as a wedge of orthonormal vectors, with its phase absorbed in one factor. Conversely every such wedge satisfies (6.2). Therefore one-copy equality holds exactly for normalized Slaters.

Since $\beta/(d+1) > 0$, equality at the maximum in (1.3) forces $s = 1$. A pure bipartite vector with a pure marginal is a product, say $\eta = u \otimes y$ with both factors unit. Its output purity is the product of the two local output purities. Equality at $\beta/2$, the positive product of their upper bounds, forces local purity $1/2$ and companion purity β. The converse is immediate.

To prove coefficient sharpness for $d \geq 3$, use $N = \Phi_d$ and

$$\eta_{\text{flat}} = D^{-1/2} \sum_e e \otimes e. \tag{6.3}$$

Then $s = 1/D$ and $\beta = 1/2$. Its physical coefficient matrix after regrouping outputs (a, c) and environments (b, f) is

$$\frac{1}{2\sqrt{D}}(\delta_{ac}\delta_{bf} - \delta_{af}\delta_{bc}).$$

Tracing the environment gives the literal output density

$$\tau_{\text{flat}} = \frac{I + (d-2)|v\rangle\langle v|}{4D}, \qquad v = \sum_{a=1}^{d} |a, a\rangle, \quad \|v\|^2 = d. \tag{6.4}$$

Its eigenvalues are $(d-1)^2/(4D)$ on the v line and $1/(4D)$ with multiplicity $d^2 - 1$. Consequently

$$\operatorname{Tr} \tau_{\text{flat}}^2 = \frac{d^2 - 3d + 4}{4d(d-1)} = \frac{1}{4} - \frac{1 - 1/D}{2(d+1)}. \tag{6.5}$$

This is equality in (1.3). As $D > 1$ for $d \geq 3$, increasing its impurity coefficient would fail at this input. At $d = 2$ every possible input has zero impurity. □

7 Tensor powers, entropy, and operational cost

Proof of Theorem 1.2. The zero-copy channel on $\mathbb{C}$ has maximum purity one. Theorem 1.1 applied successively with $N = \Phi_d^{\otimes(n-1)}$ proves the first identity in (1.7). Its equality characterization also inducts: a maximizing input factors as a first normalized Slater times a maximizing input on the remaining sites. The base case is a unit scalar, so all phases are included. Conversely every Slater tensor product attains 2^{-n}.

The isometry $V_d^{\otimes n}$ identifies every unit vector of $X_d^{\otimes n}$ with a unit physical vector in the antisymmetric support. Its Alice reduction is exactly the output of $\Phi_d^{\otimes n}$. This proves both physical purity assertions. For a density with eigenvalues λ_j, concavity of logarithm with weights λ_j gives

$$S_2(\tau) = -\sum_{\lambda_j>0} \lambda_j \log_2 \lambda_j \geq -\log_2 \sum_j \lambda_j^2.$$

The purity bound therefore gives entropy at least n on every supported pure vector.

Let P_n project onto the physical antisymmetric support. If $\alpha_d^{\otimes n} = \sum_j p_j |\psi_j\rangle\langle\psi_j|$, then

$$0 = \operatorname{Tr}(I - P_n)\alpha_d^{\otimes n} = \sum_j p_j \|(I - P_n)\psi_j\|^2.$$

Every positive-weight component lies in the support and hence has entropy at least n. This proves $E_F \geq n$ for every unrestricted finite pure ensemble. The uniform ensemble of all D^n coordinate Slater tensor products has barycenter $\alpha_d^{\otimes n}$. Each component has n independent two-point Schmidt spectra and therefore entropy n. It proves $E_F \leq n$. The empty tensor product has entropy zero, covering $n = 0$. □

For completeness, the operational quantity in Corollary 1.3 uses shared two-qubit singlets, free local ancillas, local quantum instruments, and classical communication. Convergent output limits of finite protocol prefixes are allowed. An HHT achievable rate r means that for every $\epsilon, \delta > 0$ there are integers $n \geq 1, m \geq 0$ and such a protocol with output τ for which

$$\left| r - \frac{m}{n} \right| \leq \delta, \qquad D_{\mathrm{B}}(\tau, \alpha_d^{\otimes n}) \leq \epsilon. \tag{7.1}$$

Here $D_{\mathrm{B}}(\rho, \sigma) = 2\sqrt{1 - \operatorname{Tr}\sqrt{\sqrt{\rho}\,\sigma\sqrt{\rho}}}$. The cost is the infimum of achievable rates. The same rates result from finite protocols: a convergent output can be approximated by a finite prefix on the same charged input, changing only the output error; a finite protocol is an eventually constant convergent process. Local basis embeddings and appending or discarding free local ancillas also preserve the rate and final output density.

Proof of Corollary 1.3. The regularization theorem of Hayden, Horodecki, and Terhal [1, Theorem 1] identifies this ordinary LOCC cost with

$$E_C(\rho) = \lim_{n\to\infty} \frac{E_F(\rho^{\otimes n})}{n}. \tag{7.2}$$

Its operational approximation and metric conventions are those of (7.1); equivalently one may use vanishing trace-distance error. The theorem is used here as the established operational bridge. By Theorem 1.2 the right side for α_d is one. Thus every achievable rate is at least one.

There is also an explicit rate-one finite protocol. Alice selects an edge $\{a, b\}$ uniformly, sends its classical label to Bob, and both apply the local isometry $|0\rangle \mapsto |a\rangle$, $|1\rangle \mapsto |b\rangle$ to one singlet. The resulting physical state is $V_d e_{ab}$. Forgetting the label gives α_d. Repeating independently creates $\alpha_d^{\otimes n}$ exactly from n singlets, proving that rate one itself is achievable. □

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A finite adaptive Bell strong converse for the amplitude-damping channel

Abstract

For every finite adaptive qubit amplitude-damping protocol, we bound its unconditional Bell overlap in terms of its maximum physical-call count and the sandwiched relative entropy of entanglement of its entire initial Alice–Bob state. The bound uses normalized separable comparisons, includes both damping endpoints, and holds on the output closure at fixed call, logical-dimension, order, and resource budgets. The proof constructs a complete finite phase comparison, controls coherent input-weight records, iterates on the actual adaptive tree, and removes its logarithmic block overhead using finitely many independent copies of the fixed protocol.

The rate threshold in the finite-code bound is

$$C_Z(p) = \log_2[1 + (1-p)p^{p/(1-p)}], \qquad 0 < p < 1,$$

the capacity of the classical Z-channel with damping probability p. For example $C_Z(1/2) = \log_2(5/4) \simeq 0.321928$ bits per call. Below we derive the strong-converse implication for sequences of the finite protocols defined here, including the initial resource charge. The comparison method uses the amortized-entanglement framework [5] and the relative-entropy converse approach [4]. The endpoint value $B_1(p) = \log_2(2-p)$ coincides with the amplitude-damping max-relative entanglement calculated by Rigovacca et al. [3, Proposition 2]; that endpoint is not a new channel bound.

1 States, physical protocols, and the statement

All spaces in the finite construction are finite dimensional over $\mathbb{C}$. A density operator is positive semidefinite with trace one. All logarithms are base two. For $0 \le p \le 1$, put $q = 1-p$ and

$$A_0 = |0\rangle\langle 0| + \sqrt{q}|1\rangle\langle 1|, \qquad A_1 = \sqrt{p}|0\rangle\langle 1|, \qquad \mathcal{A}_p(X) = A_0 X A_0^\dagger + A_1 X A_1^\dagger. \tag{1}$$

The physical call sends a qubit from Alice to Bob and applies this channel; both preexisting memories are acted on by identities. In particular, $\sum_{j=0}^{1} A_j^\dagger A_j = I$.

A finite adaptive protocol is a finite rooted tree. Its nodes are terminal nodes, complete Alice instruments, complete Bob instruments, or physical calls. At an Alice node, matrices $K_j : A \to A'$ satisfy $\sum_j K_j^\dagger K_j = I_A$, and public outcome j selects the continuation. The analogous definition holds for Bob. Dimensions may change from node to node; each continuation has the same prescribed terminal logical spaces. Any additional private Kraus index can be retained locally or refined into finitely many outcomes. Discarding local registers is also a complete local operation. Classical communication in either direction is represented by the outcome-dependent continuations.

Write $P[X]$ for the unconditional terminal operator. Its recursion is

$$\begin{aligned}
P[X] &= X && \text{at a terminal node,}\\
P[X] &= \sum_j P_j[(K_j \otimes I)X(K_j \otimes I)^\dagger] && \text{at an Alice node,}\\
P[X] &= \sum_j P_j[(I \otimes L_j)X(I \otimes L_j)^\dagger] && \text{at a Bob node,}\\
P[X] &= P_1[(\mathrm{id} \otimes \mathcal{A}_p)(X)] && \text{at a call node,}
\end{aligned} \tag{2}$$

where the last expression includes the change of laboratory cut caused by transmission. Its call count $c(P)$ is zero at a terminal node, the maximum of the continuation counts at a local node, and $1 + c(P_1)$ at a call node. Thus $c(P)$ is a maximum over histories, including every failure history. No output is divided by a success probability. Completeness makes P a trace-preserving completely positive (CPTP) map.

For $\alpha > 1$, let $t = (\alpha - 1)/\alpha$ and define

$$\begin{aligned} \widetilde{Q}_\alpha(\rho\|\sigma) &= \mathrm{Tr}[(\sigma^{-t/2}\rho\sigma^{-t/2})^\alpha], \\ \widetilde{D}_\alpha(\rho\|\sigma) &= \frac{1}{\alpha-1}\log\widetilde{Q}_\alpha(\rho\|\sigma), \qquad \operatorname{supp}\rho \subseteq \operatorname{supp}\sigma, \\ \widetilde{E}_\alpha(A:B)_\rho &= \inf_{\sigma\in\mathrm{SEP}(A:B)} \widetilde{D}_\alpha(\rho\|\sigma). \end{aligned} \tag{3}$$

Negative powers act on the support and vanish on the kernel. Unsupported comparisons have divergence $+\infty$. The set $\mathrm{SEP}(A:B)$ consists of finite convex combinations of normalized product densities. In finite dimensions it is closed. The infimum is over *normalized* states. The cut in (3) contains every initial register in Alice's and Bob's laboratories, including resource-correlated side registers.

For $d \geq 2$, write

$$|\Phi_d\rangle = \frac{1}{\sqrt{d}}\sum_{i=1}^{d}|ii\rangle, \qquad \Phi_d = |\Phi_d\rangle\langle\Phi_d|, \qquad G_d(\eta) = \mathrm{Tr}(\Phi_d\eta). \tag{4}$$

This is squared fidelity to the pure Bell target. For $0 < s \leq 1$ define

$$B_s(p) = \begin{cases} 1, & p = 0, \\ 0, & p = 1, \\ \log(2-p), & 0 < p < 1,\ s = 1, \\ \log\left[1 + \left(\dfrac{1-p}{[1-p^{1/(1-s)}]^{1-s}}\right)^{1/s}\right], & 0 < p < 1,\ 0 < s < 1. \end{cases} \tag{5}$$

The endpoint clauses are part of the definition and require no evaluation of the interior quotient at an endpoint.

Theorem 1 (Finite adaptive Bell converse)**.** *Let $0 \leq p \leq 1$, $0 < s \leq 1$, $m \in \mathbb{N}_0$, $d \geq 2$, and $e \in \mathbb{R}$. Let ρ be any density on any finite initial laboratory pair $A:B$, and let P be any finite protocol with terminal spaces $\mathbb{C}^d:\mathbb{C}^d$. If*

$$c(P) \leq m, \qquad \widetilde{E}_{1+s}(A:B)_\rho \leq e, \tag{6}$$

then

$$G_d(P[\rho]) \leq \min\left\{1, 2^{-\frac{s}{1+s}(\log d - e - mB_s(p))}\right\}. \tag{7}$$

For the same fixed p, s, m, d, e, (7) holds for every trace-norm limit of such terminal outputs.

The initial dimensions and the number of local rounds may vary throughout this output family. The condition in (6) is a condition on the entire initial laboratory state. A Bell pair whose two halves are both initially at Alice lies within one laboratory. A shared entangled bank lies across the cut and is included in the bound. The order of the charged quantity is exactly $1 + s$.

For fixed $0 < p < 1$, write $q = 1 - p$. Differentiation at $s = 0$ gives

$$\lim_{s\downarrow 0} \frac{\ln q - (1-s)\ln(1-p^{1/(1-s)})}{s} = \ln q + \frac{p}{q}\ln p.$$

Hence $B_s(p) \to C_Z(p)$. Consider a sequence of finite protocols with call budget $m \to \infty$ and $\log d_m \geq mR$, where $R > C_Z(p)$. Choose a fixed $s > 0$ with $B_s(p) < R$. If the initial resource charges at this order obey $e_m = o(m)$, Theorem 1 makes the Bell overlap tend to zero exponentially. In particular, this applies to separable initial laboratory states, for which $e_m = 0$. This rate conclusion uses exactly the finite-protocol model and output closures stated above.

Comparison with earlier bounds

Let h_2 denote binary entropy in bits. Two analytic benchmarks are the reverse-coherent-information achievable rate

$$L_{\mathrm{RCI}}(p) = \max_{0\leq u\leq 1} [h_2(u) - h_2(pu)]$$

and the balanced-squashing upper bound

$$U_{\mathrm{sq}}(p) = \max_{0\leq u\leq 1} [h_2((1-p/2)u) - h_2(pu/2)].$$

These appear in [6, Supplementary Note 5, Eqs. (S212), (S228)–(S231)]. The achievable entanglement rate is also an achievable secret-key rate. For comparison with the finite-code rate consequence proved above, at $p = 1/2$ the values per physical channel use are

Quantity	Value
Reverse coherent information L_{RCI}	$0.271553\ldots$
Present converse threshold $C_Z = \log_2(5/4)$	$0.321928\ldots$
Balanced-squashing bound $U_{\mathrm{sq}} = \log_2(4/3)$	$0.415037\ldots$
Max-relative bound $\log_2(3/2)$ [3]	$0.584963\ldots$

The first value is the one-variable maximum above. The third is exact: $h_2(3u/4) - h_2(u/4)$ is strictly concave on $(0, 1)$, and its derivative vanishes at $u = 4/9$. The substitution $u = 1/2$ in [6, Eq. (S232)] instead gives $0.410869\ldots$; it does not maximize the preceding expression. Pirandola already identified this issue and evaluated the retained maximization [9, footnote 3 and Theorem IV.1]. Thus $C_Z(1/2)$ improves both analytic upper benchmarks in the table while leaving a gap above the achievable rate. Fawzi, Shayeghi, and Ta also provide a hierarchy of semidefinite strong-converse bounds for assisted quantum capacity [7, Sec. 6]; the table does not rank that hierarchy.

There is a particularly small asymptotic gap at high damping. Abdelhadi et al. give the triple-rail achievable rate $L_3(p) = (1-p)\log_2(3)/3$ [8, Sec. 2.2]. Writing $q = 1 - p \downarrow 0$, the elementary expansion $p^{p/q} = e^{-1}(1 + O(q))$ gives

$$C_Z(p) = \frac{q}{e\ln 2} + O(q^2), \qquad \lim_{p\uparrow 1}\left(\frac{C_Z(p)}{L_3(p)} - 1\right) = \frac{3}{e\ln 3} - 1 = 0.00457489\ldots.$$

The converse threshold is therefore about 0.4575% above this known achievable rate in the high-damping limit. This is a relative asymptotic comparison, not a uniform gap for every p.

2 Order and support estimates

We use the sandwiched-divergence data-processing theorem: for finite CPTP Λ and $\alpha > 1$,

$$\widetilde{Q}_\alpha(\Lambda(\rho)\|\Lambda(\sigma)) \le \widetilde{Q}_\alpha(\rho\|\sigma) \tag{8}$$

on supported positive inputs. The corresponding extended-real divergence inequality holds on all density inputs. This theorem is proved in [1, 2]. It also applies to unnormalized positive inputs: normalize the two inputs separately and use preservation of both traces and the scaling $\widetilde{Q}_\alpha(a\rho\|b\sigma) = a^\alpha b^{1-\alpha}\widetilde{Q}_\alpha(\rho\|\sigma)$.

Lemma 2 (Comparison order)**.** *If $\rho, \sigma, \tau \ge 0$, $\sigma \le c\tau$ with $c > 0$, and $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$, then*

$$\widetilde{Q}_\alpha(\rho\|\tau) \le c^{\alpha-1}\widetilde{Q}_\alpha(\rho\|\sigma), \qquad \alpha > 1. \tag{9}$$

Proof. Put $t = (\alpha-1)/\alpha \in (0,1)$. The Löwner–Heinz inequality gives $\sigma^t \le c^t\tau^t$. Therefore $C = c^{-t/2}\tau^{-t/2}\sigma^{t/2}$ satisfies $CC^\dagger \le P_\tau$, where P_τ is the support projection, and $\|C\|_\infty \le 1$. The supports obey $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma \subseteq \operatorname{supp}\tau$. Hence

$$\tau^{-t/2}\rho\tau^{-t/2} = c^t C(\sigma^{-t/2}\rho\sigma^{-t/2})C^\dagger.$$

The Schatten-α ideal inequality $\|CXC^\dagger\|_\alpha \le \|C\|_\infty^2\|X\|_\alpha$ proves (9), since $t\alpha = \alpha - 1$. □

Full-rank comparisons on the original input space suffice for every separable infimum used below. Given a supported separable comparison σ and a full-rank product density ζ, set $\sigma_\delta = (1-\delta)\sigma + \delta\zeta$ for $0 < \delta < 1$. Lemma 2 gives

$$\widetilde{D}_\alpha(\rho\|\sigma_\delta) \le \widetilde{D}_\alpha(\rho\|\sigma) - \log(1-\delta). \tag{10}$$

Thus the restricted infimum equals the original one. For full-rank σ there is a finite C_0 with $\rho \le C_0\sigma$. Every CP branch then preserves the necessary support containment. This argument does not require full rank on a subsequently enlarged record space.

Two consequences of (8) will be needed for adaptive recombination. A complete instrument (R_j) gives

$$\sum_j \widetilde{Q}_\alpha(R_j\rho R_j^\dagger\|R_j\sigma R_j^\dagger) \le \widetilde{Q}_\alpha(\rho\|\sigma), \tag{11}$$

by recording the outcome in a direct sum. Discarding a direct-sum flag gives

$$\widetilde{Q}_\alpha\left(\sum_j X_j\middle\|\sum_j Y_j\right) \le \sum_j \widetilde{Q}_\alpha(X_j\|Y_j). \tag{12}$$

These statements include zero-mass branches, with their contributions zero. They do not require normalizing individual outcomes.

3 A complete finite phase comparison

Fix $n \ge 1$. Identify computational basis strings with subsets of $[n] = \{1, \dots, n\}$. Let H_w be the span of $|S\rangle$ with $|S| = w$. For $0 \le r \le w \le n$, define

$$K_L|S\rangle = \begin{cases} \binom{w}{r}^{-1/2}|S \setminus L\rangle, & L \subseteq S,\ |L| = w-r, \\ 0, & \text{otherwise,} \end{cases} \qquad \mathcal{N}_{w,r}(X) = \sum_{|L|=w-r} K_L X K_L^\dagger. \tag{13}$$

Distinct inputs containing a fixed L have distinct survivors. Thus $K_L^\dagger K_L$ is diagonal. Each S contains exactly $\binom{w}{r}$ such loss sets, so $\sum_L K_L^\dagger K_L = I_{H_w}$. The tensor damping map restricted to H_w is exactly

$$\mathcal{A}_p^{\otimes n}\big|_{H_w} = \bigoplus_{r=0}^{w} \pi_{w,r}\mathcal{N}_{w,r}, \qquad \pi_{w,r} = \binom{w}{r} q^r p^{w-r}. \tag{14}$$

The output subspaces H_r are orthogonal. These binomial probabilities sum to one and are independent of the input and all side memories. Zero-probability sectors are retained with zero mass.

Let $\omega = e^{2\pi i/3}$ and let $z \in \{0,1,2\}^n$ be uniform. Define

$$|\chi_z\rangle = \sum_{|S|=w} \omega^{-\sum_{i\in S} z_i}|S\rangle, \qquad |b_z\rangle = \binom{n}{r}^{-1/2} \sum_{|T|=r} \omega^{-\sum_{i\in T} z_i}|T\rangle, \tag{15}$$

and

$$\mathcal{S}_{w,r}(X) = 3^{-n}\sum_z \langle\chi_z|X|\chi_z\rangle |b_z\rangle\langle b_z|. \tag{16}$$

Character orthogonality gives $3^{-n}\sum_z |\chi_z\rangle\langle\chi_z| = I_{H_w}$. Each b_z is normalized. Consequently (16) is a complete finite measure-and-prepare channel. Its output can be placed at Bob using a classical message and local preparation. It preserves separability across the entire laboratory cut with arbitrary old memories.

Lemma 3 (Physical CP domination)**.** *For $0 \le r \le w \le n$,*

$$\mathcal{N}_{w,r} \le_{\mathrm{CP}} c_{n,w,r}\mathcal{S}_{w,r}, \qquad c_{n,w,r} = \frac{\binom{n}{r}}{\binom{w}{r}} \ge 1. \tag{17}$$

The inequality remains valid after tensoring any memory identity.

Proof. Use an unnormalized maximally entangled vector to define Choi matrices. A matrix entry in the Choi vector of $3^{-n/2}|b_z\rangle\langle\chi_z|$, labeled by (S,T), has phase $\omega^{\sum_i z_i(1_S(i)-1_T(i))}$ and magnitude $(3^n\binom{n}{r})^{-1/2}$. Averaging over z separates the signatures $1_S - 1_T$ coordinatewise. The three values $-1, 0, 1$ are distinct modulo three, so this separation has no signature collisions.

All signatures with no negative coordinate have $T \subseteq S$ and are exactly 1_L, where $L = S \setminus T$ and $|L| = w - r$. Their positive rank-one blocks equal the Choi blocks of the K_L in (13), with coefficient $1/\binom{n}{r}$ in place of $1/\binom{w}{r}$. Every other signature contributes a positive block disjoint from these blocks. Multiplication by $\binom{n}{r}/\binom{w}{r}$ therefore leaves the Choi matrix of $\mathcal{N}_{w,r}$ plus a positive remainder. Choi positivity is equivalent to complete positivity, proving (17) and its memory extensions. The cases $w = 0$ and $r = 0$ follow from the same formulas. □

For any supported positive pair X, Y on the input including side memories, Lemma 2, (17), and data processing imply

$$\widetilde{Q}_\alpha(\mathcal{N}_{w,r}(X)\|\mathcal{S}_{w,r}(Y)) \le c_{n,w,r}^{\alpha-1}\widetilde{Q}_\alpha(X\|Y). \tag{18}$$

Here the intermediate comparison is $\mathcal{N}_{w,r}(Y)$, whose support contains that of $\mathcal{N}_{w,r}(X)$; CP domination then supplies the support in $\mathcal{S}_{w,r}(Y)$.

4 Coherent recording and an actual block comparison map

Suppose Alice holds the n qubits to be transmitted. Adjoin a local record W by the isometry $V|S\rangle = |S\rangle|S\rangle_W$, leaving all old memories unchanged. Apply the physical channel to the first factor and call the recorded output Ω_X. Measure W in the binary Fourier basis indexed by $a \in \{0,1\}^n$ and apply the announced phase correction Z^a to Bob's new output qubits. Denote this complete recovery map by $\mathcal{R}$.

$$\mathcal{R}(\Omega_X) = (\mathrm{id} \otimes \mathcal{A}_p^{\otimes n})(X). \tag{19}$$

To verify the identity with arbitrary memory coefficients, expand X in $|S\rangle\langle U|$. A loss set $L \subseteq S \cap U$ gives Fourier factor $2^{-n}(-1)^{a\cdot(S-U)}$. Its surviving strings are $S \setminus L$ and $U \setminus L$, whose phase difference is the same. Bob's correction cancels it. Summing the 2^n outcomes restores each original matrix entry. Thus recovery is an equality of linear maps, including all physical loss terms.

Let $\mathcal{P}$ pinch the local record W by its $n+1$ weight projectors. The weight-w input operators are

$$X_w = (I \otimes \Pi_w) V X V^\dagger (I \otimes \Pi_w), \qquad Y_w = (I \otimes \Pi_w) V Y V^\dagger (I \otimes \Pi_w).$$

Their channel inputs lie in H_w, while the record stays in Alice's side memory. For a separable Y, the Y_w are positive separable operators. Define the recorded comparison and its recovered version by

$$\kappa_Y = \bigoplus_{w,r} \pi_{w,r} \mathcal{S}_{w,r}(Y_w), \qquad \mathcal{T}_{n,p}(Y) = \mathcal{R}(\kappa_Y). \tag{20}$$

The direct sum denotes Alice's orthogonal w record and Bob's orthogonal survivor spaces. Any explicitly attached flags are discarded by complete local maps before recovery. Recording, weight measurement, the finite phase measurement and preparation, and recovery are complete operations. Hence $\mathcal{T}_{n,p}$ is a concrete CPTP separability-preserving map with the same input and output laboratory spaces as the physical block.

Put $\alpha = 1 + s$ and

$$M_{n,s}(p) = \max_{0 \le w \le n} \sum_{r=0}^{w} \pi_{w,r} c_{n,w,r}^s, \qquad F_{n,s}(p) = (n+1)^{1+s} M_{n,s}(p). \tag{21}$$

Both quantities are at least one. Exact direct-sum scaling and (18) give

$$\begin{aligned} \widetilde{Q}_{1+s}(\mathcal{P}\Omega_X \| \kappa_Y) &\le \sum_w \widetilde{Q}_{1+s}(X_w \| Y_w) \sum_r \pi_{w,r} c_{n,w,r}^s \\ &\le M_{n,s}(p) \widetilde{Q}_{1+s}(X \| Y). \end{aligned} \tag{22}$$

The last step is (11) for the recorded weight instrument. The physical and comparison probabilities are matched, so $\pi_{w,r}^{1+s} \pi_{w,r}^{-s} = \pi_{w,r}$ on occupied sectors.

Pinching satisfies $\Omega_X \le (n+1)\mathcal{P}\Omega_X$. The support containments above and this order imply $\operatorname{supp} \Omega_X \subseteq \operatorname{supp} \kappa_Y$. Congruence by $\kappa_Y^{-s/(2(1+s))}$ preserves the order. For positive ordered matrices, each ordered eigenvalue is no larger, so the trace of their $(1+s)$ powers is ordered. Therefore

$$\widetilde{Q}_{1+s}(\Omega_X \| \kappa_Y) \le (n+1)^{1+s} \widetilde{Q}_{1+s}(\mathcal{P}\Omega_X \| \kappa_Y). \tag{23}$$

This uses scalar eigenvalue powers, not operator monotonicity of a power greater than one. Applying recovery data processing to (22)–(23) proves the actual block estimate

$$\widetilde{Q}_{1+s}((\mathrm{id} \otimes \mathcal{A}_p^{\otimes n})(X) \| \mathcal{T}_{n,p}(Y)) \le F_{n,s}(p) \widetilde{Q}_{1+s}(X \| Y). \tag{24}$$

It holds with arbitrary finite side memories and supported positive inputs, including the subnormalized branches of an adaptive protocol.

5 The finite type bound

Define the bit charge

$$c_{n,s}(p) = \frac{1}{s}\log F_{n,s}(p) = \frac{1+s}{s}\log(n+1) + \frac{1}{s}\log M_{n,s}(p). \tag{25}$$

We next prove, throughout $0 \le p \le 1$ and $0 < s \le 1$,

$$c_{n,s}(p) \le nB_s(p) + \frac{2+s}{s}\log(n+1). \tag{26}$$

For interior p, let $h(x) = -x\log x - (1-x)\log(1-x)$, with the continuous endpoint convention, and let $D_{\mathrm{bin}}(v\|q) = v\log(v/q) + (1-v)\log((1-v)/p)$. For $w > 0$, set $u = w/n$ and $v = r/w$. The standard binomial coefficient bound $\binom{N}{k} \le 2^{Nh(k/N)}$ yields

$$\begin{aligned}\pi_{w,r}c^s_{n,w,r} &= \binom{n}{r}^s \binom{w}{r}^{1-s} q^r p^{w-r} \\ &\le 2^{ns[,h(uv)-uh(v)-(u/s)D_{\mathrm{bin}}(v\|q)\,]}.\end{aligned} \tag{27}$$

To see the coefficient bound directly, the kth term of the binomial distribution at parameter k/N is at most one; rearranging gives the inequality, including $k = 0, N$. Both coefficient exponents in (27) are nonnegative. The $w = 0$ sum equals one. Summing at most $n+1$ terms gives $M_{n,s}(p) \le (n+1)2^{ns\beta_s(p)}$, where

$$\beta_s(p) = \max_{0\le u,v\le 1}\{h(uv) - uh(v) - (u/s)D_{\mathrm{bin}}(v\|q)\}. \tag{28}$$

For $v > 0$ put $J(v) = [h(v) + D_{\mathrm{bin}}(v\|q)/s]/v$ and $x = uv$. The expression is $h(x) - xJ(v)$ with $0 \le x \le v$. Its derivative at $u = 0$ is positive infinite, whereas at $u = 1$ it is $\log(1-v) - D_{\mathrm{bin}}(v\|q)/s < 0$, interpreted by a limit at $v = 1$. The unconstrained entropy-conjugate maximizer $x = (1 + 2^{J(v)})^{-1}$ is therefore admissible, and the maximum is $\log(1 + 2^{-J(v)})$. Differentiation gives

$$J'(v) = \frac{\log p + (s-1)\log(1-v)}{sv^2}. \tag{29}$$

For $0 < s < 1$ its unique minimum is $v_s = 1 - p^{1/(1-s)}$, at which $J(v_s) = ((1-s)\log v_s - \log q)/s$. This evaluates (28) as the interior clause in (5). At $s = 1$, J decreases to its minimum at $v = 1$ and gives $\log(1+q)$. At $v = 0$ the original expression is nonpositive, so that boundary cannot increase the maximum. Thus $\beta_s = B_s$, proving (26) for interior p.

At $p = 0$, only $r = w$ has nonzero mass, and $M_{n,s}(0) = \max_w \binom{n}{w}^s \le 2^{ns}$ by the binomial theorem. At $p = 1$, only $r = 0$ has nonzero mass and $c_{n,w,0} = 1$, so $M_{n,s}(1) = 1$. These facts prove (26) with the stated endpoint constants, for every allowed s.

6 Iteration on the complete adaptive tree

Consider a protocol R whose calls are n-qubit physical blocks. Evaluate the *same* tree on a comparison input Y, using the same local instruments and the same outcome-dependent continuations, but replacing each physical block by $\mathcal{T}_{n,p}$. Call the resulting output $R^{\mathrm{cmp}}[Y]$. It is normalized and separable whenever Y is normalized and separable. Support containment propagates by CP branch maps, by (24), and by positive sums.

Lemma 4 (Unconditional tree iteration)**.** *For supported positive X, Y,*

$$\widetilde{Q}_{1+s}(R[X]\|R^{\mathrm{cmp}}[Y]) \le F_{n,s}(p)^{c(R)}\widetilde{Q}_{1+s}(X\|Y). \tag{30}$$

Consequently, if $c(R) \le m$ and $\widetilde{E}_{1+s}(X) \le e$, then

$$\widetilde{E}_{1+s}(R[X]) \le e + mc_{n,s}(p) \tag{31}$$

for density X on the entire input cut.

Proof. Induct on the finite tree. The terminal case is equality. At a local node, (12) first bounds the recombined output quantity by the sum of the continuation quantities. Apply induction to each continuation, replace its exponent by the maximum count using $F_{n,s} \ge 1$, and then apply (11). The resulting factor is exactly $F_{n,s}^{\max_j c(R_j)}$. At a block-call node, induction followed by (24) multiplies the continuation factor by one further $F_{n,s}$. This proves (30) with every branch retained.

Taking logarithms gives $\widetilde{D}_{1+s}(R[X]\|R^{\mathrm{cmp}}[Y]) \le \widetilde{D}_{1+s}(X\|Y) + c(R)c_{n,s}$. For any $\varepsilon > 0$, choose a supported separable comparison with input divergence less than $e + \varepsilon$, using (10) if needed. Its comparison output is a normalized separable state, so the same bound holds for the output infimum. Let ε decrease to zero and use $c_{n,s} \ge 0$ to replace $c(R)$ by m. This proves (31). □

7 Independent finite lanes and the Bell test

Fix the original P, ρ, p, s, m, d, e. For any integer $n \ge 1$, run n independent copies of this same tree and initial state. Pad each history after its last physical call to m calls: prepare a fresh qubit in $|0\rangle$ locally, send it, and discard the received $|0\rangle$ locally. Since $\mathcal{A}_p(|0\rangle\langle 0|) = |0\rangle\langle 0|$, this padding leaves the original output unchanged. It is complete on every history.

Schedule all copies by call number. Before block call j, execute in each lane its own local tree segment up to the next call; then send all n current qubits through $\mathcal{A}_p^{\otimes n}$. This gives a finite m-block protocol retaining every original feedback chronology. Its decoded output is exactly $P[\rho]^{\otimes n}$, including failures. Independent copies of any classical choices are used in distinct lanes.

Tensor products of supported separable comparisons give

$$\widetilde{E}_{1+s}(\rho^{\otimes n}) \le n\widetilde{E}_{1+s}(\rho) \le ne. \tag{32}$$

Indeed functional-calculus powers factor on tensor products, so $\widetilde{Q}_{1+s}(\rho^{\otimes n}\|\sigma^{\otimes n}) = \widetilde{Q}_{1+s}(\rho\|\sigma)^n$; approximate the input infimum. Lemma 4 therefore gives

$$\widetilde{E}_{1+s}(P[\rho]^{\otimes n}) \le ne + mc_{n,s}(p). \tag{33}$$

Across the total two-laboratory cut, $\Phi_d^{\otimes n}$ is a Bell projector of dimension d^n after local relabeling. For any normalized product vector $|a\rangle|b\rangle$, its acceptance is at most d^{-n} by Cauchy–Schwarz. Convexity proves the same bound for every normalized separable comparison, including comparisons with correlations among all lanes inside each laboratory. The actual acceptance is G^n, where $G = G_d(P[\rho])$.

For $G > 0$, binary-test data processing and omission of the nonnegative failure term give, for every separable comparison with acceptance a,

$$\widetilde{Q}_{1+s}(P[\rho]^{\otimes n}\|\sigma) \ge (G^n)^{1+s}a^{-s} \ge G^{n(1+s)}d^{ns}.$$

If $a = 0$, the supported divergence is infinite whenever $G > 0$, so the inequality remains valid. Infimizing gives

$$\widetilde{E}_{1+s}(P[\rho]^{\otimes n}) \geq n \log d + \frac{n(1+s)}{s} \log G. \tag{34}$$

The Bell test is a CPTP map used for a mathematical data-processing bound; it need not be a local measurement within the physical protocol.

Combining (33), (34), and (26), then dividing by n, yields for *every* finite $n \geq 1$

$$\log d + \frac{1+s}{s} \log G \leq e + mB_s(p) + \frac{m(2+s)}{s} \frac{\log(n+1)}{n}. \tag{35}$$

Keep the original code and all its parameters fixed and let n increase. Since $\log(n+1)/n \to 0$, exponentiation gives (7). The extra cap by one follows from density normalization. If $G = 0$, the bound holds directly. This proves the finite part of Theorem 1, including $m = 0$ and both damping endpoints.

8 Fixed-budget output closure

Let $\mathcal{O}_{p,s,m,d,e}$ be the set of all $P[\rho]$ with arbitrary finite initial dimensions and with the conditions (6). Its closure is taken in the trace norm on the fixed terminal space $\mathbb{C}^d \otimes \mathbb{C}^d$. Every member has the same numerical upper bound in (7). If $\eta_j \to \eta$ in trace norm,

$$|G_d(\eta_j) - G_d(\eta)| \leq \|\Phi_d\|_\infty \|\eta_j - \eta\|_1 = \|\eta_j - \eta\|_1 \longrightarrow 0.$$

Positivity and trace one also pass to the limit. Hence (7) holds throughout $\overline{\mathcal{O}_{p,s,m,d,e}}$, proving the closure part of Theorem 1. The common resource bound e remains a condition on every approximating code. This closure specifies a set of finite-dimensional logical outputs; no separate membership theorem for infinite-memory protocols is used.

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Finite adaptive secret-key converses for the amplitude-damping channel

Abstract

We prove strong, weak, binary-entropy, and correctness/secrecy converses for finite adaptive qubit amplitude-damping protocols with a charged independent classical seed. The final key fidelity is optimized against a state on the actual complete Eve register, containing every channel environment, the entire immutable public transcript, and the prior Eve register. Optimized normalized separable comparison flags give the strong exponent. An extensive privacy-test bound gives the weak and entropy inequalities. All four conclusions extend to the actual output closure at fixed budgets and fixed finite complete Eve.

The converse has rate threshold $C_Z(p) = \log_2[1 + (1-p)p^{p/(1-p)}]$, the capacity of the classical Z-channel. In particular $C_Z(1/2) = \log_2(5/4) \simeq 0.321928$. The argument builds on the private-state formulation of Horodecki, Horodecki, Horodecki, and Oppenheim [7], the privacy-test converse of Wilde, Tomamichel, and Berta [6], and the amortized framework of Kaur and Wilde [3]. The limit of the comparison rate as its order parameter tends to one is $\log_2(2-p)$, the known max-relative entanglement value of Rigovacca et al. [5, Proposition 2]. The improvement here comes from the order-dependent finite-code bounds and their $C_Z(p)$ limit at order parameter zero, with the seed and Eve specified in the model below.

1 The finite physical model and the four converses

All Hilbert spaces in the construction are finite dimensional over $\mathbb{C}$, and all logarithms are base two. A density operator is positive semidefinite and has trace one. Fix $0 < p < 1$, put $q = 1 - p$, and define

$$A_0 = |0\rangle\langle 0| + \sqrt{q}|1\rangle\langle 1|, \qquad A_1 = \sqrt{p}|0\rangle\langle 1|, \qquad \mathcal{A}_p(X) = \sum_{j=0}^{1} A_j X A_j^\dagger. \tag{1}$$

A physical call moves Alice's input qubit to Bob while leaving both old memories untouched. Its actual dilation is

$$V_p = \sum_{j=0}^{1} A_j \otimes |j\rangle_{E_{\mathrm{call}}}. \tag{2}$$

The environment in (2) is retained by Eve. It is a quantum register: off-diagonal terms between different physical loss labels are not dephased.

A protocol is a finite tree of complete local instruments, public outcome-dependent continuations, and physical calls. At a local Alice node the operators K_j satisfy $\sum_j K_j^\dagger K_j = I$; outcome j is sent publicly and selects the continuation. The Bob definition is analogous. Each private Kraus index can be retained coherently in the owning laboratory before a public outcome is sent. Local dimensions may change. The maximum call count is zero at a terminal node, the maximum of the continuation counts at a local node, and one plus the continuation count at a physical call. It is bounded by an integer $m \geq 0$ on *every* history. Complete instruments include all failure outcomes.

Eve's history space is defined recursively. It is a singleton at a terminal node, the direct sum $\bigoplus_j E(P_j)$ at a public local outcome, and $\mathbb{C}^2 \otimes E(P_1)$ at a physical call. A prior Eve register E_0 is included as an additional tensor factor. The actual honest–Eve output is obtained by summing all public branches with a classical copy of j in Eve's transcript, and by applying (2) at each call. Equivalently, the call recursion acts on the block $A_j X A_{j'}^\dagger$ for each pair of Eve labels (j, j'). Publicly

distinct histories have zero off-diagonal blocks. Physically distinct loss histories within the same public history retain their quantum coherences. This fixes the full Eve register rather than a selected marginal of it.

The initial state is exactly

$$\left(\sum_{j=1}^{D_0} a_j |jj\rangle\langle jj|_{J_A J_B}\right) \otimes \rho_{A_0} \otimes \rho_{B_0} \otimes \rho_{E_0}, \quad a_j \geq 0, \quad \sum_j a_j = 1, \quad D_0 \leq 2^k. \tag{3}$$

Here D_0 is a positive integer, $k \in \mathbb{R}$ is the alphabet budget, and the seed is independent of both local ancillas and prior Eve. There is no additional shared entangled input. Any local mixed ancilla or local private randomness can be purified within its own laboratory. The actual seed value is not announced as a public outcome.

At the terminal node, Alice and Bob choose basis-label functions with values in $\{1, \ldots, d\}$, measure those labels privately, and discard local work. The full output is a normalized classical–classical–quantum state

$$\rho_{K_A K_B E} = \sum_{i,j=1}^{d} |ij\rangle\langle ij| \otimes \rho_E^{ij}, \qquad \rho_E^{ij} \geq 0, \quad \sum_{i,j} \operatorname{Tr} \rho_E^{ij} = 1. \tag{4}$$

Arbitrary finite local processing can precede this readout. Every outcome, including a failure or unequal key pair, is included in (4). Define

$$\begin{aligned} C_d &= \frac{1}{d} \sum_{i=1}^{d} |ii\rangle\langle ii|, \\ F(\rho, \sigma) &= \|\sqrt{\rho}\sqrt{\sigma}\|_1^2, \qquad G(\rho) = \sup_{\xi_E \text{ density}} F(\rho, C_d \otimes \xi_E). \end{aligned} \tag{5}$$

The ideal comparison ξ_E is normalized on the *same complete* Eve space. The actual disagreement and Alice-key secrecy errors are

$$\epsilon(\rho) = \sum_{i \neq j} \operatorname{Tr} \rho_E^{ij}, \qquad \delta(\rho) = \frac{1}{2} \left\| \rho_{K_A E} - \frac{I_{K_A}}{d} \otimes \rho_E \right\|_1. \tag{6}$$

The marginal ρ_E in (6) is the actual one.

For $0 < z < 1$, set

$$B_z(p) = \log\left[1 + \left(\frac{q}{[1 - p^{1/(1-z)}]^{1-z}}\right)^{1/z}\right], \qquad C_Z(p) = \log[1 + qp^{p/q}]. \tag{7}$$

For $x \in [0, 1]$, write

$$L_d(x) = \sup_{\lambda \geq 0} \left\{ \lambda x - \log\left[1 + \frac{2^\lambda - 1}{d}\right] \right\}, \qquad h(x) = -x \log x - (1 - x) \log(1 - x). \tag{8}$$

The function h uses its continuous endpoint values.

Theorem 1 (Finite seed-key converses)**.** *Let $0 < p < 1$, $m \in \mathbb{N}_0$, $d \geq 2$, and $k \in \mathbb{R}$. For every finite code in the model above, let ρ be its complete-Eve output and let $G = G(\rho)$. Then for every*

$0 < z < 1$,

$$G \le \min\{1, 2^{-z(\log d - k - mB_z(p))}\}, \tag{9}$$

$$L_d(G) \le k + mC_Z(p), \tag{10}$$

$$G \log d - h(G) \le k + mC_Z(p), \tag{11}$$

$$\epsilon(\rho) + \delta(\rho) \ge 1 - \min\{1, 2^{-(z/2)(\log d - k - mB_z(p))}\}. \tag{12}$$

Consequently (12) *also holds with any larger explicit correctness and secrecy tolerances in place of the actual errors.*

Theorem 2 (Actual fixed-budget finite-Eve output closure)**.** *Fix the parameters of Theorem 1 and a finite Eve space E. Let $\mathcal{F}_{p,m,d,k}(E)$ consist of the actual code outputs placed in E by a* bijective *register relabeling of the complete space $E(P) \otimes E_0$. Define*

$$\mathcal{O}_{p,m,d,k}(E) = \{\rho : \ \forall \eta > 0 \ \exists \tau \in \mathcal{F}_{p,m,d,k}(E), \ \|\rho - \tau\|_1 < \eta\}. \tag{13}$$

All four inequalities of Theorem 1 hold for every density $\rho \in \mathcal{O}_{p,m,d,k}(E)$, with its own actual full-Eve marginals and errors. The parameters p, m, d, k, E are fixed in (13)*, and z is fixed when applying a strong inequality.*

For a sequence of these finite codes with $m \to \infty$, seed budgets $k_m = o(m)$, and correctness and secrecy errors tending to zero, $G_m \to 1$ by the fidelity–trace-distance inequalities. Equation (11) and $h(G_m) \le 1$ therefore give

$$\limsup_{m\to\infty} \frac{\log d_m}{m} \le C_Z(p).$$

The bound is thus a rate converse for the stated seed-assisted finite-code model, as well as a finite-error inequality.

Comparison with earlier bounds

Let h_2 denote binary entropy in bits. Two analytic benchmarks are the reverse-coherent-information achievable rate

$$L_{\mathrm{RCI}}(p) = \max_{0\le u\le 1} [h_2(u) - h_2(pu)]$$

and the balanced-squashing upper bound

$$U_{\mathrm{sq}}(p) = \max_{0\le u\le 1} [h_2((1-p/2)u) - h_2(pu/2)].$$

These appear in [8, Supplementary Note 5, Eqs. (S212), (S228)–(S231)]. The achievable entanglement rate is also an achievable secret-key rate. For comparison with the finite-code rate consequence proved above, at $p = 1/2$ the values per physical channel use are

Quantity	Value
Reverse coherent information L_{RCI}	$0.271553\ldots$
Present converse threshold $C_Z = \log_2(5/4)$	$0.321928\ldots$
Balanced-squashing bound $U_{\mathrm{sq}} = \log_2(4/3)$	$0.415037\ldots$
Max-relative bound $\log_2(3/2)$ [5]	$0.584963\ldots$

The first value is the one-variable maximum above. The third is exact: $h_2(3u/4) - h_2(u/4)$ is strictly concave on $(0,1)$, and its derivative vanishes at $u = 4/9$. The substitution $u = 1/2$ in [8, Eq. (S232)] instead gives $0.410869\ldots$; it does not maximize the preceding expression. Pirandola already identified this issue and evaluated the retained maximization [11, footnote 3 and Theorem IV.1]. Thus $C_Z(1/2)$ improves both analytic upper benchmarks in the table while leaving a gap above the achievable rate. Fawzi, Shayeghi, and Ta also provide a hierarchy of semidefinite strong-converse bounds for assisted quantum capacity [9, Sec. 6]; the table does not rank that hierarchy.

There is a particularly small asymptotic gap at high damping. Abdelhadi et al. give the triple-rail achievable rate $L_3(p) = (1-p)\log_2(3)/3$ [10, Sec. 2.2]. Writing $q = 1 - p \downarrow 0$, the elementary expansion $p^{p/q} = e^{-1}(1 + O(q))$ gives

$$C_Z(p) = \frac{q}{e \ln 2} + O(q^2), \qquad \lim_{p\uparrow 1}\left(\frac{C_Z(p)}{L_3(p)} - 1\right) = \frac{3}{e\ln 3} - 1 = 0.00457489\ldots.$$

The converse threshold is therefore about 0.4575% above this known achievable rate in the high-damping limit. This is a relative asymptotic comparison, not a uniform gap for every p.

2 Sandwiched quantities and supported comparison order

For $\alpha > 1$, write $t = (\alpha - 1)/\alpha$ and define

$$\begin{aligned} \widetilde{Q}_\alpha(X\|Y) &= \mathrm{Tr}[(Y^{-t/2}XY^{-t/2})^\alpha], \\ \widetilde{D}_\alpha(\rho\|\sigma) &= \frac{1}{\alpha-1}\log \widetilde{Q}_\alpha(\rho\|\sigma), \\ \widetilde{E}_\alpha(A:B)_\rho &= \inf_{\sigma\in \mathrm{SEP}(A:B)} \widetilde{D}_\alpha(\rho\|\sigma). \end{aligned} \tag{14}$$

Inverse powers act on the support. Unsupported comparisons have divergence $+\infty$. The separable set consists of normalized finite convex combinations of product densities across the entire honest cut. The sandwiched data-processing theorem for every $\alpha > 1$ is [1, 2]

$$\widetilde{Q}_\alpha(\Lambda(X)\|\Lambda(Y)) \le \widetilde{Q}_\alpha(X\|Y) \tag{15}$$

for CPTP Λ and supported positive X, Y. Normalizing each positive input separately establishes the unnormalized version, using trace preservation and $\widetilde{Q}_\alpha(aX\|bY) = a^\alpha b^{1-\alpha}\widetilde{Q}_\alpha(X\|Y)$.

For positive X, Y, Z, if $Y \le cZ$ with $c > 0$ and $\mathrm{supp}\, X \subseteq \mathrm{supp}\, Y$, then

$$\widetilde{Q}_\alpha(X\|Z) \le c^{\alpha-1}\widetilde{Q}_\alpha(X\|Y). \tag{16}$$

Indeed Löwner–Heinz gives $Y^t \le c^t Z^t$ and makes $C = c^{-t/2}Z^{-t/2}Y^{t/2}$ a contraction. Support containment implies $Z^{-t/2}XZ^{-t/2} = c^t C(Y^{-t/2}XY^{-t/2})C^\dagger$. The Schatten-$\alpha$ ideal inequality proves (16). This uses no commutativity. Mixing any supported separable σ with a full-rank product density ζ gives

$$\widetilde{D}_\alpha(\rho\|(1-\eta)\sigma + \eta\zeta) \le \widetilde{D}_\alpha(\rho\|\sigma) - \log(1-\eta), \quad 0 < \eta < 1. \tag{17}$$

Thus full-rank comparisons on the original input space approximate the infimum. A finite scalar domination $\rho \le C_0\sigma$ then propagates all input support conditions through CP branches.

For later use, recording a complete instrument (R_j) and then discarding a direct-sum flag gives

$$\sum_j \widetilde{Q}_\alpha(R_j X R_j^\dagger \| R_j Y R_j^\dagger) \le \widetilde{Q}_\alpha(X\|Y),$$

$$\widetilde{Q}_\alpha\left(\sum_j X_j \Big\| \sum_j Y_j\right) \le \sum_j \widetilde{Q}_\alpha(X_j\|Y_j). \tag{18}$$

Both inequalities follow from (15) and the exact direct-sum formula. Zero branches are retained with zero contribution.

3 Finite loss sectors and the phase channel

Fix $n \ge 1$. Identify computational strings with subsets of $[n]$ and let $H_w = \mathrm{span}\{|S\rangle : |S| = w\}$. For $0 \le r \le w \le n$, put

$$K_L|S\rangle = \begin{cases} \binom{w}{r}^{-1/2} |S \setminus L\rangle, & L \subseteq S,\ |L| = w-r, \\ 0, & \text{otherwise}, \end{cases} \qquad \mathcal{N}_{w,r}(X) = \sum_{|L|=w-r} K_L X K_L^\dagger. \tag{19}$$

Each input S occurs for $\binom{w}{r}$ loss sets. For a fixed L, distinct inputs containing L have distinct survivors. It follows that $\sum_L K_L^\dagger K_L = I_{H_w}$ and

$$\mathcal{A}_p^{\otimes n}\big|_{H_w} = \bigoplus_{r=0}^{w} \pi_{w,r} \mathcal{N}_{w,r}, \qquad \pi_{w,r} = \binom{w}{r} q^r p^{w-r}. \tag{20}$$

The survivor spaces are orthogonal, and the probabilities are input-independent and sum to one. This is a formula for the honest channel after tracing its quantum loss environment; Eve's physical dilation remains (2) throughout the actual code.

Let $\omega = e^{2\pi i/3}$ and choose $a \in \{0,1,2\}^n$ uniformly. Set

$$|\chi_a\rangle = \sum_{|S|=w} \omega^{-\sum_{i\in S} a_i} |S\rangle, \qquad |b_a\rangle = \binom{n}{r}^{-1/2} \sum_{|T|=r} \omega^{-\sum_{i\in T} a_i} |T\rangle,$$

and define the finite measure-and-prepare channel

$$\mathcal{S}_{w,r}(X) = 3^{-n} \sum_a \langle\chi_a|X|\chi_a\rangle |b_a\rangle\langle b_a|. \tag{21}$$

Character orthogonality gives $3^{-n}\sum_a |\chi_a\rangle\langle\chi_a| = I_{H_w}$, and b_a is normalized. Thus this is a complete CPTP channel. Measuring at Alice, sending the finite classical outcome, and preparing b_a at Bob implements it with only local operations and classical communication. It preserves the separable cone on every whole-memory cut.

Its unnormalized Choi matrix separates into positive rank-one blocks labeled by $1_S - 1_T$ modulo three. The coordinates $-1, 0, 1$ are distinct modulo three. Signatures without negative coordinates occur exactly when $T \subseteq S$, with $L = S \setminus T$; their coefficients are $1/\binom{n}{r}$, whereas the physical normalized loss coefficients are $1/\binom{w}{r}$. The remaining signatures give disjoint positive blocks. Consequently

$$\mathcal{N}_{w,r} \le_{\mathrm{CP}} c_{n,w,r} \mathcal{S}_{w,r}, \qquad c_{n,w,r} = \frac{\binom{n}{r}}{\binom{w}{r}} \ge 1. \tag{22}$$

Choi positivity proves complete positivity of the remainder, so (22) remains true with arbitrary memory identities. Combining (22), (16), and data processing gives

$$\widetilde{Q}_\alpha(\mathcal{N}_{w,r}(X)\|\mathcal{S}_{w,r}(Y)) \le c_{n,w,r}^{\alpha-1}\widetilde{Q}_\alpha(X\|Y) \tag{23}$$

on supported positive inputs including all side memories.

4 Optimized normalized flags and coherent input records

Fix $0 < z < 1$ and set

$$\alpha = \frac{1}{1-z} > 1, \qquad \alpha - 1 = \alpha z. \tag{24}$$

The following comparison flags are functions only of n, w, p, z:

$$Z_w(z) = \sum_{r=0}^{w} \pi_{w,r} c_{n,w,r}^z, \qquad \mu_{w,r} = \frac{\pi_{w,r} c_{n,w,r}^z}{Z_w(z)}. \tag{25}$$

They are positive and sum to one; $Z_w(z) \ge 1$. The physical flag law continues to be $\pi_{w,r}$. For these *comparison* masses,

$$\sum_r \pi_{w,r}^\alpha \mu_{w,r}^{1-\alpha} c_{n,w,r}^{\alpha-1} = Z_w(z)^\alpha. \tag{26}$$

To verify it, substitute (25): the power of c becomes $(\alpha-1)(1-z) = z$, and the power of π becomes one. More generally Hölder gives $\sum_r a_r \mu_r^{1-\alpha} \ge (\sum_r a_r^{1/\alpha})^\alpha$ for $a_r = \pi_r^\alpha c_r^{\alpha-1}$, with equality at $\mu_r \propto a_r^{1/\alpha} = \pi_r c_r^z$. Only this explicit normalized comparison is needed below.

For an arbitrary whole-cut input, Alice makes the local basis record $V|S\rangle = |S\rangle|S\rangle_W$ before the block channel. Write Ω_X for the recorded physical output. Fourier measurement of W in the binary basis, public announcement of its label b, and Bob's Z^b phase correction define a complete recovery map $\mathcal{R}$ with

$$\mathcal{R}(\Omega_X) = (\mathrm{id} \otimes \mathcal{A}_p^{\otimes n})(X). \tag{27}$$

For the matrix unit $|S\rangle\langle T|$ and a common lost set L, the measured branch has factor $2^{-n}(-1)^{b\cdot(S-T)}$; the surviving strings $S \setminus L, T \setminus L$ have the same phase difference. Bob's correction cancels it and the sum of all 2^n outcomes is exactly the original output. Operator-valued coefficients on old memories cause no change to this calculation.

Pinch W by its $n+1$ weight projections and call this map $\mathcal{P}$. Let X_w, Y_w be the recorded weight branches of supported positive inputs X, Y. Their channel inputs are in H_w. Define

$$\kappa_Y = \bigoplus_{w,r} \mu_{w,r} \mathcal{S}_{w,r}(Y_w), \qquad \mathcal{T}_{n,p,z}(Y) = \mathcal{R}(\kappa_Y). \tag{28}$$

The w record is Alice's; survivor flags are Bob's. These flags may be retained explicitly or represented by their orthogonal subspaces, then discarded by complete local maps. All constituent maps in (28) are finite and complete. Thus $\mathcal{T}_{n,p,z}$ is a literal CPTP separability-preserving channel on the block's original laboratory coordinates.

The direct-sum formula, (23), and (26) give

$$\begin{aligned} \widetilde{Q}_\alpha(\mathcal{P}\Omega_X\|\kappa_Y) &\le \sum_w \widetilde{Q}_\alpha(X_w\|Y_w) Z_w(z)^\alpha \\ &\le \big(\max_w Z_w(z)\big)^\alpha \widetilde{Q}_\alpha(X\|Y). \end{aligned} \tag{29}$$

The last step is the complete recorded input instrument in (18). In particular, actual and comparison input-weight probabilities are charged by this same direct-sum formula rather than by selective averages of an entanglement measure.

The pinching order $\Omega_X \le (n+1)\mathcal{P}\Omega_X$ gives support containment in κ_Y. Congruence by $\kappa_Y^{-z/2}$ and positive eigenvalue monotonicity yield $\widetilde{Q}_\alpha(\Omega_X\|\kappa_Y) \le (n+1)^\alpha \widetilde{Q}_\alpha(\mathcal{P}\Omega_X\|\kappa_Y)$. Use recovery data processing to obtain

$$\widetilde{Q}_\alpha((\mathrm{id}\otimes\mathcal{A}_p^{\otimes n})(X)\|\mathcal{T}_{n,p,z}(Y)) \le F_{n,z}\widetilde{Q}_\alpha(X\|Y), \quad F_{n,z} = (n+1)^\alpha(\max_w Z_w(z))^\alpha. \tag{30}$$

The bit charge is consequently

$$C_{n,z}(p) = \frac{\log F_{n,z}}{\alpha-1} = \frac{1}{z}\big[\log(n+1) + \log\max_w Z_w(z)\big]. \tag{31}$$

All statements hold on arbitrary finite side-memory spaces and all supported positive subnormalized branches. The recovery and comparison operations concern honest states used in the bound; they never replace the actual channel environment in the key criterion.

5 The scalar estimate and the order limit

For $w > 0$, put $u = w/n$, $v = r/w$, and let $D_{\mathrm{bin}}(v\|q) = v\log(v/q) + (1-v)\log((1-v)/p)$. The binomial bound $\binom{N}{k} \le 2^{Nh(k/N)}$ follows by bounding a binomial probability at parameter k/N by one. Therefore

$$\begin{aligned}\pi_{w,r}c_{n,w,r}^z &= \binom{n}{r}^z\binom{w}{r}^{1-z} q^r p^{w-r} \\ &\le 2^{nz[\,,h(uv)-uh(v)-(u/z)D_{\mathrm{bin}}(v\|q)\,]}.\end{aligned} \tag{32}$$

Both binomial exponents are nonnegative. The $w = 0$ sum is one. There are at most $n+1$ terms, so

$$C_{n,z}(p) \le nB_z(p) + \frac{2}{z}\log(n+1), \tag{33}$$

once the maximum of the expression in brackets is evaluated.

For $v > 0$, set $J(v) = [h(v) + D_{\mathrm{bin}}(v\|q)/z]/v$ and $x = uv$. The objective is $h(x) - xJ(v)$ on $0 \le x \le v$. Its derivative in u is positive infinite at zero and equals $\log(1-v) - D_{\mathrm{bin}}(v\|q)/z < 0$ at one. Thus the unconstrained entropy-conjugate maximum is admissible and is $\log(1 + 2^{-J(v)})$. Direct differentiation gives

$$J'(v) = \frac{\log p + (z-1)\log(1-v)}{zv^2}.$$

Its unique minimum is $v_z = 1 - p^{1/(1-z)}$, and $J(v_z) = ((1-z)\log v_z - \log q)/z$. Substitution gives exactly (7). The $v = 0$ boundary contributes at most zero. This proves (33) analytically.

For the weak bound the needed order limit is

$$\lim_{z\downarrow 0} B_z(p) = C_Z(p). \tag{34}$$

Indeed let $f(z) = \ln q - (1-z)\ln[1 - p^{1/(1-z)}]$. It satisfies $f(0) = 0$ and $f'(0) = \ln q + (p/q)\ln p$. The inner power in (7) is $\exp(f(z)/z)$, which tends to $qp^{p/q}$. This proves (34). The block overhead in (33) is removed at each fixed $z > 0$ before this order limit is taken.

6 Actual adaptive iteration and the charged preparation

For a finite n-qubit block tree R, evaluate a comparison tree using exactly the same local instruments and continuations, with each physical block replaced by $\mathcal{T}_{n,p,z}$. Call its output $R^{\mathrm{cmp}}[Y]$. It is a normalized separable state on the output cut whenever Y is normalized and separable. Support is preserved at each CP branch and block, and upon recombination.

Induction on the complete tree, using (18) at every local node and (30) at every block-call node, gives

$$\widetilde{Q}_\alpha(R[X]\|R^{\mathrm{cmp}}[Y]) \le F_{n,z}^{c(R)}\widetilde{Q}_\alpha(X\|Y). \tag{35}$$

At a local node first sum the continuation bounds, then replace their call exponents by the maximum using $F_{n,z} \ge 1$, and finally apply the complete input instrument bound. At a call node multiply the continuation factor by one additional $F_{n,z}$. Thus the induction uses the maximum call budget on every public history and all unnormalized branch masses. Approximating the input separable infimum using (17) and taking logarithms yields

$$\widetilde{E}_\alpha(R[\sigma]) \le e + mC_{n,z}(p) \quad \text{if } \widetilde{E}_\alpha(\sigma) \le e, \quad c(R) \le m. \tag{36}$$

We now construct the coherent upper preparation for (3). Use the vector

$$|a\rangle_{J_A J_B} = \sum_j \sqrt{a_j}|j\rangle|j\rangle. \tag{37}$$

Alice privately copies j into a local work register. Tracing that work recovers the classical seed of (3) exactly, while retaining it keeps a pure upper preparation. Bob and Alice retain local purifications of their independent ancillas. An independent prior Eve density may be purified with its additional purifier placed in Alice's work; tracing that purifier recovers precisely the given prior Eve. All subsequent original operations act as identities on these new work registers. Tracing them at the end gives the original honest–Eve output, entry by entry, including all its physical-loss coherences. No seed label is added to the public transcript.

The explicit normalized separable comparison for (37) is $\tau_0 = D_0^{-1}\sum_j |jj\rangle\langle jj|$. On the support of the seed vector it acts as I/D_0. The sandwiched operator is $D_0^t|a\rangle\langle a|$, and therefore

$$\widetilde{D}_\alpha(|a\rangle\langle a|\|\tau_0) = \log D_0 \le k. \tag{38}$$

The private copy is an Alice isometry; independent local factors do not increase this exhibited comparison cost. Tracing prior Eve is local to the Alice–Eve side for this input calculation and leaves the comparison separable on the honest cut. Hence the actual retained initial honest state has $\widetilde{E}_\alpha \le k$. For n independent preparations the tensor comparison gives an initial bound nk. This charges the actual seed alphabet without an entropy assumption on its law.

7 A coherent lift of the complete-Eve key output

Retain all local work and local copies of public records. At a public Alice node the global dilation is the sum over j of the original local operator times $|j\rangle_{H_A}|j\rangle_{H_B}|j\rangle_E$; analogously for Bob. Tracing Eve dephases the honest copies, so this honest retained operation is a complete local instrument followed by an ordinary public message. The copies do not supply coherent nonlocal communication. Channel loss labels are retained exactly through (2).

Coherently implement the terminal local labels and retain orthogonal local key copies C_A, C_B. A purification of the complete output can be written

$$|\psi\rangle = \sum_{i,j} |ij\rangle_{K_A K_B} |ij\rangle_{C_A C_B} |\chi_{ij}\rangle_{S'E}. \tag{39}$$

The vectors are subnormalized and $\rho_E^{ij} = \mathrm{Tr}_{S'} |\chi_{ij}\rangle\langle\chi_{ij}|$. Tracing honest work gives the original (4); tracing Eve gives the retained coherent honest state $\widehat{\rho}$. Public record copies and private key copies are part of the honest shields S_A, S_B. The lift is the usual finite private-state construction, with squared fidelity as in [3, Sec. 3.3].

Fix a normalized ξ_E. Direct-sum fidelity gives

$$f(\rho, C_d \otimes \xi_E) = \frac{1}{\sqrt{d}} \sum_i \| \sqrt{\rho_E^{ii}} \sqrt{\xi_E} \|_1, \qquad f = \sqrt{F}. \tag{40}$$

After a finite local enlargement of the shields, Uhlmann's theorem chooses for each i a normalized purification $|\eta_i\rangle$ of the *same* ξ_E, in the key-copy subspace labeled ii, whose overlap with the actual branch is the corresponding nonnegative summand in (40). Fix a purification $|\theta\rangle_{SE}$ of ξ_E. Extending the purifying isometries to finite shield unitaries gives $|\eta_i\rangle = U_i|\theta\rangle$. Define a controlled twisting unitary $U = \sum_{i,j} |ij\rangle\langle ij| \otimes U_{ij}$ with $U_{ii} = U_i$ and arbitrary unitary choices on the other blocks. Then

$$|\Gamma\rangle = \frac{1}{\sqrt{d}} \sum_i |ii\rangle U_i |\theta\rangle, \qquad \gamma = \mathrm{Tr}_E |\Gamma\rangle\langle\Gamma| = U(\Phi_d \otimes \theta_S) U^\dagger \tag{41}$$

is a private state. Its overlap with (39) is exactly (40), so fidelity monotonicity under tracing Eve gives

$$F(\widehat{\rho}, \gamma) \geq F(\rho, C_d \otimes \xi_E). \tag{42}$$

The twisting unitaries define an ideal mathematical target and are not additional operations performed by the original protocol. The shield enlargement is finite because ξ_E has finite rank. Taking candidates approaching the supremum in (5) is sufficient throughout.

The privacy-test projection for this target is $\Pi = U(\Phi_d \otimes I_S) U^\dagger$. It passes γ with probability one. Measuring this projection in (42) gives

$$\mathrm{Tr}\, \Pi \widehat{\rho} \geq F(\rho, C_d \otimes \xi_E). \tag{43}$$

For a normalized product vector with local expansions $|a\rangle = \sum_i |i\rangle|a_i\rangle$ and $|b\rangle = \sum_j |j\rangle|b_j\rangle$, its acceptance is

$$\frac{1}{d} \left\| \sum_i U_{ii}^\dagger (|a_i\rangle \otimes |b_i\rangle) \right\|^2 \leq \frac{1}{d} \left(\sum_i \|a_i\| \|b_i\| \right)^2 \leq \frac{1}{d}. \tag{44}$$

The first inequality is the triangle inequality and the second is Cauchy–Schwarz. Convexity extends (44) to every normalized separable state, with no factor depending on the shield dimensions. It also applies to tensor products of privacy tests as a private test of the product key dimension.

8 Finite independent lanes and the strong bound

Fix one original code and prepare n independent copies, each with its own seed and local preparations. Pad every history to m physical calls with a fresh ground-state input. A padding call has output ground state and ground-state environment, because $V_p|0\rangle = |0\rangle|0\rangle$. Its local preparation and

discard have only a constant public outcome. The expanded public history is determined by the original history; adding its padding labels and the zero loss labels embeds the original full Eve space isometrically into the padded space.

Optimal ideal-key fidelity is unchanged by this Eve isometry. For an arbitrary candidate on the larger space, compress it to the isometry's image and normalize. If its compressed trace is $r \le 1$, its root fidelity to the supported actual state equals $\sqrt{r}$ times the normalized candidate fidelity. Normalizing therefore cannot decrease that candidate fidelity. The reverse inequality follows by embedding an original candidate. This proves equality without discarding any actual Eve information.

Schedule the padded copies by scalar call number: execute each lane's original local segment until its next call, then apply one $\mathcal{A}_p^{\otimes n}$ block to their n inputs. All local feedback depends on the same original public outcomes. The resulting tree is finite and has at most m block calls. Retain both honest public copies. Local removal of unused padding ports and regrouping of the retained records recover $\widehat{\rho}^{\otimes n}$ from its honest output. All these changes are local coordinate isometries or complete local discards. A basis injection is decoded by its inverse on its image; on the orthogonal complement a fixed local output completes the decoder to a trace-preserving channel. The complement has zero probability on the constructed state. Thus the decoding does not introduce a resource premise or postselection.

Applying (36) with the independently charged input (38), and then local decoding data processing, gives

$$\widetilde{E}_\alpha(\widehat{\rho}^{\otimes n}) \le nk + mC_{n,z}(p). \tag{45}$$

The actual decoded key–Eve outputs are the independent product of the original full outputs, up to the described Eve isometry. For any fixed candidate ξ_E, let $g = F(\rho, C_d \otimes \xi_E)$ and use its private test from (43). The product test on the n retained outputs has acceptance at least g^n. Every separable comparison across the total two-laboratory cut has acceptance at most d^{-n} by (44). For $g > 0$, binary sandwiched data processing gives

$$\widetilde{E}_\alpha(\widehat{\rho}^{\otimes n}) \ge n \log d + \frac{n}{z} \log g. \tag{46}$$

The positive failure term in the binary $\widetilde{Q}_\alpha$ was omitted. A zero comparison acceptance gives infinite divergence and satisfies the same bound. The zero-fidelity case is immediate.

Take the supremum of candidates, combine (45)–(46), and use (33). For $G > 0$ the result is, for every finite $n \ge 1$,

$$\log d + \frac{1}{z} \log G \le k + mB_z(p) + \frac{2m}{z} \frac{\log(n+1)}{n}. \tag{47}$$

Let n increase with the original code and z fixed. The remainder vanishes. Exponentiation and $G \le 1$ prove (9), including $m = 0$. Every construction before this numerical limit is finite.

9 The extensive weak privacy-test bound

For the same private target, let $a = \operatorname{Tr} \Pi \widehat{\rho}$, so $a \ge g$. On n independent honest outputs measure the individual commuting projections $\Pi_1, \dots, \Pi_n$, and let T count successes. For any normalized separable comparison τ across the *total* honest cut and any subset $J \subseteq [n]$,

$$\operatorname{Tr} \tau \prod_{i \in J} \Pi_i \le d^{-|J|}. \tag{48}$$

The product is a private test of key dimension $d^{|J|}$; tracing the remaining lanes preserves separability. For $\lambda \geq 0$ the nonnegative expansion of the exponential moment then gives

$$\operatorname{Tr} \tau \, 2^{\lambda T} = \sum_{J \subseteq [n]} (2^{\lambda} - 1)^{|J|} \operatorname{Tr} \tau \prod_{i \in J} \Pi_i \leq \left[1 + \frac{2^{\lambda} - 1}{d} \right]^n . \tag{49}$$

No conditioning on a successful nonlocal test is involved.

Let P, Q be the two measured classical distributions. Their ordinary relative entropy obeys

$$D(P\|Q) \geq \lambda \, \mathbb{E}_P T - \log \mathbb{E}_Q 2^{\lambda T}. \tag{50}$$

For example, normalize $Q_x 2^{\lambda T(x)}$ to a probability law Q'; nonnegativity of $D(P\|Q')$ gives the identity and inequality in (50). If Q vanishes on positive P mass, the inequality holds with infinite divergence. For supported classical distributions and $\alpha > 1$, Jensen's inequality also gives

$$D(P\|Q) = \sum_x P_x \log(P_x/Q_x) \leq \frac{1}{\alpha - 1} \log \sum_x P_x (P_x/Q_x)^{\alpha - 1} = \widetilde{D}_{\alpha}(P\|Q). \tag{51}$$

The mean success count on the actual product output is $na \geq ng$. Thus sandwiched data processing, (49), and (50)–(51) prove

$$\widetilde{D}_{\alpha}(\widehat{\rho}^{\otimes n}\|\tau) \geq n \left\{ \lambda g - \log \left[1 + \frac{2^{\lambda} - 1}{d} \right] \right\}.$$

Infimize over the same normalized separable comparisons, take candidates approaching G, and then take the supremum in λ. This yields the extensive lower bound

$$n L_d(G) \leq \widetilde{E}_{\alpha}(\widehat{\rho}^{\otimes n}). \tag{52}$$

It is a direct ordinary-divergence test estimate bounded above by the sandwiched resource through (51).

Combine (52), (45), and (33). At each fixed $z > 0$, let n increase to obtain $L_d(G) \leq k + m B_z(p)$. Only after this finite-lane remainder has vanished, let z decrease to zero and apply (34). This proves (10).

For completeness the scalar conjugate in (8) is

$$L_d(x) = \begin{cases} 0, & 0 \leq x \leq 1/d, \\ x \log(dx) + (1 - x) \log \dfrac{1 - x}{1 - 1/d}, & 1/d < x \leq 1. \end{cases} \tag{53}$$

Differentiating the objective gives $x - 2^{\lambda}/(d - 1 + 2^{\lambda})$. If $x \leq 1/d$, its maximum is at zero. If $1/d < x < 1$, the maximum satisfies $2^{\lambda} = x(d-1)/(1-x)$; substitution gives (53). The case $x = 1$ follows as $\lambda \to \infty$. For $x > 1/d$, expanding the second clause gives $L_d(x) = x \log d - h(x) - (1 - x) \log(1 - 1/d) \geq x \log d - h(x)$. For $0 < x \leq 1/d$, $h(x) \geq x \log(1/x) \geq x \log d$; at zero it is immediate. Thus

$$L_d(x) \geq x \log d - h(x), \qquad 0 \leq x \leq 1. \tag{54}$$

Equations (10) and (54) prove (11).

10 Actual correctness and secrecy against whole Eve

For (4), correct Bob's label to Alice's mathematically by

$$\rho^{\mathrm{corr}} = \sum_i |ii\rangle\langle ii| \otimes \sum_j \rho_E^{ij}.$$

The wrong-label blocks contribute total trace ϵ; the added diagonal blocks contribute the same trace. Both differences are positive on their respective orthogonal key blocks, so

$$\tfrac{1}{2}\|\rho - \rho^{\mathrm{corr}}\|_1 = \epsilon(\rho). \tag{55}$$

The isometry $|i\rangle \mapsto |ii\rangle$ identifies ρ^{corr} with the actual Alice–Eve marginal. It also sends the uniform Alice state tensored with actual ρ_E to $C_d \otimes \rho_E$. Consequently

$$\tfrac{1}{2}\|\rho^{\mathrm{corr}} - C_d \otimes \rho_E\|_1 = \delta(\rho), \qquad \tfrac{1}{2}\|\rho - C_d \otimes \rho_E\|_1 \le \epsilon + \delta. \tag{56}$$

Fuchs–van de Graaf [4] gives $\sqrt{F(\rho, C_d \otimes \rho_E)} \ge (1 - \epsilon - \delta)_+$. The actual ρ_E is one of the candidates in (5), so

$$\epsilon + \delta \ge 1 - \sqrt{G}. \tag{57}$$

Taking the square root of (9) gives $\sqrt{G} \le \min\{1, 2^{-(z/2)(\log d - k - mB_z(p))}\}$. Combining this with (57) proves (12). The factor $1/2$ in the exponent follows from the squared fidelity convention; the Eve marginal in the secrecy error was never optimized or reduced. This completes Theorem 1.

11 Continuity on the actual fixed-Eve output family

Bijective relabeling of a complete Eve register preserves trace norm, fidelity, the actual marginal, and both errors. The ideal candidates are in bijection under the same relabeling. Thus every member of $\mathcal{F}_{p,m,d,k}(E)$ satisfies Theorem 1.

For densities ρ, τ on the same complete output space, let $u = \|\rho - \tau\|_1$. For every fixed density σ,

$$|F(\rho, \sigma) - F(\tau, \sigma)| \le 2\sqrt{u}, \qquad |G(\rho) - G(\tau)| \le 2\sqrt{u}. \tag{58}$$

To prove the first bound, Uhlmann and Fuchs–van de Graaf give purifications $|r\rangle, |t\rangle$ with $\|r - t\|^2 = 2(1 - f(\rho, \tau)) \le u$. For a purification of σ optimal against r, the difference of overlaps is at most $\|r - t\| \le \sqrt{u}$; the other overlap is bounded by $f(\tau, \sigma)$. Interchanging the states gives $|f(\rho, \sigma) - f(\tau, \sigma)| \le \sqrt{u}$. Since the two root fidelities are in $[0, 1]$, squaring gives the first inequality in (58). It is uniform in σ, so taking the two suprema over $C_d \otimes \xi_E$ gives the second inequality.

Every finite output is ccq. If $\tau_j \to \rho$ in trace norm, each off-diagonal key matrix entry of ρ is zero because its absolute value is bounded by $\|\rho - \tau_j\|_\infty \le \|\rho - \tau_j\|_1$. Thus ρ is also ccq. Its G is the limit of the finite-output values by (58). The functions L_d and h are continuous on $[0, 1]$ by (53). Hence (9), (10), and (11) pass to every output in (13). Finally (57) applies to this ccq limit with its actual full-Eve marginal, proving its (12). This proves Theorem 2.

The closure is explicitly the norm closure of actual finite outputs with fixed complete finite Eve and fixed numerical budgets. No continuity of entanglement measures in a growing shield dimension, and no membership claim for an additional infinite protocol model, is required.

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A finite-ensemble obstruction to Holevo-rate visible compression

Abstract

We construct a finite ensemble of rank-eight states on $\mathbb{C}^9$ whose visible quantum compression rate is strictly larger than its Holevo information, even with arbitrary finite shared classical randomness at each blocklength. The model excludes initially shared entanglement and averages over the shared seed before the trace-norm error. A finite-block inequality gives a strong converse and a strict rate gap above $\log_2(9/8)$.

All logarithms below are base two. The shared seed is independent of the source and is averaged before the trace norm, as specified below.

1 Statement

The coding model permits shared classical randomness and a quantum message, but no initially shared entanglement or additional communication. The shared seed can have arbitrary finite size at each blocklength.

A length-n visible code has a message space M_n of dimension d_n, a finite shared seed s of source-independent probability $q_n(s)$, encoded density matrices $\tau_{x^n,s}$ on M_n, and CPTP decoding maps $\mathcal{D}_{n,s}$. Its reproduced state and error are

$$\widehat{W}_{x^n} = \sum_s q_n(s)\mathcal{D}_{n,s}(\tau_{x^n,s}),$$
$$\varepsilon_n = \sum_{x^n} p^n(x^n)\frac{1}{2}\|\widehat{W}_{x^n} - W_{x^n}\|_1.$$

The rate $R_{\mathrm{vis},q,\mathrm{cr}}$ is the infimum of $\limsup_n n^{-1}\log_2 d_n$ over sequences of such codes with $\varepsilon_n \to 0$. The Holevo information is $\chi(p,W) = S(\sum_x p_x W_x) - \sum_x p_x S(W_x)$, with $S(\rho) = -\operatorname{Tr}(\rho\log_2\rho)$.

There is an explicit finite ensemble of rank-eight mixed states on $\mathbb{C}^9$, with full-support probabilities, for which

$$\chi(p,W) = \log_2(9/8) \quad \text{and} \quad R_{\mathrm{vis},q,\mathrm{cr}}(p,W) \geq \frac{2}{3}\log_2(45/37) = \log_2(9/8) + \frac{1}{3}\log_2(12800/12321). \tag{A}$$

The last summand is strictly positive. Thus the identity $R_{\mathrm{vis},q,\mathrm{cr}}(p,W) = \chi(p,W)$ does not hold for every finite ensemble in this model. This contradicts the communication-rate assertion of Winter's Conjecture 23 [Win02, Section VII, p. 11], even after removing its bound on the amount of shared randomness. Equations (15)–(16) of [Win02] define the unassisted quantum visible-coding model. Section III introduces shared randomness for classical sources; Conjecture 23 proposes the quantum extension. Here the reproduced state is averaged over that randomness before the error norm.

Hayashi [Hay06, Section 10.8, Lemma 10.5] gives the Holevo lower bound in the shared-randomness model. Equality in his Theorem 10.6 assumes commuting signal states. The noncommuting ensemble constructed below lies outside that hypothesis.

In fact, every n-block code for this ensemble, with $d = \dim M_n$ and the seed-averaged block error ε_n, obeys

$$(1-\varepsilon_n)^2 \leq d^{3/2}(37/45)^n. \tag{B}$$

Thus the lower bound is a strong converse.

2 An explicit finite ensemble

Let $\zeta = e^{2\pi i/3}$, and fix the standard basis $e_1, \dots, e_9$. There are two families of labels:

- The nine basis vectors e_j, each with probability $1/90$.
- For every $r = (r_2, \dots, r_9) \in \{0, 1, 2\}^8$, the vector

$$v_r = \frac{1}{3}\left(e_1 + \sum_{j=2}^{9} \zeta^{r_j} e_j\right),$$

  each with probability $1/7290$.

There are $9 + 3^8 = 6570$ labels, all with positive probability. Their probabilities sum to $9/90 + 3^8/7290 = 1/10 + 9/10 = 1$. For the vector u_x associated with a label, put

$$Q_x = |u_x\rangle\langle u_x|, \qquad P_x = I_9 - Q_x, \qquad W_x = P_x/8. \tag{C}$$

Every W_x is a density operator of rank eight. This one-letter dimension, label set and probability distribution are fixed, independent of n.

Write F for the swap on $\mathbb{C}^9 \otimes \mathbb{C}^9$, and $Z = \sum_{j=1}^{9} |jj\rangle\langle jj|$. The pure projectors have moments

$$\mathbb{E}_x Q_x = I_9/9, \qquad \mathbb{E}_x(Q_x \otimes Q_x) = \frac{I + F}{90}. \tag{D}$$

Here is a direct calculation of the second identity. In the phase family, set $r_1 = 0$. Its matrix entry with row indices (i, j) and column indices (k, l) is

$$\frac{1}{81}\mathbb{E}_r \zeta^{r_i + r_j - r_k - r_l}.$$

For every coordinate $h \geq 2$, the exponent of r_h is an integer in $[-2, 2]$. Averaging over a cube root gives zero unless that exponent is zero. If all these exponents vanish, the exponent for coordinate 1 vanishes as well, since the total exponent over coordinates is zero. Therefore the entry is $1/81$ precisely when the index multisets $\{i, j\}$ and $\{k, l\}$ agree, and is zero otherwise. Equivalently,

$$\mathbb{E}_{\text{phase}}(Q \otimes Q) = \frac{I + F - Z}{81}.$$

The basis-family moment is $Z/9$. With respective total weights $9/10$ and $1/10$, these give

$$\frac{9}{10}\frac{I + F - Z}{81} + \frac{1}{10}\frac{Z}{9} = \frac{I + F}{90}.$$

Taking a partial trace gives the first identity in (D), because $\mathrm{Tr}_2\, I = 9I$ and $\mathrm{Tr}_2\, F = I$. No enumeration of the phase vectors is needed.

It follows from (D) and (C) that

$$\mathbb{E}_x W_x = I_9/9, \qquad \chi(p, W) = \log_2 9 - \log_2 8 = \log_2(9/8). \tag{E}$$

Expanding $(I - Q_x) \otimes (I - Q_x)$ gives

$$\mathbb{E}_x(P_x \otimes P_x) = aI + bF, \qquad a = \frac{71}{90}, \quad b = \frac{1}{90}. \tag{F}$$

Indeed, the coefficient of I is $1 - 2/9 + 1/90 = 71/90$.

3 A channel-dimension lemma

Let $\Psi : \mathcal{L}(\mathbb{C}^d) \to \mathcal{L}(\mathbb{C}^h)$ be any CPTP map. For a fixed basis of its input, let $E_{ij} = |i\rangle\langle j|$. Its unnormalized Choi operator is

$$J_\Psi = \sum_{i,j=1}^{d} E_{ij} \otimes \Psi(E_{ij}).$$

Complete positivity and trace preservation give

$$J_\Psi \geq 0, \qquad \mathrm{Tr}_{\mathrm{out}}\, J_\Psi = I_d, \qquad \mathrm{Tr}\, J_\Psi = d. \tag{G}$$

We claim

$$\mathrm{Tr}\, J_\Psi^2 \leq d \min\{d, h\}. \tag{H}$$

Positivity and trace d give $\|J_\Psi\|_\infty \leq d$, so $\mathrm{Tr}\, J_\Psi^2 \leq d^2$.

For the other bound, write $J_\Psi = (J_{bb'})_{b,b'=1}^{h}$ in an output basis. The positive diagonal blocks sum to I_d, so $0 \leq J_{bb} \leq I_d$. For a vector $v = \sum_b v_b \otimes |b\rangle$, positivity gives

$$|\langle v_b, J_{bb'} v_{b'}\rangle| \leq \sqrt{\langle v_b, J_{bb} v_b\rangle}\sqrt{\langle v_{b'}, J_{b'b'} v_{b'}\rangle}.$$

This follows by applying Cauchy–Schwarz to $J_\Psi^{1/2}(v_b \otimes |b\rangle)$ and $J_\Psi^{1/2}(v_{b'} \otimes |b'\rangle)$. Consequently

$$\begin{aligned}\langle v, J_\Psi v\rangle &\leq \left(\sum_b \sqrt{\langle v_b, J_{bb} v_b\rangle}\right)^2 \\ &\leq h \sum_b \langle v_b, J_{bb} v_b\rangle \leq h \sum_b \|v_b\|^2 = h\|v\|^2.\end{aligned}$$

Thus $J_\Psi \leq hI$ and $\mathrm{Tr}\, J_\Psi^2 \leq h\, \mathrm{Tr}\, J_\Psi = hd$. Combining the two bounds proves (H).

This lemma applies to every channel, with arbitrary ancillary environments already included in the map.

4 A tensor-product support-score bound

Fix any CPTP decoder $\Phi : \mathcal{L}(\mathbb{C}^d) \to \mathcal{L}((\mathbb{C}^9)^{\otimes n})$. For x^n, put $P_{x^n} = \bigotimes_{t=1}^n P_{x_t}$ and

$$A_{x^n} = \Phi^*(P_{x^n}), \qquad L(\Phi) = \mathbb{E}_{x^n} \lambda_{\max}(A_{x^n}).$$

Since Φ^* is positive and unital, $0 \leq A_{x^n} \leq I_d$. For any density operator τ_{x^n},

$$\mathbb{E}_{x^n}\, \mathrm{Tr}[P_{x^n} \Phi(\tau_{x^n})] \leq L(\Phi). \tag{I}$$

Cauchy–Schwarz and the eigenvalue inequality give

$$L(\Phi)^2 \leq \mathbb{E}_{x^n} \lambda_{\max}(A_{x^n})^2 \leq \mathbb{E}_{x^n}\, \mathrm{Tr}\, A_{x^n}^2. \tag{J}$$

We compute the last expression explicitly. The adjoint identity implies

$$[A_{x^n}]_{ij} = \mathrm{Tr}[P_{x^n} \Phi(E_{ji})],$$

and therefore

$$\operatorname{Tr} A_{x^n}^2 = \sum_{i,j} \operatorname{Tr}[P_{x^n}\Phi(E_{ij})] \operatorname{Tr}[P_{x^n}\Phi(E_{ji})]. \tag{K}$$

Let $F_d = \sum_{i,j} E_{ij} \otimes E_{ji}$ be the input swap. Equation (K) is equivalent to

$$\operatorname{Tr} A_{x^n}^2 = \operatorname{Tr}\left[(\Phi \otimes \Phi)(F_d)(P_{x^n} \otimes P_{x^n})\right]. \tag{L}$$

For each $T \subseteq [n]$, let F_T swap the two output copies at positions in T and act as identity elsewhere. The source is i.i.d.; hence (F) implies

$$\mathbb{E}_{x^n}(P_{x^n} \otimes P_{x^n}) = \prod_{t=1}^{n}(aI + bF_{\{t\}}) = \sum_{T\subseteq[n]} a^{n-|T|}b^{|T|}F_T. \tag{M}$$

This is the moment of product signals from the fixed finite ensemble. It does not assert that the n-letter supports form a Haar ensemble on the whole output space.

Let $\Phi_T = \operatorname{Tr}_{[n]\setminus T} \circ \Phi$. It is CPTP from a d-dimensional input to a $9^{|T|}$-dimensional output. For $T = \varnothing$, take its output to be the one-dimensional scalar system. The swap identity $\operatorname{Tr}[(X \otimes Y)F] = \operatorname{Tr}(XY)$ gives

$$\begin{aligned} \operatorname{Tr}[(\Phi \otimes \Phi)(F_d)F_T] &= \sum_{i,j} \operatorname{Tr}[\Phi_T(E_{ij})\Phi_T(E_{ji})] \\ &= \operatorname{Tr} J_{\Phi_T}^2. \end{aligned} \tag{N}$$

The second equality follows by multiplying the matrix units in $J_{\Phi_T} = \sum_{i,j} E_{ij} \otimes \Phi_T(E_{ij})$: $\operatorname{Tr}(E_{ij}E_{kl}) = \delta_{jk}\delta_{il}$. In particular, no partial transpose is present. When $T = \varnothing$, $J_{\Phi_T} = I_d$ and this term is exactly d.

Combining (L)–(N) and the channel lemma,

$$\begin{aligned} \mathbb{E}_{x^n} \operatorname{Tr} A_{x^n}^2 &= \sum_{T\subseteq[n]} a^{n-|T|}b^{|T|} \operatorname{Tr} J_{\Phi_T}^2 \\ &\le d \sum_{T\subseteq[n]} a^{n-|T|}b^{|T|} \min\{d, 9^{|T|}\} \\ &\le d^{3/2} \sum_{T\subseteq[n]} a^{n-|T|}b^{|T|}3^{|T|} \\ &= d^{3/2}(a+3b)^n = d^{3/2}(37/45)^n. \end{aligned} \tag{O}$$

Here $\min\{d, h\} \le \sqrt{dh}$ for positive d, h. All expansion coefficients $a^{n-|T|}b^{|T|}$ are nonnegative. The operator $(\Phi \otimes \Phi)(F_d)$ need not be positive; no order comparison involving it was used. Instead each scalar term in (N) is exactly a nonnegative Choi purity.

Equations (J) and (O) establish

$$L(\Phi) \le d^{3/4}(37/45)^{n/2}. \tag{P}$$

5 Arbitrary free shared randomness and the exact error criterion

For an arbitrary code with source-independent seed probabilities $q_n(s)$, encoded states $\tau_{x^n,s}$ and CPTP decoders $D_{n,s}$, put

$$\begin{aligned} V_n &= \sum_{x^n} p^n(x^n) \operatorname{Tr}[P_{x^n}\widehat{W}_{x^n}] \\ &= \sum_s q_n(s) \sum_{x^n} p^n(x^n) \operatorname{Tr}[P_{x^n}\mathcal{D}_{n,s}(\tau_{x^n,s})]. \end{aligned} \tag{Q}$$

For each seed, (I) and (P) apply with the same $d = \dim M_n$. Because the seed is independent of the labels, the conditional source distribution is p^n. Thus, without any restriction on the finite seed's size or probabilities,

$$V_n \le \sum_s q_n(s) L(\mathcal{D}_{n,s}) \le d^{3/4}(37/45)^{n/2}. \tag{R}$$

The target $W_{x^n} = P_{x^n}/8^n$ has $\mathrm{Tr}[P_{x^n} W_{x^n}] = 1$. For two trace-one states ρ, σ and any effect $0 \le E \le I$,

$$|\,\mathrm{Tr}[E(\rho - \sigma)]| \le \tfrac{1}{2}\|\rho - \sigma\|_1.$$

For example, this follows by writing the traceless Hermitian difference as its positive part minus its negative part, each of trace $\frac{1}{2}\|\rho - \sigma\|_1$. Apply it to the effect P_{x^n} and the seed-averaged state $\widehat{W}_{x^n}$. Averaging over labels gives

$$V_n \ge 1 - \varepsilon_n. \tag{S}$$

The error is computed after averaging the seed; no error condition on individual seeds is used.

Equations (R) and (S), followed by squaring nonnegative quantities, prove (B).

6 The strict asymptotic gap

For a vanishing-error code, eventually $\varepsilon_n < 1$. Taking logarithms in (B) gives

$$\frac{1}{n}\log_2 d \ge \frac{2}{3}\log_2(45/37) + \frac{4}{3n}\log_2(1 - \varepsilon_n).$$

The final term tends to zero, so even the lower limit of every valid code's rate is at least the first term. Taking the infimum of upper-limit rates proves the lower bound in (A).

Finally,

$$\begin{aligned}\frac{2}{3}\log_2(45/37) - \log_2(9/8) &= \frac{1}{3}\log_2\left[\frac{(45/37)^2}{(9/8)^3}\right] \\ &= \frac{1}{3}\log_2(12800/12321) > 0,\end{aligned}$$

since $12800 > 12321$. This proves the strict gap for a fixed finite mixed-state ensemble, allowing every independent finite shared seed and every CPTP decoder in the stated model.

7 References

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Finite PR-box simulation of partially entangled qubits

Abstract

For every fixed partially entangled pure two-qubit state, we give an exact simulation of all local projective-measurement correlations using a fixed finite number of binary PR boxes. The construction allows Borel local wirings and unrestricted measurement-independent shared randomness. An exact finite-list sampling lemma yields an explicit, though quantitatively loose, worst-case bound on the number of boxes.

Theorem. Fix $0 < \theta < \pi/4$. Under the measure-theoretic model in which local wirings are Borel stochastic maps and measurement-independent shared randomness is unrestricted, there is a finite integer N_θ and a fixed local wiring of N_θ independent binary PR boxes reproducing

$$P_\theta(a, b \mid x, y) = \mathrm{Tr}\Big[|\psi_\theta\rangle\langle\psi_\theta|(M_{a|x} \otimes M_{b|y})\Big]$$

for every $x, y \in S^2$, where

$$|\psi_\theta\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle, \qquad M_{a|x} = \frac{I + (-1)^a x \cdot \sigma}{2}.$$

The number of boxes is a fixed worst-case bound. One possible choice, with $c = \cos(2\theta)$, is

$$k = \left\lceil \left(\frac{12\pi(1+c)}{1-c} \right)^2 \right\rceil + 1, \qquad N_\theta = k^2. \tag{1}$$

"Fixed wiring" means one preselected map whose arguments include its own local measurement direction. Neither the map, the resource nor its number of copies is chosen afresh as a function of the pair of directions.

The resource restriction is finiteness of the PR-box alphabet and number of uses; it does not impose finite precision or a computational time bound on the Borel local maps or shared randomness. The bound (1) is only a sufficient bound, not a lower bound on the required number of boxes. As θ approaches $\pi/4$ from below it gives $1423^2 = 2{,}024{,}929$ boxes, so it is quantitatively very loose.

A single PR box suffices for projective measurements on a maximally entangled qubit pair [CGMP05], whereas some partially entangled pairs require more than one [BGS05]. Brunner, Gisin, Popescu and Scarani [BGPS08, Section IV] simulate these correlations using PR boxes together with a continuous-input millionaire box, and ask whether finite-input resources suffice. The theorem proves finite binary PR-box sufficiency in the stated Borel model. Its proof uses the local-plus-unbiased decomposition from [RQ23] and the finite-list selection argument below; it does not simulate the millionaire box itself.

1 An exact finite-list sampling lemma

Let S and X be compact metric spaces, let q be a Borel probability on S, and let $f : X \times S \to [0, \infty)$ be continuous with

$$\int_S f(x,t)\, q(dt) = 1, \qquad f(x,t) \le M \tag{2}$$

for every x, t. Suppose S admits a sequence of finite Borel partitions of equal positive q-mass, with mesh tending to zero and cardinality tending to infinity.

Then there is a fixed finite list $Z_1, \dots, Z_N$, drawn independently of x, and a single Borel function

$$I : X \times S^N \times [0,1] \longrightarrow \{1, \dots, N\} \tag{3}$$

such that, with U independent uniform randomness, $Z_{I(x,Z,U)}$ has law $\nu_x(dt) = f(x,t)q(dt)$ for every x.

Proof: the capacity bound. Since each density averages to one, it is at least one somewhere. Uniform continuity on $X \times S$ gives a radius $r > 0$, independent of x, such that each f_x is at least $\epsilon = 1/2$ throughout a ball of radius r. Take an equal-mass partition $C_1, \dots, C_N$ with mesh less than r. For every x, the cell containing a point where $f_x \geq 1$ therefore satisfies

$$f_x \geq \epsilon \quad \text{on that entire cell.} \tag{4}$$

The point need not be selected measurably in x; it is used only to prove the inequality below.

Draw the Z_i independently, with law Nq restricted to C_i. Put $\delta = 1/(2M)$. Choose the partition sufficiently fine and N sufficiently large that

$$1 - e^{-N\delta} \geq \tfrac{1}{2}, \qquad e^{\,1-N\delta} \leq \frac{\epsilon}{N}. \tag{5}$$

These conditions are possible along the partition sequence.

For any Borel $B \subseteq S$, set

$$t = q(B), \quad \beta_i = Nq(B \cap C_i), \quad H(B) = \Pr\{\exists i : Z_i \in B\} = 1 - \prod_i (1 - \beta_i).$$

Here $0 \leq \beta_i \leq 1$ and $\sum_i \beta_i = Nt$. If $t \leq \delta$, AM–GM and concavity give

$$\begin{aligned} H(B) &\geq 1 - (1-t)^N \geq 1 - e^{-Nt} \\ &\geq \frac{t}{\delta}(1 - e^{-N\delta}) \geq Mt \geq \nu_x(B). \end{aligned} \tag{6}$$

If $t \geq \delta$, choose a cell C_d satisfying (4). Then

$$\nu_x(B^c) \geq \epsilon q(C_d \setminus B) = \frac{\epsilon(1 - \beta_d)}{N},$$

whereas

$$\begin{aligned} 1 - H(B) &= \prod_i (1 - \beta_i) \\ &\leq (1 - \beta_d) \exp\left(- \sum_{i \neq d} \beta_i \right) \\ &\leq (1 - \beta_d) e^{1-N\delta} \leq \frac{\epsilon(1 - \beta_d)}{N} \leq \nu_x(B^c). \end{aligned} \tag{7}$$

No division by $1 - \beta_d$ is used, so the case $\beta_d = 1$ is included. Thus, for every x and every Borel set,

$$\nu_x(B) \leq H(B). \tag{8}$$

Proof: sufficiency. This is the finite-list instance of the random-set selection criterion of Strassen [S65, Theorem 5] and Artstein [A83, Theorems 3.1–3.2]. We give the compact-space

argument, including the additional parameter measurability. Write $\Omega = S^N$, let K be the fixed list law, and put

$$E = \{(z_1, \dots, z_N; t) : t = z_i \text{ for at least one } i\}.$$

The relation E is closed in the compact metric space $\Omega \times S$. The probability measures π on this space with $\pi_\Omega = K$ and $\pi(E) = 1$ form a nonempty compact convex set. Nonemptiness follows, for example, by selecting z_1. Compactness follows from compactness of the space of probability measures and the closed constraints; for the support constraint, use Portmanteau on the closed E. Their second marginals form a compact convex set Γ.

For $h \in C(S)$,

$$\sup_{\eta \in \Gamma} \int h \, d\eta = \int_\Omega \max_i h(z_i) \, K(dz). \tag{9}$$

The upper bound is pointwise; equality uses the first index maximizing $h(z_i)$, a Borel choice. After adding a constant, assume $h \geq 0$. Apply (8) to the sets $\{h > t\}$ and integrate over t. Layer-cake integration yields

$$\int h \, d\nu_x \leq \int_\Omega \max_i h(z_i) \, K(dz). \tag{10}$$

The same holds for every real $h \in C(S)$. If $\nu_x \notin \Gamma$, Hahn–Banach separation in the weak topology of finite measures would give a continuous h contradicting (9)–(10). Therefore a coupling with second marginal ν_x exists.

Proof: one Borel selection kernel for all x. The map $x \mapsto \nu_x$ is weakly continuous by (2) and uniform continuity of f. Consequently

$$G = \{(x, \pi) : \pi_\Omega = K, \ \pi_S = \nu_x, \ \pi(E) = 1\} \tag{11}$$

is compact, with nonempty compact fibers G_x, in $X \times \mathcal{P}(\Omega \times S)$. These are compact metric spaces.

Here is an explicit measurable-selection argument for these exact hypotheses. Take a countable dense sequence t_j in $\mathcal{P}(\Omega \times S)$. Recursively choose the smallest $j_n(x)$ for which G_x, intersected with all previously chosen closed balls, intersects $\overline{B}(t_{j_n(x)}, 2^{-n})$. Some such index exists by density. For any fixed finite history and any candidate index, this test is a projection of a compact subset of G, hence is closed in X. Countably partitioning according to the histories and taking the first successful index shows that each j_n is Borel. The resulting nested compact intersections have diameters at most 2^{1-n}. The centers $t_{j_n(x)}$ converge to their unique common point $\pi_x \in G_x$; pointwise limits into a metric space are Borel. Thus $x \mapsto \pi_x$ is Borel.

On E assign $J(z, t)$ to the first index with $t = z_i$, and define the finite subprobability kernels

$$\mu_i^x(D) = \pi_x\{(z, t) \in E : z \in D, \ J(z, t) = i\} \quad (D \subseteq \Omega \text{ Borel}). \tag{12}$$

The sets in (12) are Borel. Evaluation of a Borel set by a probability measure is a Borel function of that measure, so these are Borel kernels in x. They satisfy

$$0 \leq \mu_i^x \leq K, \qquad \sum_i \mu_i^x = K. \tag{13}$$

To obtain jointly Borel Radon–Nikodym densities, take refining finite Borel partitions $\mathcal{D}_n$ of Ω generating its Borel sigma algebra. Such partitions are the atoms generated by the first n sets of a countable base. On an atom D of positive K-mass, define

$$w_i^n(x, z) = \frac{\mu_i^x(D)}{K(D)} \quad (z \in D); \tag{14}$$

put zero on zero-mass atoms. These functions are jointly Borel and lie in $[0,1]$. For each fixed x, they are the conditional expectations of $d\mu_i^x/dK$ on $\mathcal{D}_n$. Bounded martingale convergence and the generating property imply their almost-sure convergence to that derivative. Take their pointwise limsups, normalize the finite nonnegative vector when its sum is positive, and otherwise use $(1,0,\dots,0)$. The resulting functions $w_i(x,z)$ are jointly Borel, sum to one everywhere, and satisfy

$$\mu_i^x(D) = \int_D w_i(x,z)\, K(dz) \tag{15}$$

for every fixed x and every Borel D.

Use $U \sim \mathrm{Unif}[0,1)$ to choose the first i whose cumulative weight exceeds U; define $I = 1$ at $U = 1$. This is a single Borel rule. For any Borel $B \subseteq S$,

$$\Pr(Z_I \in B \mid x) = \int_\Omega \sum_i w_i(x,z)\mathbf{1}_B(z_i)\, K(dz) = \pi_x(\Omega \times B) = \nu_x(B). \tag{16}$$

This completes the lemma.

The measure π_x is used to define the local rule; it is not the shared prior. The actual list law is always the same K. The independent seed U matters: Artstein's counterexample shows that capacity inequalities need not permit deterministic selection from the original list alone. Exceptional sets for (15) may depend on x. Equation (16) nevertheless holds for every fixed x, and the rule is jointly Borel. No simultaneous uncountable intersection of full-measure sets is being assumed.

2 The known local-plus-unbiased decomposition

Put

$$c = \cos(2\theta), \quad s = \sin(2\theta), \quad z = (0,0,1), \quad w_x = (sx_1, -sx_2, x_3).$$

For sign outcomes $A = (-1)^a, B = (-1)^b$, direct expansion of the state gives

$$P_\theta(A,B \mid x,y) = \frac{1 + Acx_3 + Bcy_3 + AB\, w_x \cdot y}{4}. \tag{17}$$

In particular the x_2y_2 term has coefficient $-s$. Throughout, $\mathrm{sgn}(t) = +1$ for $t \geq 0$ and -1 otherwise. Write $\Theta(t) = \max(t,0)$, and use sphere surface measure $d\omega$.

For $\eta \in \{+1,-1\}$, define

$$r_\eta = \frac{cz + \eta w_x}{2}, \qquad p_\eta = \frac{1 + \eta c x_3}{2}.$$

Since

$$\|cz + \eta w_x\|^2 = (1 + \eta c x_3)^2,$$

we have $\|r_\eta\| = p_\eta \geq (1-c)/2 > 0$. Set

$$\rho_x(\lambda) = \frac{\Theta(r_+ \cdot \lambda) + \Theta(r_- \cdot \lambda)}{\pi}, \qquad T_x(\eta \mid \lambda) = \frac{\Theta(r_\eta \cdot \lambda)}{\pi \rho_x(\lambda)}. \tag{18}$$

At $\rho_x = 0$, define $T_x(\pm 1 \mid \lambda) = 1/2$.

For unit vectors v, y,

$$\int_{S^2} \Theta(v \cdot \lambda)\, d\omega = \pi, \qquad \int_{S^2} \frac{\Theta(v \cdot \lambda)}{\pi}\, \mathrm{sgn}(y \cdot \lambda)\, d\omega = v \cdot y. \tag{19}$$

For the second identity, write $\Theta(t) = (t+|t|)/2$. The absolute-value term times the sign is odd almost everywhere. Rotational symmetry and one-dimensional integration give $\int \lambda \operatorname{sgn}(y \cdot \lambda) d\omega = 2\pi y$. The first identity follows by integrating over a hemisphere.

Thus ρ_x is normalized. Sampling it, producing Alice's output with response T_x, and setting Bob's output to $\operatorname{sgn}(y \cdot \lambda)$ gives joint probabilities

$$p_\eta \frac{1 + B(r_\eta/p_\eta) \cdot y}{2} = \frac{1 + \eta c x_3 + B c y_3 + \eta B\, w_x \cdot y}{4}, \tag{20}$$

which are exactly (17).

Now define

$$\rho_z(\lambda) = \frac{\Theta(z \cdot \lambda)}{\pi}, \qquad \widetilde{\rho}_x(\lambda) = \rho_x(\lambda) - c\rho_z(\lambda).$$

Using $|u+v| + |u-v| = 2\max(|u|, |v|)$ in (18) yields

$$\widetilde{\rho}_x(\lambda) = \frac{\big(|w_x \cdot \lambda| - c|z \cdot \lambda|\big)_+}{2\pi}. \tag{21}$$

It is nonnegative, even and continuous, with integral $1 - c$.

Let L use the input-independent density ρ_z, Alice's response T_x, and Bob's sign response. This is local. Let Q use density $\widetilde{\rho}_x/(1-c)$ and the same responses. Then

$$P_\theta = cL + (1-c)Q. \tag{22}$$

Where $\widetilde{\rho}_x > 0$, $|w_x \cdot \lambda| > c|z \cdot \lambda|$, so exactly one of $r_+ \cdot \lambda, r_- \cdot \lambda$ is positive. Therefore the residual's responses are deterministic:

$$A = \operatorname{sgn}(w_x \cdot \lambda), \qquad B = \operatorname{sgn}(y \cdot \lambda). \tag{23}$$

The density in Q is even, while both responses are odd outside null great circles. Since $s > 0$, $w_x \neq 0$ for every x. Consequently both residual marginals are exactly uniform, for every x, y. In particular Q is nonsignalling and is determined by

$$C_Q(x,y) = \int_{S^2} \frac{\widetilde{\rho}_x(\lambda)}{1-c} \operatorname{sgn}(w_x \cdot \lambda) \operatorname{sgn}(y \cdot \lambda)\, d\omega. \tag{24}$$

The local weight $c = \cos(2\theta)$ appears in the earlier local-content analysis of Portmann, Branciard and Gisin [PBG12], whose angle convention uses $\theta/2$ in the state. The analytic Kochen–Specker construction used here is from Renner–Quintino [RQ23, Appendices C and E], with $p = \cos^2\theta$ and $2p - 1 = c$. The calculation above records its marginals in the present conventions.

3 A fixed finite list for the residual

Let $q = d\omega/(4\pi)$ and define the normalized density

$$f_x(\lambda) = \frac{2}{1-c}\big(|w_x \cdot \lambda| - c|z \cdot \lambda|\big)_+. \tag{25}$$

It is jointly continuous on $S^2 \times S^2$, bounded by $M = 2/(1-c)$, and is Lipschitz in λ for Euclidean distance with constant

$$L_0 = \frac{2(1+c)}{1-c}. \tag{26}$$

Thus any point where $f_x \geq 1$ has a neighborhood of radius $1/(2L_0)$ on which $f_x \geq 1/2$, uniformly in x.

An explicit equal-area partition is obtained by splitting the height coordinate into k intervals of length $2/k$ and longitude into k intervals of length $2\pi/k$. Off the poles, sphere area is $d\phi\, dh$, so each cell has q-mass $1/k^2$. Use half-open intervals, with the final height endpoint included, and assign each pole to a cell in its adjacent height band. This gives a disjoint Borel partition of the entire sphere.

Its geodesic mesh is at most

$$\frac{\pi}{\sqrt{k}} + \frac{2\pi}{k} \leq \frac{3\pi}{\sqrt{k}}. \tag{27}$$

Indeed, for polar angles $\alpha, \beta \in [0, \pi]$,

$$|\cos\alpha - \cos\beta| = 2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{|\alpha-\beta|}{2}\right) \geq 2\sin^2\left(\frac{|\alpha-\beta|}{2}\right).$$

Height difference at most $2/k$ therefore gives polar-angle difference at most $2\arcsin(1/\sqrt{k}) \leq \pi/\sqrt{k}$. Joining points by a meridian and a parallel adds at most their longitude difference $2\pi/k$. The pole assignments obey the same bound.

Choose k, N as in (1). Then (27) is strictly less than $(1-c)/(4(1+c)) = 1/(2L_0)$, so every x has an entire positive cell, including its assigned boundary points. With $\epsilon = 1/2$ and $\delta = (1-c)/4$, this choice also satisfies (5). To check this explicitly, $k \geq 8$ and $k\delta > 36\pi^2(1+c)^2/(1-c) > 1$, whence $N\delta \geq \sqrt{N}$. For $N \geq 64$, $\sqrt{N} \geq 1 + \log(2N)$: it holds at 64 and the difference has positive derivative for $N > 4$. Hence

$$e^{1-N\delta} \leq \frac{1}{2N}, \qquad 1 - e^{-N\delta} \geq \frac{1}{2}.$$

The lemma supplies a fixed Borel index rule $I(x, \lambda_1, \ldots, \lambda_N, U)$ whose selected point has exactly density (25).

4 The finite nonsignalling wiring

The common resource R is the ordinary PR box:

$$R(\alpha, \beta \mid u, v) = \begin{cases} \frac{1}{2}, & \alpha \oplus \beta = uv, \\ 0, & \text{otherwise}, \end{cases} \qquad u, v, \alpha, \beta \in \{0, 1\}. \tag{28}$$

Its local marginals are fair and input-independent. Use $N = N_\theta$ independent copies. The same binary box can be used for every θ.

Share in advance the stratified list $(\lambda_1, \ldots, \lambda_N)$, an independent uniform U, an independent point λ_0 with density ρ_z, a uniform seed V for Alice's stochastic local response, and a branch bit with local-branch probability c. All these have one fixed joint law independent of x, y, and are independent of the PR resources.

In the local branch, Alice uses V to sample $T_x(\cdot \mid \lambda_0)$; Bob outputs $\mathrm{sgn}(y \cdot \lambda_0)$. They input $u_i = v_i = 0$ into all N boxes and ignore their outputs. This branch gives L.

In the residual branch Alice computes

$$I = I(x, \lambda_1, \ldots, \lambda_N, U), \qquad d = \frac{1 - \mathrm{sgn}(w_x \cdot \lambda_I)}{2}.$$

Bob computes the N bits

$$e_i = \frac{1 - \operatorname{sgn}(y \cdot \lambda_i)}{2}.$$

They query all boxes in parallel, using only their own local inputs:

$$u_i = \mathbf{1}_{\{I=i\}}, \qquad v_i = e_i. \tag{29}$$

Their final output bits are

$$a = d \oplus \bigoplus_{i=1}^{N} \alpha_i, \qquad b = \bigoplus_{i=1}^{N} \beta_i. \tag{30}$$

Bob does not receive I or the selected point.

The PR relations imply

$$a \oplus b = d \oplus \bigoplus_i u_i v_i = d \oplus e_I. \tag{31}$$

Conditioned on the shared data and the pair x, y, the vector of Alice's PR outputs is uniform on $\{0,1\}^N$; this follows directly from the product of (28). Its parity is fair. Equation (31) then makes both final marginals fair, while their sign-product expectation is

$$\mathbb{E}[\operatorname{sgn}(w_x \cdot \lambda_I) \operatorname{sgn}(y \cdot \lambda_I)] = C_Q(x, y), \tag{32}$$

by the selected-point law. An unbiased binary-output joint distribution has probabilities $(1+ABC)/4$. Hence this branch gives the entire Q, and not merely its correlator. The branch coin and (22) give exactly P_θ for all directions.

All preprocessing and output functions in this protocol are Borel. The shared prior, resource, number of queries and local maps are fixed for θ. Every run queries exactly N_θ independent boxes and returns a pair of output bits whose distribution is P_θ. This proves the theorem.

5 Zeros and denominators

The only denominator in the local response is ρ_x; its zero set was given an explicit fallback in (18). Whenever $\rho_z > 0$, (21) implies $\rho_x \geq c\rho_z > 0$, so the fallback is unused on the positive-density part of the local branch. In the residual branch, $\widetilde{\rho}_x = 0$ has zero probability under its selected-point law for each x. Great circles $y \cdot \lambda = 0$ have zero probability because this law is absolutely continuous with respect to sphere area. The sign convention still defines the wiring on every point.

Every zero of a target joint probability is preserved. For example, conditional on Alice's sign η, the target conditional Bloch vector is r_η/p_η, and its zero outcome occurs at $y = -Br_\eta/p_\eta$. More generally (22) is a convex decomposition with strictly positive weights and nonnegative branches. If a target event has probability zero, both branch probabilities are zero. The finite sampler reproduces Q exactly for that fixed x, so input-dependent kernel null sets do not weaken this conclusion.

6 References

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Parallel and nested adaptive Stein exponents for superchannel discrimination

Abstract

For every ordered pair of compatible finite-dimensional deterministic superchannels, the parallel and nested adaptive Stein exponents coincide when the type-I error tends to zero after the block-length limit inferior. Accessible input and output memories may have arbitrary finite dimensions; the two superchannels may have different private realization memories. The proof transfers a channel-uniform CP, trace-nonincreasing approximation to one common flagged parallel strategy. It derives an entropy increment bound, applies ordinary-channel amortization, and inducts on the complete nested circuit. The approximation lemma and its uniformity over entangled inputs are established in an appendix.

The equality proved here resolves the nested-versus-parallel conjecture in Hirche's analysis of superchannel discrimination [1, Sec. 5.4], for the Stein exponents and order of limits defined below. Hirche identified a channel asymptotic-equipartition condition sufficient for this conclusion and developed the amortization route [1, Appendix A]. We follow that route and establish the uniform approximation needed here, rather than assume it. Nested strategies use each superchannel as a whole; the result does not settle the advantage of the fully general adaptive strategies considered in [1, Sec. 5.5].

1 Superchannels and the two strategy families

All Hilbert spaces in this paper are nonzero and finite dimensional. Channels are completely positive and trace preserving (CPTP). Two compatible deterministic superchannels have realizations

$$\Theta_i(N) = D_i \circ (N \otimes \mathrm{id}_{S_i}) \circ E_i, \qquad E_i : C \to A \otimes S_i, \qquad D_i : B \otimes S_i \to D, \quad i = 1, 2. \tag{1}$$

The private memories S_1, S_2 need not have the same dimension. For accessible side ports X, Y and a channel $N : A \otimes X \to B \otimes Y$, write $\Theta_i^{X,Y}(N)$ for the same realization with identity action on X before the slot and on Y after it, using the canonical wire permutations. The private memory is never an accessible port.

A parallel n-use strategy consists of arbitrary finite X, Y, one common channel $N_n : A^{\otimes n} \otimes X \to B^{\otimes n} \otimes Y$, one common density operator on $C^{\otimes n} \otimes X$, and one common final binary effect on $D^{\otimes n} \otimes Y$. It inserts N_n into $\Theta_i^{\otimes n}$ under hypothesis i.

The nested family is defined recursively, as whole-channel nesting [1]. A known channel is a circuit of cost zero. A common known deterministic superchannel may wrap any circuit, without changing its number of tested uses. Finally, if an n-use circuit has channel type $A \otimes X \to B \otimes Y$, applying $\Theta_i^{X,Y}$ to that entire channel produces an $(n+1)$-use circuit of type $C \otimes X \to D \otimes Y$. A strategy uses one such circuit, a common density input, and a common binary effect. This recursion specifies the allowed nesting. All known wrappers, states, and effects are fixed independently of the unknown hypothesis. Finite classical records and feedforward may be carried in the accessible wrapper memory. Its finite dimension may depend on the block length. The recursion provides no separate access to the components E_i, D_i or the private memory S_i.

For an acceptance effect Q, let α be rejection under Θ_1 and β acceptance under Θ_2. For $\mathsf{S} \in \{\mathrm{par}, \mathrm{nest}\}$ put

$$\beta_{\varepsilon,n}^{\mathsf{S}} = \inf\{\beta : \text{an } n\text{-use } \mathsf{S} \text{ strategy has } \alpha \le \varepsilon\}, \qquad \zeta_{\mathsf{S}} = \lim_{\varepsilon \downarrow 0} \liminf_{n \to \infty} -\frac{1}{n} \log_2 \beta_{\varepsilon,n}^{\mathsf{S}}. \tag{2}$$

We use $-\log_2 0 = +\infty$. The accept-all effect is available, so $0 \le \beta^{\mathsf{S}}_{\varepsilon,n} \le 1$ and each exponent is in $[0, +\infty]$. Monotonicity in ε gives the outer limit. No optimizer is required.

Theorem 1. *For every pair (1), with the strategy families and order of limits in (2),*

$$\zeta_{\mathrm{nest}}(\Theta_1 \| \Theta_2) = \zeta_{\mathrm{par}}(\Theta_1 \| \Theta_2).$$

2 Ordinary-channel and state estimates

For density operators define the relative entropy in bits by

$$D(\rho\|\sigma) = \operatorname{Tr} \rho(\log_2 \rho - \log_2 \sigma) \quad \text{if } \operatorname{supp} \rho \subseteq \operatorname{supp} \sigma,$$

and $D(\rho\|\sigma) = +\infty$ otherwise. For compatible channels define

$$D_{\mathrm{ch}}(N\|M) = \sup_{T,\xi} D((N \otimes \mathrm{id}_T)(\xi)\|(M \otimes \mathrm{id}_T)(\xi)), \qquad d(N, M) = \sup_{m \ge 1} \frac{1}{m} D_{\mathrm{ch}}(N^{\otimes m}\|M^{\otimes m}), \quad (3)$$

where T ranges over finite reference spaces and ξ over common density inputs. The stabilized reference can be restricted to a copy of the channel input, by purifying the input and using data processing on its purification. The sequence in the regularization is superadditive, so its supremum is also its asymptotic limit. In particular $D_{\mathrm{ch}}(N\|M) \le d(N, M)$. Appending an identity side channel preserves both quantities, by combining references for one inequality and appending a fixed state for the reverse inequality.

The diamond norm here is unhalved:

$$\|T\|_\diamond = \sup_{R,Z:\|Z\|_1 \le 1} \|(T \otimes \mathrm{id}_R)(Z)\|_1.$$

Unsubscripted norms of operators are Hilbert-space operator norms. The notation $L \le_{\mathrm{CP}} cM$ means that $cM - L$ is completely positive.

Lemma 2 (Uniform dominated approximation)**.** *Let N, M be finite-dimensional channels with $d(N, M) < \infty$. For every real $a > d(N, M)$ there exist CP, trace-nonincreasing maps L_m for every positive integer m and numbers $\delta_m \ge 0$ tending to zero such that, for every sufficiently large m,*

$$L_m \le_{\mathrm{CP}} 2^{ma} M^{\otimes m}, \qquad \|L_m - N^{\otimes m}\|_\diamond \le \delta_m. \quad (4)$$

One map L_m works before any input state or entangled reference is chosen.

The stronger exponential version is Corollary 5.2 of Gao–Ji–Liu [3]. Appendix A proves the vanishing-error version needed here, retaining CP domination, trace nonincrease, and every sufficiently large integer block length.

Lemma 3 (Normalized state smoothing)**.** *If $s = D(\rho\|\sigma) < \infty$ for density operators on a fixed finite space and $b > s$, there are density operators ν_m with*

$$\|\nu_m - \rho^{\otimes m}\|_1 \longrightarrow 0, \qquad \nu_m \le 2^{mb} \sigma^{\otimes m} \quad \textit{for every sufficiently large } m. \quad (5)$$

Proof. The finite-state smooth-max estimate [2, Lemma 3.4], with purified-distance radius $1/m$, supplies subnormalized ρ'_m with

$$P(\rho'_m, \rho^{\otimes m}) \le 1/m, \qquad \rho'_m \le 2^{ms + O(\sqrt{m \log m})} \sigma^{\otimes m}.$$

Its block-length requirement is eventually satisfied. Let $t_m = \operatorname{Tr} \rho'_m$. Since the comparator in the purified distance is normalized, its fidelity squared is at most t_m, and $t_m \geq 1 - m^{-2}$. Normalization $\nu_m = \rho'_m / t_m$ increases fidelity with $\rho^{\otimes m}$, hence $\|\nu_m - \rho^{\otimes m}\|_1 \leq 2/m$. The additional logarithmic domination cost $-\log_2 t_m$ is $o(m)$. The margin $b - s > 0$ absorbs both costs for all large m. Choose arbitrary normalized states at the finitely many earlier lengths. □

We also use the finite-dimensional ordinary-channel amortization identity [2, Theorem 3.5 and Corollary 3.7]:

$$d(P, Q) = \sup_{T,\rho,\sigma:\, D(\rho\|\sigma)<\infty} \left[D((P \otimes \mathrm{id}_T)(\rho) \| (Q \otimes \mathrm{id}_T)(\sigma)) - D(\rho\|\sigma) \right]. \tag{6}$$

We apply it with $\operatorname{supp} J_P \subseteq \operatorname{supp} J_Q$, equivalently finite channel max-relative entropy, and verify this support condition at the point of use. Here J_P is the Choi matrix. Finite input relative entropy makes the subtracted quantity finite.

For finite density operators τ, ω, the direct quantum Stein lemma [4] gives, for every real $r < D(\tau\|\omega)$, effects Q_m with null rejection tending to zero and

$$\operatorname{Tr}(Q_m \omega^{\otimes m}) \leq 2^{-mr} \quad \text{for all sufficiently large } m. \tag{7}$$

In the infinite-relative-entropy case this can be seen directly. If Π projects onto $\ker \omega$ and $u = \operatorname{Tr}(\Pi\tau) > 0$, the effect $I - (I - \Pi)^{\otimes m}$ has alternative acceptance zero and null rejection $(1-u)^m$. Thus singular pairs are included.

3 A common flagged parallel strategy

Put $\mathcal{B} = \zeta_{\mathrm{par}}(\Theta_1 \| \Theta_2)$. Fix channels $N, M : A \otimes X \to B \otimes Y$ and density operators ρ, σ on $C \otimes X$. Denote

$$\tau = \Theta_1^{X,Y}(N)(\rho), \qquad \omega = \Theta_2^{X,Y}(M)(\sigma), \qquad d_0 = d(N, M), \qquad s = D(\rho\|\sigma).$$

Lemma 4. *If d_0, s are finite, then, as an extended-real inequality,*

$$D(\tau\|\omega) \leq \mathcal{B} + d_0 + s. \tag{8}$$

Proof. Choose $a > d_0$ and $b > s$ and take L_m, ν_m from Lemmas 2 and 3. For a CP slot T, let

$$\Gamma_{i,m}(T) = (D_i^{\otimes m} \otimes \mathrm{id}_{Y^{\otimes m}}) \circ (T \otimes \mathrm{id}_{S_i^{\otimes m}}) \circ (E_i^{\otimes m} \otimes \mathrm{id}_{X^{\otimes m}}), \tag{9}$$

with the fixed wire permutations. This construction preserves CP order and preserves trace nonincrease. The latter assertion follows because pre- and postprocessing are trace preserving and a CP trace-nonincreasing map remains so after every reference extension.

For each m, the effect $H_m = I - L_m^*(I)$ is positive. Choose any density operator η_m on $B^{\otimes m} \otimes Y^{\otimes m}$ and define $J_m(Z) = \operatorname{Tr}(H_m Z)\eta_m$. This is CP and $L_m + J_m$ is CPTP. Let $F = \mathbb{C}^2$ be a new accessible flag, and use the single common inserted channel

$$\widehat{L}_m(Z) = L_m(\operatorname{Tr}_F Z) \otimes |0\rangle\langle 0|_F + J_m(\operatorname{Tr}_F Z) \otimes |1\rangle\langle 1|_F. \tag{10}$$

It is CPTP on its entire domain, including all reference extensions. Insert it with the same input $\nu_m \otimes |0\rangle\langle 0|_F$ under both hypotheses. The flag travels through the accessible identity side port. Expanding the physical composition gives the normalized output

$$\Omega_{i,m} = \Gamma_{i,m}(L_m)(\nu_m) \otimes |0\rangle\langle 0|_F + \Gamma_{i,m}(J_m)(\nu_m) \otimes |1\rangle\langle 1|_F. \tag{11}$$

This formula also holds when the success branch is correlated with the private memory: the postprocessing acts on each flag block separately.

Choose the state-test effects Q_m for τ, ω at any strict rate $r < D(\tau\|\omega)$. Accept only on the effect $Q_m \otimes |0\rangle\langle 0|_F$. Its actual unconditional probabilities p'_m, q'_m satisfy, for all large m,

$$p'_m \geq \mathrm{Tr}(Q_m \tau^{\otimes m}) - \delta_m - \|\nu_m - \rho^{\otimes m}\|_1, \tag{12}$$

$$q'_m \leq 2^{m(a+b)} \,\mathrm{Tr}(Q_m \omega^{\otimes m}). \tag{13}$$

For (12), first use diamond-norm contraction through the common pre- and postprocessing in (9), with the private memory treated only as a reference for the diamond estimate. Then use trace-norm contraction for the input-state change; $\Gamma_{1,m}(L_m)$ is CP and trace nonincreasing. In both steps an effect has operator norm at most one. For (13), positivity and CP domination give

$$\Gamma_{2,m}(L_m)(\nu_m) \leq 2^{mb} \Gamma_{2,m}(L_m)(\sigma^{\otimes m}) \leq 2^{m(a+b)} \omega^{\otimes m}.$$

The two domination bounds concern the same physical success branch. There is no domination requirement on J_m.

The repair branch is rejected exactly by the final flag effect. Accordingly the probabilities in (12)–(13) are neither conditioned on success nor divided by its probability. All coefficients of the inserted channel, input, and test may be chosen using the two specified hypotheses, but they are common to the two experiments. The ideal $M^{\otimes m}, \sigma^{\otimes m}$ occur only as comparators in the alternative-probability bound.

By (7) the resulting actual parallel strategies have $\alpha'_m = 1 - p'_m \to 0$ and

$$\beta'_m = q'_m \leq 2^{-m(r-a-b)}.$$

At every fixed $0 < \varepsilon < 1$, these tests are feasible at all sufficiently large integer m. Passing to the type-II infimum and then the limits in (2) gives $\mathcal{B} \geq r - a - b$. Let $a \downarrow d_0$, $b \downarrow s$, and let the strict real rates r increase to $D(\tau\|\omega)$. If this output entropy is infinite, the same inequalities hold for arbitrarily large r and force $\mathcal{B} = \infty$. Thus the argument proves the extended inequality (8) in every case. □

4 The channel increment and the nested converse

Suppose first $\mathcal{B} < \infty$. For fixed N, M with $d(N, M) < \infty$, put $P = \Theta_1^{X,Y}(N)$ and $Q = \Theta_2^{X,Y}(M)$. Any additional finite reference T can be absorbed into the accessible ports as $X \otimes T, Y \otimes T$. The same public reassociation is used under both hypotheses. Identity extension leaves $d(N, M)$ unchanged. Lemma 4 therefore gives, for every finite-cost input pair on that reference extension,

$$D((P \otimes \mathrm{id}_T)(\rho)\|(Q \otimes \mathrm{id}_T)(\sigma)) - D(\rho\|\sigma) \leq \mathcal{B} + d(N, M). \tag{14}$$

To check the support hypothesis for (6), take the same maximally entangled input on the domain of P, Q and its copy. Its input cost is zero. Lemma 4 makes the relative entropy between the normalized Choi outputs finite. Hence $\mathrm{supp}\, J_P \subseteq \mathrm{supp}\, J_Q$, and finite dimensions imply $J_P \leq c J_Q$ for a finite c. Applying (6) to (14) proves

$$d(\Theta_1^{X,Y}(N), \Theta_2^{X,Y}(M)) \leq \mathcal{B} + d(N, M). \tag{15}$$

The approximation thresholds may depend on each fixed reference dimension. The exact pointwise inequality is obtained first, with a right-hand side independent of that dimension, and only then is the reference supremum taken.

For a common known deterministic superchannel Ψ, stabilization and relative-entropy data processing give

$$d(\Psi(P), \Psi(Q)) \le d(P, Q). \tag{16}$$

At each tensor block, absorb its common preprocessing memory into the reference and apply its common postprocessing by data processing. Optimize the common input, divide by the block length, and take the regularization to obtain (16).

Structural induction on the nested circuit now proves that its evaluated channels $K_{n,1}, K_{n,2}$ satisfy

$$d(K_{n,1}, K_{n,2}) \le n\mathcal{B}. \tag{17}$$

A known seed has identical channels and cost zero. A known wrapper preserves the bound by (16). A tested use increases it by at most $\mathcal{B}$ by (15). In particular every intermediate channel cost is finite. This induction covers every finite accessible memory dimension in the recursively defined circuit.

For its final common input and effect, let $p = 1 - \alpha$ and $q = \beta$. State and binary-measurement data processing, followed by (17), give

$$D\big((p, 1-p)\|(q, 1-q)\big) \le D_{\mathrm{ch}}(K_{n,1}\|K_{n,2}) \le d(K_{n,1}, K_{n,2}) \le n\mathcal{B}.$$

If $\alpha \le \varepsilon < 1$, the binary expression obeys

$$D\big((p, 1-p)\|(q, 1-q)\big) \ge p \log_2(1/q) - h_2(p) \ge (1-\varepsilon)\log_2(1/\beta) - 1.$$

Finite $n\mathcal{B}$ excludes $q = 0$ when $p > 0$. Hence every feasible nested test obeys

$$\beta \ge 2^{-(n\mathcal{B}+1)/(1-\varepsilon)}, \qquad \beta_{\varepsilon,n}^{\mathrm{nest}} \ge 2^{-(n\mathcal{B}+1)/(1-\varepsilon)}. \tag{18}$$

The second bound passes to the infimum without an optimizer. It implies

$$\limsup_{n\to\infty} -\frac{1}{n}\log_2 \beta_{\varepsilon,n}^{\mathrm{nest}} \le \frac{\mathcal{B}}{1-\varepsilon}.$$

Taking the ordered limits in (2) yields $\zeta_{\mathrm{nest}} \le \mathcal{B}$.

5 Embedding a parallel circuit in the nested family

We give the routing that supplies the reverse inequality. Let the common joint inserted channel be $N : A_1 \cdots A_n X \to B_1 \cdots B_n Y$. Start with the known channel $J_{n+1} = N$. For $j = n, n-1, \ldots, 1$, set

$$X_j = C_{j+1} \cdots C_n A_1 \cdots A_{j-1} X, \qquad Y_j = D_{j+1} \cdots D_n \mathcal{B}_1 \cdots B_{j-1} Y.$$

The empty products at the endpoints are one-dimensional. Known public wire permutations view $J_{j+1,i}$ as a channel $A_j \otimes X_j \to B_j \otimes Y_j$. Apply the whole tested use $\Theta_i^{X_j, Y_j}$ to obtain $J_{j,i}$, with domain $C_j \otimes X_j$ and codomain $D_j \otimes Y_j$. This is exactly the nested recursion and uses n tested superchannels.

Expanding its realization applies the preprocessings in the order $E_1, \ldots, E_n$, then the joint N, then the postprocessings in the order $D_n, \ldots, D_1$. Preprocessings on disjoint external factors commute, as do the postprocessings on their respective output and private memory factors. The expanded circuit consequently equals the parallel tensor realization. Each private S_j remains inside its own tested use; the common permutations act only on accessible external systems. The final common input and effect have exactly the parallel Born probabilities. This routing is the parallel-to-nested inclusion [1, Appendix C] with its side-port dimensions made explicit.

Therefore $\beta_{\varepsilon,n}^{\mathrm{nest}} \le \beta_{\varepsilon,n}^{\mathrm{par}}$ and $\zeta_{\mathrm{nest}} \ge \mathcal{B}$. Together with the converse, this proves Theorem 1 for finite $\mathcal{B}$. If $\mathcal{B} = +\infty$, inclusion already forces the nested exponent to be $+\infty$, proving the extended-real equality as well.

A A uniform CP, trace-nonincreasing approximation

We prove Lemma 2 using the testing-to-dilation and tensor rate-reduction mechanism of [3, Theorem 2.2]. The maps constructed below are chosen in fixed environments and precede all choices of input or reference.

Fix finite Stinespring isometries $V : X_0 \to Y_0 \otimes E$ and $W : X_0 \to Y_0 \otimes F$ of N, M. Let $d = d(N, M) < \infty$. The normalized Choi states are outputs on a common maximally entangled input, so their relative entropy is at most $D_{\mathrm{ch}}(N\|M) \le d$. Thus $\operatorname{supp} J_N \subseteq \operatorname{supp} J_M$ and, for some finite real L, $J_N \le 2^L J_M$. Kraus-column factorization gives an environmental map $C_* : F \to E$ with

$$V = (I \otimes C_*)W, \qquad \|C_*\|^2 \le 2^L. \tag{19}$$

To verify this in the prescribed, possibly nonminimal environments, write $J_N = F_N F_N^\dagger$, $J_M = F_M F_M^\dagger$. The operator inequality permits $F_N = F_M D$ with $\|D\| \le 2^{L/2}$; this follows by the support pseudoinverse or Douglas factorization. In the fixed Kraus bases take $C_* = D^T$. Complex rectangular transpose preserves singular values. This proof allows redundant and zero Kraus columns. Enlarge L freely when needed.

A.1 Weak approximation uniform over inputs

For $t > 0$, let $E_t(N, M)$ be the maximum of $p - tq$ over common entangled inputs and binary effects. With unnormalized Choi matrices, its dual is

$$E_t(N, M) = \min_{Z \ge 0,\ Z \ge J_N - tJ_M} \|\mathrm{Tr}_{Y_0} Z\|_\infty. \tag{20}$$

Here is the finite SDP justification. The primal variables are a density matrix ξ on X_0 and $0 \le T \le I_{Y_0} \otimes \xi$, with objective $\mathrm{Tr}[(J_N - tJ_M)T]$. The dual also has the inequality $\mathrm{Tr}_{Y_0} Z \le \lambda I$ and minimizes λ. The choices $\xi = I/\dim X_0$, $T = I/(2 \dim X_0)$ are strictly feasible. A sufficiently large $Z = cI$, followed by $\lambda > c \dim Y_0$, is strictly dual feasible. Slater duality yields (20). Bounded dual sublevels have $\mathrm{Tr}\, Z \le \lambda \dim X_0$, so positivity gives compactness and attainment. The primal captures arbitrary references by purification and the input-copy reduction stated after (3).

For an attaining Z, put $A_0 = tJ_M$ and $H = A_0 + Z \ge J_N$. With the Moore–Penrose inverse H^+, define $F_0 = A_0 H^+ F_N$ and $F_1 = Z H^+ F_N$. Support inclusion gives $F_0 + F_1 = F_N$, and

$$F_0 F_0^\dagger \le A_0 H^+ A_0 \le A_0, \qquad F_1 F_1^\dagger \le Z H^+ Z \le Z.$$

Factor $F_0 = F_M D$ with $\|D\| \le \sqrt{t}$ and take $C = D^T$. The residual Stinespring operator has Kraus columns F_1, and therefore

$$\|V - (I \otimes C)W\|^2 = \|(\mathrm{Tr}_{Y_0} F_1 F_1^\dagger)^T\|_\infty \le E_t(N, M), \qquad \|C\| \le \sqrt{t}. \tag{21}$$

All inequalities involve the full operators, rather than one chosen input vector.

For tensor powers at $t = 2^{mr}$ and $r > d$, binary data processing bounds the testing advantage by

$$E_{2^{mr}}(N^{\otimes m}, M^{\otimes m}) \le \min\{1, (md + 1)/(mr)\}. \tag{22}$$

Indeed, positive advantage implies $q < 2^{-mr}$. The binary entropy bound $D((p, 1-p)\|(q, 1-q)) \ge p \log_2(1/q) - 1$ gives $p \le (md + 1)/(mr)$, and the advantage is at most p. If $q = 0$, finite channel entropy forces $p = 0$. Combining (21)–(22), for any $\sqrt{d/r} < \delta < 1$, supplies maps C_m^{w} at every large integer m with

$$\|C_m^{\mathrm{w}}\| \le 2^{mr/2}, \qquad \|V^{\otimes m} - (I \otimes C_m^{\mathrm{w}})W^{\otimes m}\| \le \delta. \tag{23}$$

A.2 Finite rate descent with vanishing error

Choose $d < r < a$ and enlarge L so $L > r$. Suppose accurate environmental maps C_m^{a} are available at a rate $S \in [r, L]$, with

$$\|C_m^{\mathrm{a}}\| \le 2^{mS/2}, \qquad e_m := \|V^{\otimes m} - (I \otimes C_m^{\mathrm{a}})W^{\otimes m}\| \le K \exp(-\gamma m^{\alpha}), \quad 0 < \alpha \le 1,$$

for all large m. The exact maps $C_*^{\otimes m}$ supply the initial case $S = L$ with zero error. Let

$$X_m = (I \otimes C_m^{\mathrm{w}})W^{\otimes m}, \qquad Y_m = (I \otimes C_m^{\mathrm{a}})W^{\otimes m}, \qquad H_m = Y_m - X_m.$$

Eventually $\|X_m\| \le A := 1 + \delta$ and $\|H_m\| \le B_0 := (1 + \delta)/2 < 1$. Choose $0 < \beta < 1$ close enough to one that for some $\kappa > 0$

$$h(u) + (1 - u) \ln A + u \ln B_0 \le -\kappa \quad (\beta \le u \le 1),$$

where h is binary entropy with natural logarithms. This is possible by continuity, since the expression at $u = 1$ is $\ln B_0 < 0$.

Expand $Y_m^{\otimes k} = (X_m + H_m)^{\otimes k}$ and retain only words with at most $\lfloor \beta k \rfloor$ factors H_m. The omitted physical operator has norm at most

$$\sum_{j > \beta k} \binom{k}{j} A^{k-j} B_0^j \le (k + 1) e^{-\kappa k}, \tag{24}$$

using $\binom{k}{j} \le e^{kh(j/k)}$. The retained environmental coefficient has norm at most

$$\sum_{j \le \beta k} \binom{k}{j} 2^j 2^{m[(k-j)r + jS]/2} \le 3^k 2^{mk[(1-\beta)r + \beta S]/2}. \tag{25}$$

This follows because H_m is obtained from the coefficient $C_m^{\mathrm{a}} - C_m^{\mathrm{w}}$, of norm at most $2\, 2^{mS/2}$, in the same fixed alternative environment.

For every sufficiently large total integer length M, take $m = \lfloor \sqrt{M} \rfloor$, $k = \lfloor M/m \rfloor$, and $\ell = M - mk < m$. Tensor the retained coefficient with the exact $C_*^{\otimes \ell}$, and reindex tensor factors publicly. Its squared norm has base-two logarithm at most

$$mk[(1 - \beta)r + \beta S] + 2k \log_2 3 + \ell L.$$

Dividing by M makes the last two terms vanish. Thus any fixed $R > (1 - \beta)r + \beta S$ bounds that logarithm by MR at all large integer M. The physical error is at most

$$(k + 1)e^{-\kappa k} + k e_m (1 + e_m)^{k-1}.$$

The second term is the telescoping difference between $Y_m^{\otimes k}$ and $V^{\otimes mk}$. Since m, k are of order $\sqrt{M}$, it and the first term are bounded, after enlarging and decreasing constants, by $K' \exp(-\gamma' M^{\alpha/2})$. Exact remainder padding has operator norm one after action on $W^{\otimes \ell}$ and adds no error. This constructs an accurate family at rate R for every large integer M, not only a subsequence of block lengths.

Let $\theta = (1 + \beta)/2 \in (\beta, 1)$ and set $S_j = r + \theta^j (L - r)$. Each S_{j+1} is strictly above $(1 - \beta)r + \beta S_j$. Starting from the exact family at $S_0 = L$, the preceding step gives accurate families at all S_j. Choose a finite j with $S_j < a$. Although each step halves the stretched-exponential power, after finitely many steps it remains positive. We have therefore constructed C_m before any input or reference choice with

$$\|C_m\| \le 2^{ma/2}, \qquad e_m = \|V^{\otimes m} - (I \otimes C_m)W^{\otimes m}\| \longrightarrow 0. \tag{26}$$

A.3 CP order, trace nonincrease, and the diamond estimate

Put $\widehat{V}_m = (I \otimes C_m)W^{\otimes m}$ and $\widetilde{V}_m = \widehat{V}_m/(1+e_m)$. Since $V^{\otimes m}$ is an isometry and $\|\widehat{V}_m - V^{\otimes m}\| = e_m$, the operator $\widetilde{V}_m$ is a contraction and $\|\widetilde{V}_m - V^{\otimes m}\| \le 2e_m$. Define

$$L_m(Z) = \mathrm{Tr}_{E^{\otimes m}} \widetilde{V}_m Z \widetilde{V}_m^\dagger.$$

This is CP and trace nonincreasing on the entire input algebra. In the fixed Kraus-column matrices, its environmental coefficient is $C_m/(1+e_m)$. Its squared norm is at most 2ma, so $J_{L_m} \le 2^{ma} J_M^{\otimes m}$, which is exactly CP domination.

For every reference R and every complex Z of trace norm at most one, the two-term product difference and contraction of partial trace give

$$\|[(L_m - N^{\otimes m}) \otimes \mathrm{id}_R](Z)\|_1 \le \|\widetilde{V}_m - V^{\otimes m}\|(\|\widetilde{V}_m\| + \|V^{\otimes m}\|)\|Z\|_1 \le 4e_m.$$

All operator norms are unchanged by tensoring with I_R. Thus (4) holds with $\delta_m = 4e_m$ eventually. At the finitely many earlier lengths take $L_m = 0$ and any finite error bound. This supplies CP/TNI maps at every length and proves the lemma.

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Entanglement of formation for two isotropic factors

Abstract

Entanglement of formation is additive for a product of two isotropic states, with independently chosen dimensions at least two and arbitrary fidelities. A two-factor fidelity bound for unrestricted complex pure states passes to the convex roof and yields the equality. The result treats two factors; it does not assert additivity for arbitrary tensor powers.

Terhal and Vollbrecht determined the entanglement of formation of a single isotropic state [1]. The theorem here proves additivity for a product of two isotropic factors, allowing different dimensions and fidelities. An earlier additivity claim by Terhal was withdrawn because its argument for Theorem 1 was insufficient [3]; we do not use that claim. The present proof treats two factors only. It does not establish additivity for arbitrary tensor powers or identify the regularized entanglement cost.

1 Definitions and theorem

All spaces are finite-dimensional complex Hilbert spaces. Traces and Schatten norms are unnormalized. All entropies in this paper use natural logarithms, with $0 \log 0 = 0$. On $\mathbb{C}^d \otimes \mathbb{C}^d$, $d \geq 2$, set

$$|\phi_d\rangle = \frac{1}{\sqrt{d}} \sum_{a=1}^{d} |a, a\rangle, \qquad \Phi_d = |\phi_d\rangle\langle\phi_d|, \qquad \sigma_{d,F} = F\Phi_d + \frac{1-F}{d^2-1}(I - \Phi_d), \quad 0 \leq F \leq 1. \tag{1.1}$$

This is the usual isotropic family [1]. Its parameter is the actual Bell fidelity: $\operatorname{Tr} \sigma_{d,F}\Phi_d = F$. For a bipartite density ρ, define the unrestricted finite pure-state convex roof

$$E_F(\rho) = \inf_{\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|} \sum_j p_j S(\operatorname{Tr}_B |\psi_j\rangle\langle\psi_j|), \qquad S(\tau) = -\operatorname{Tr} \tau \log \tau. \tag{1.2}$$

The probabilities are nonnegative and sum to one, every ψ_j is a unit complex vector, and the barycenter equality is the actual density equality. Set $G_d(F) = E_F(\sigma_{d,F})$.

For two factors of independent dimensions $d_1, d_2 \geq 2$, regroup the physical registers as $A_1A_2 : B_1B_2$. Every normalized pure vector has a coefficient matrix M on $\mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2}$:

$$|\psi_M\rangle = \sum_{a,b,c,e} M_{ab,ce} |a, b\rangle_A |c, e\rangle_B, \qquad \|M\|_{\mathrm{HS}}^2 = \operatorname{Tr} M^* M = 1. \tag{1.3}$$

Its pure entanglement entropy is $S(MM^*) = S(M^*M)$. Partial traces $\operatorname{Tr}_1 M$ and $\operatorname{Tr}_2 M$ here are traces of the coefficient operator, not of the density $|\psi_M\rangle\langle\psi_M|$. Direct Bell contraction gives

$$f_1(M) = \langle\psi_M|(\Phi_{d_1} \otimes I)|\psi_M\rangle = \frac{\|\operatorname{Tr}_1 M\|_{\mathrm{HS}}^2}{d_1}, \qquad f_2(M) = \frac{\|\operatorname{Tr}_2 M\|_{\mathrm{HS}}^2}{d_2}. \tag{1.4}$$

Each lies in $[0, 1]$.

Theorem 1.1 (Two-factor fidelity bound)**.** *Every normalized complex matrix M in (1.3) satisfies*

$$S(M^*M) \geq G_{d_1}(f_1(M)) + G_{d_2}(f_2(M)). \tag{1.5}$$

For every mixed density ρ on these registers, with actual pair fidelities $F_i = \operatorname{Tr} \rho(\Phi_{d_i} \otimes I)$, one has

$$E_F(\rho) \geq G_{d_1}(F_1) + G_{d_2}(F_2). \tag{1.6}$$

Corollary 1.2 (Exact two-factor additivity)**.** *For every $d_1, d_2 \geq 2$ and independently chosen $F_1, F_2 \in [0,1]$,*

$$E_F(\sigma_{d_1,F_1} \otimes \sigma_{d_2,F_2}) = E_F(\sigma_{d_1,F_1}) + E_F(\sigma_{d_2,F_2}). \tag{1.7}$$

The bipartition is $A_1A_2 : B_1B_2$, and all three roofs are the unrestricted complex roofs in (1.2).

2 Finite one-factor preparations

For a probability vector $q = (q_1, \ldots, q_d)$, define

$$H(q) = -\sum_a q_a \log q_a, \qquad t_d(q) = \frac{1}{d}\left(\sum_a \sqrt{q_a}\right)^2, \qquad M_q = \operatorname{diag}(\sqrt{q_a}). \tag{2.1}$$

The pure vector of M_q has entropy $H(q)$ and Bell fidelity $t_d(q)$. Cauchy–Schwarz gives $0 \leq t_d(q) \leq 1$, with equality $t_d(q) = 1$ if and only if q is uniform.

Lemma 2.1 (Actual finite preparations)**.** *For every probability vector q,*

$$G_d(t_d(q)) \leq H(q). \tag{2.2}$$

Also $G_d(0) = 0$, $0 \leq G_d(F) \leq \log d$, and G_d is convex on $[0,1]$.

Proof. We construct a finite ensemble of local-unitary images with the stated barycenter. Write $D = \sum_a |a,a\rangle\langle a,a|$, the diagonal-pair projector, and define

$$R(F,Q) = F\Phi_d + \frac{Q}{d-1}(D - \Phi_d) + \frac{1-F-Q}{d(d-1)}(I - D). \tag{2.3}$$

Average $|\psi_{M_q}\rangle\langle\psi_{M_q}|$ over the $d!$ simultaneous physical permutations $P \otimes \overline{P}$. The diagonal-pair diagonal entries become $1/d$. Its distinct diagonal-pair entries become

$$\frac{(\sum_a \sqrt{q_a})^2 - 1}{d(d-1)}.$$

All other entries vanish. Hence this finite average is $R(t_d(q), 1 - t_d(q))$.

Let U be the normalized Fourier matrix, $U_{ab} = d^{-1/2}\exp(2\pi iab/d)$, with indices modulo d. It is unitary and $|U_{ab}|^2 = 1/d$. Let $\mathcal{T}$ be the average over the 4^d independent fourth-root diagonal phases L, acting physically as $L \otimes \overline{L}$. This average retains just the ordinary diagonal entries and the diagonal-pair block. In particular,

$$\mathcal{T}\big((U \otimes \overline{U})D(U \otimes \overline{U})^*\big) = \Phi_d + \frac{I - D}{d}. \tag{2.4}$$

To check the formula, the $(a,a;c,c)$ entry is $\sum_j |U_{aj}|^2|U_{cj}|^2 = 1/d$; the $(a,b;a,b)$ entry for $a \neq b$ is also $1/d$. Every other entry averages to zero because its independent phase character is nonconstant. Both I and Φ_d are fixed by these operations. Substituting (2.4) into (2.3) therefore gives

$$\mathcal{T}\big((U \otimes \overline{U})R(F,Q)(U \otimes \overline{U})^*\big) = R\left(F, \frac{1-F-Q}{d}\right). \tag{2.5}$$

Mix the first ensemble with weight $1/(d+1)$ and the Fourier–phase ensemble with weight $d/(d+1)$. The resulting Q coefficient is

$$\frac{Q}{d+1} + \frac{d}{d+1}\frac{1-F-Q}{d} = \frac{1-F}{d+1}.$$

Both complementary coefficients of (2.3) are then $(1-F)/(d^2-1)$, so the barycenter is exactly $\sigma_{d,F}$. All ensemble members are local-unitary images of the original pure vector, with its normalization and entropy unchanged. This proves (2.2) by the finite convex-roof definition.

For $G_d(0)$ start instead with the product vector $|a,b\rangle$, $a \neq b$. Its permutation average is $(I-D)/(d(d-1)) = R(0,0)$. The same Fourier–phase construction and mixture give $\sigma_{d,0}$ as an ensemble of product vectors of zero entropy. Thus $G_d(0) = 0$.

Every density has a finite spectral pure decomposition, each pure entropy lies in $[0, \log d]$, and concatenating finite near-optimal ensembles proves convexity of E_F. Since $\sigma_{d,F}$ is affine in F, these observations give the remaining assertions about G_d. □

3 Scalar entropy supports including the endpoints

Lemma 3.1 (Universal approximate scalar supports)**.** *For every $F \in [0,1]$ and $\epsilon > 0$, there are real a, b with $b \geq 0$ such that*

$$a + bF \geq G_d(F) - \epsilon, \tag{3.1}$$

and, for every probability vector q of length d,

$$a + bt_d(q) \leq H(q). \tag{3.2}$$

Proof. For $0 < F < 1$, a finite convex function has a supporting line on its interval: choose $a + bF = G_d(F)$ and $a + bx \leq G_d(x)$ for every $x \in [0,1]$. This elementary convex fact follows by choosing a slope between the supremum of left secant slopes and the infimum of right secant slopes. Evaluation at zero gives $a \leq G_d(0) = 0$. Since $G_d(F) \geq 0$, the identity $bF = G_d(F) - a$ gives $b \geq 0$. Lemma 2.1 then proves (3.2). For $F = 0$, the zero line has both required properties.

For $F = 1$, put $c = \log d - \epsilon$. If $c \leq 0$, the zero line works because $G_d(1) \leq \log d$. If $c > 0$, consider the compact set

$$B_c = \{q : q_a \geq 0,\ \sum_a q_a = 1,\ H(q) \leq c\}.$$

It contains probability vertices. Continuity of t_d and H gives $r = \max_{q \in B_c} t_d(q)$. One has $r < 1$: fidelity one forces the uniform vector, whose entropy is $\log d > c$. Choose

$$b = \frac{c}{1-r}, \qquad a = c - b.$$

For $q \in B_c$, $a + bt_d(q) \leq a + br = 0 \leq H(q)$. For $q \notin B_c$, $a + bt_d(q) \leq a + b = c < H(q)$. Thus (3.2) holds, while $a + b = c \geq G_d(1) - \epsilon$. No differentiability or finite endpoint tangent is needed. □

4 The product-diagonal entropy estimate

Fix two scalar supports $\ell_i(x) = a_i + b_i x$, $b_i \geq 0$, satisfying (3.2) in dimensions d_i. We will prove the affine estimate

$$S(M^*M) \geq a_1 + a_2 + b_1 f_1(M) + b_2 f_2(M) \tag{4.1}$$

for every normalized complex M.

Lemma 4.1 (Product diagonal)**.** *The estimate* (4.1) *holds for* $M = \operatorname{diag}(\sqrt{p_{xy}})$*, where* p *is any probability distribution on* $\{1, \dots, d_1\} \times \{1, \dots, d_2\}$*.*

Proof. Its entropy is $H(X, Y)$. Let $p_y = \sum_x p_{xy}$. For every $p_y > 0$, apply the first scalar support to the conditional distribution $p_{x|y} = p_{xy}/p_y$:

$$H(X \mid Y = y) \geq a_1 + \frac{b_1}{d_1} \left(\sum_x \sqrt{p_{x|y}} \right)^2 .$$

After weighting by p_y and summing,

$$H(X \mid Y) \geq a_1 + b_1 f_1(M), \qquad H(Y \mid X) \geq a_2 + b_2 f_2(M). \tag{4.2}$$

Zero-weight conditionals are omitted; their entire column or row is zero. The fidelity identities are

$$f_1(M) = \frac{1}{d_1} \sum_y \left(\sum_x \sqrt{p_{xy}} \right)^2 , \qquad f_2(M) = \frac{1}{d_2} \sum_x \left(\sum_y \sqrt{p_{xy}} \right)^2 .$$

Finally $H(X, Y) - H(X \mid Y) - H(Y \mid X) = I(X : Y) \geq 0$. Adding (4.2) proves the result. This conditional entropy argument is the product-basis argument represented in [2, Section III, Eqs. (18)–(32)]. □

5 Weighted spectral comparison for positive coefficients

Lemma 5.1 (Two-site spectral comparator)**.** *Let* $P \geq 0$ *be a coefficient matrix on* $\mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2}$*, and let* $w_1, w_2 \geq 0$*. There is a positive matrix* P' *diagonal in a product basis, with the same eigenvalues as* P*, such that*

$$w_1 \| \operatorname{Tr}_1 P \|_{\mathrm{HS}}^2 + w_2 \| \operatorname{Tr}_2 P \|_{\mathrm{HS}}^2 \leq w_1 \| \operatorname{Tr}_1 P' \|_{\mathrm{HS}}^2 + w_2 \| \operatorname{Tr}_2 P' \|_{\mathrm{HS}}^2 . \tag{5.1}$$

Proof. Put $A = \operatorname{Tr}_1 P$, $B = \operatorname{Tr}_2 P$, and define the Hermitian site field

$$K = w_1 I_1 \otimes A + w_2 B \otimes I_2 .$$

The two summands act on separate factors. Diagonalize A and B in their own orthonormal eigenbases. This makes K diagonal in a product basis, with entries $h_{xy} = w_1 \mu_y + w_2 \nu_x$. If λ_j are the eigenvalues of P, its diagonal entries in this basis are $\sum_j |U_{ij}|^2 \lambda_j$, for a unitary U. The matrix $(|U_{ij}|^2)$ is doubly stochastic. A linear functional on doubly stochastic matrices attains its maximum at a permutation matrix, by the Birkhoff decomposition. Hence some assignment of the λ_j to the product basis gives P' with

$$\operatorname{Tr} P' K \geq \operatorname{Tr} P K = w_1 \|A\|_{\mathrm{HS}}^2 + w_2 \|B\|_{\mathrm{HS}}^2 =: C(P). \tag{5.2}$$

For clarity, this assignment depends on the given matrix and the chosen weights; it retains its complete spectrum.

Define $C(Q) = w_1 \| \operatorname{Tr}_1 Q \|_{\mathrm{HS}}^2 + w_2 \| \operatorname{Tr}_2 Q \|_{\mathrm{HS}}^2$. Completing the two independent Hilbert–Schmidt squares gives

$$\begin{aligned} C(P') &\geq 2 w_1 \operatorname{Tr}(\operatorname{Tr}_1 P') A + 2 w_2 \operatorname{Tr}(\operatorname{Tr}_2 P') B - C(P) \\ &= 2 \operatorname{Tr} P' K - C(P) \geq C(P). \end{aligned}$$

This proves (5.1). Vanishing weights and zero eigenvalues are allowed throughout. □

Lemma 5.2 (Positive affine estimate)**.** *The estimate* (4.1) *holds for every* $M = P \geq 0$ *with* $\|P\|_{\mathrm{HS}} = 1$.

Proof. Apply Lemma 5.1 with $w_i = b_i/d_i$. The resulting P' has the same squared spectrum and therefore $S(P'^2) = S(P^2)$ and $\|P'\|_{\mathrm{HS}} = 1$, while

$$b_1 f_1(P) + b_2 f_2(P) \leq b_1 f_1(P') + b_2 f_2(P').$$

Its nonnegative diagonal eigenvalues can be written $\sqrt{p_{xy}}$ with $\sum p_{xy} = 1$. Lemma 4.1 applies in that product basis. Product coefficient conjugations preserve both partial-trace norms and the squared spectrum; thus its estimate applies to P' in the original coordinates as well. Combining the inequalities proves (4.1) for P. □

6 Both polar moduli and unrestricted complex coefficients

Lemma 6.1 (Partial-trace modulus bound)**.** *For every complex square coefficient matrix* M*, write* $L = \sqrt{MM^*}$ *and* $R = \sqrt{M^*M}$*. Then*

$$f_i(M) \leq \frac{f_i(L) + f_i(R)}{2}, \qquad i = 1, 2. \tag{6.1}$$

If $\|M\|_{\mathrm{HS}} = 1$*, both moduli have Hilbert–Schmidt norm one and their pure entropies equal* $S(M^*M)$.

Proof. The literal block matrix

$$\begin{pmatrix} L & M \\ M^* & R \end{pmatrix} \geq 0 \tag{6.2}$$

is positive. For example, a singular-value decomposition $M = U\Sigma V^*$ writes it as $\binom{U}{V}\Sigma \begin{pmatrix} U^* & V^* \end{pmatrix}$. Applying the completely positive partial trace to its blocks gives

$$\begin{pmatrix} A_i & X_i \\ X_i^* & B_i \end{pmatrix} \geq 0, \qquad A_i = \mathrm{Tr}_i\, L, \quad B_i = \mathrm{Tr}_i\, R, \quad X_i = \mathrm{Tr}_i\, M.$$

A positive block matrix admits $X_i = A_i^{1/2} T_i B_i^{1/2}$ with $\|T_i\|_\infty \leq 1$. This includes singular diagonal blocks: restrict to their supports, or obtain the statement by adding δI and taking a convergent subsequence of contractions as $\delta \downarrow 0$. Schatten Hölder gives

$$\|X_i\|_{\mathrm{HS}} \leq \|A_i^{1/2}\|_4 \|T_i\|_\infty \|B_i^{1/2}\|_4, \qquad \|X_i\|_{\mathrm{HS}}^2 \leq \|A_i\|_{\mathrm{HS}} \|B_i\|_{\mathrm{HS}}.$$

Arithmetic–geometric mean and division by d_i give (6.1). The equal norm and entropy claims follow from the common singular values, including all zero values, of M. □

Proof of the affine estimate (4.1). Apply Lemma 5.2 to L and R. Average its two inequalities. Their left sides have the same entropy $S(M^*M)$. Since $b_1, b_2 \geq 0$, Lemma 6.1 gives

$$\begin{aligned} S(M^*M) &\geq a_1 + a_2 + \frac{b_1}{2}\big(f_1(L) + f_1(R)\big) + \frac{b_2}{2}\big(f_2(L) + f_2(R)\big) \\ &\geq a_1 + a_2 + b_1 f_1(M) + b_2 f_2(M). \end{aligned}$$

Thus the estimate holds for every complex coefficient, without a positivity, normality, rank, or product-basis hypothesis on M. □

7 Pure and mixed roofs

Proof of Theorem 1.1. Choose supports from Lemma 3.1 independently at $f_1(M)$ and $f_2(M)$, each within ϵ of the corresponding one-factor value. The affine estimate yields

$$S(M^*M) \geq G_{d_1}(f_1(M)) + G_{d_2}(f_2(M)) - 2\epsilon.$$

Letting $\epsilon \downarrow 0$ proves (1.5), including all fidelity endpoints.

For the mixed statement, let $\rho = \sum_j p_j |\psi_{M_j}\rangle\langle\psi_{M_j}|$ be any finite normalized complex ensemble. Physical Bell expectations are linear in the actual density barycenter:

$$\sum_j p_j f_i(M_j) = F_i. \tag{7.1}$$

Average (1.5) and use convexity of each G_{d_i}:

$$\sum_j p_j S(M_j^* M_j) \geq \sum_j p_j \big(G_{d_1}(f_1(M_j)) + G_{d_2}(f_2(M_j))\big) \geq G_{d_1}(F_1) + G_{d_2}(F_2).$$

Only the ensemble means are fixed in (7.1); individual components can have arbitrary fidelities. Taking the infimum over every such ensemble proves (1.6). □

Proof of Corollary 1.2. The pair fidelities of the actual tensor product $\sigma_{d_1,F_1} \otimes \sigma_{d_2,F_2}$ are exactly F_1, F_2, since each factor has trace one. Thus (1.6) gives the lower inequality in (1.7).

For the upper inequality choose, for each factor, a finite ensemble whose cost is within ϵ of its roof. Their product ensemble has the stated tensor density as its actual barycenter. Each product pure vector has entropy equal to the sum of its two entropies across $A_1A_2 : B_1B_2$, because its Alice density is a tensor product. Therefore the product ensemble cost is at most $G_{d_1}(F_1) + G_{d_2}(F_2) + 2\epsilon$. Taking the infimum and letting $\epsilon \downarrow 0$ proves the upper inequality. No minimizing ensemble is needed. □

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Square-root remainders with logarithmic factors for generalized quantum equipartition

Abstract

Under compactness, convexity, permutation symmetry, primal and positive-polar tensor closure, and uniform growth assumptions, we prove a finite-error quantum equipartition estimate with remainder $O_\varepsilon(\sqrt{n}\,[\log(n+1)]^{3/2})$. The estimate uses the regularized Umegaki rate and is accompanied by an exponentially weighted likelihood-moment bound for every exact entropy-minimizing pair.

We prove a finite-error equipartition estimate for two families of matrices on tensor powers of a fixed finite-dimensional Hilbert space. The testing quantity is the infimum of the usual pairwise hypothesis-testing divergence over both families. The same regularized Umegaki rate controls both sides of the estimate and the exponentially weighted likelihood moment of every exact joint entropy minimizer.

Fang, Fawzi, and Fawzi established generalized equipartition with a finite-error remainder of order $n^{2/3}\log n$ [1]. Under the assumptions displayed below, our remainder is $O_\varepsilon(\sqrt{n}\,[\log(n+1)]^{3/2})$. Thus the improvement is in the power of n, with logarithmic factors remaining; this is neither a purely logarithmic remainder nor an $O(\sqrt{n})$ estimate.

For a fixed i.i.d. pair, Tomamichel–Hayashi [3] and Li [4] give the second-order expansion

$$D_{\mathrm{H}}^{\varepsilon}(\rho^{\otimes n}\|\sigma^{\otimes n}) = nD(\rho\|\sigma) + \sqrt{nV(\rho\|\sigma)}\,\Phi^{-1}(\varepsilon) + O(\log n),$$

where V is the relative-entropy variance and Φ the standard normal distribution function. For full-rank pairs with $V > 0$ and $\varepsilon \neq 1/2$, these singleton families already require a remainder of order $\sqrt{n}$. The present estimate approaches that scale but retains logarithmic factors. Generalized Stein lemmas instead identify first-order rates against composite alternative families; see Lami [5]. The result below concerns a quantitative remainder under the stated primal and polar tensor-closure assumptions.

1 Families, divergences, and the theorem

Let $H = \mathbb{C}^d$, with $d \geq 1$. For each integer $n \geq 1$, let $\mathcal{A}_n$ be a nonempty compact convex set of density matrices on $H^{\otimes n}$, and let $\mathcal{B}_n$ be a nonempty compact convex set of positive semidefinite matrices on that space. Both families are invariant under permutations of the tensor factors. For a set S of positive matrices, define its positive polar by

$$S^{\circ,+} = \{X \geq 0 : \operatorname{Tr}(YX) \leq 1 \text{ for every } Y \in S\}.$$

Assume, for all $m, n \geq 1$, the four inclusions

$$\mathcal{A}_m \otimes \mathcal{A}_n \subseteq \mathcal{A}_{m+n}, \qquad \mathcal{B}_m \otimes \mathcal{B}_n \subseteq \mathcal{B}_{m+n}, \tag{1.1}$$

$$\mathcal{A}_m^{\circ,+} \otimes \mathcal{A}_n^{\circ,+} \subseteq \mathcal{A}_{m+n}^{\circ,+}, \qquad \mathcal{B}_m^{\circ,+} \otimes \mathcal{B}_n^{\circ,+} \subseteq \mathcal{B}_{m+n}^{\circ,+}. \tag{1.2}$$

Finally, assume there is a finite constant C such that every $n \geq 1$ and every $\rho \in \mathcal{A}_n$, $\sigma \in \mathcal{B}_n$ satisfy

$$\rho \leq 2^{Cn}\sigma, \qquad \log_2 \operatorname{Tr}\sigma \leq Cn. \tag{1.3}$$

We may enlarge C so that $C \geq 1$. In particular $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$. Alternatives need not be normalized or invertible.

All divergences below use base-two logarithms. For a supported pair define

$$D(\rho\|\sigma) = \operatorname{Tr} \rho(\log_2 \rho - \log_2 \sigma).$$

The logarithms are evaluated on their positive spectral supports; zero-weight terms are zero. Set $D = +\infty$ when support inclusion fails. Write

$$\begin{aligned} D_n &= \inf_{\rho\in\mathcal{A}_n,\,\sigma\in\mathcal{B}_n} D(\rho\|\sigma), \\ \beta_\varepsilon(\rho\|\sigma) &= \min\{\operatorname{Tr}\sigma Q : 0 \le Q \le I,\ \operatorname{Tr}\rho Q \ge 1-\varepsilon\}, \\ D_{\mathrm{H}}^{\varepsilon}(\rho\|\sigma) &= -\log_2 \beta_\varepsilon(\rho\|\sigma), \\ H_n^{\varepsilon} &= \inf_{\rho\in\mathcal{A}_n,\,\sigma\in\mathcal{B}_n} D_{\mathrm{H}}^{\varepsilon}(\rho\|\sigma). \end{aligned}$$

Thus the minimizing test may depend on the pair. Define

$$r = \inf_{n\ge 1} \frac{D_n}{n}, \qquad \ell_n = \log_2(n+1). \tag{1.4}$$

For spectral decompositions $\rho = \sum_i p_i |u_i\rangle\langle u_i|$ and $\sigma = \sum_j q_j |v_j\rangle\langle v_j|$, use the Nussbaum–Szkoła law [2]

$$w_{ij} = p_i |\langle u_i, v_j\rangle|^2, \qquad Z_{ij} = \log_2 p_i - \log_2 q_j. \tag{1.5}$$

It is a probability law. If $p_i = 0$ or $q_j = 0$, its relevant weight is zero: in the second case this follows from support inclusion. Assign any finite value to Z on those zero-weight atoms. Expectations in the theorem refer to this law, for the particular pair in question.

Theorem 1.1. *Under (1.1)–(1.3), the following statements hold.*

1. *Each D_n is finite, its infimum is attained, and the entire sequence D_n/n converges to the finite number r in (1.4).*
2. *For every fixed $\varepsilon \in (0,1)$ there is a finite $K_\varepsilon > 0$ such that, for every $n \ge 1$, H_n^ε is finite and*

$$\left|H_n^\varepsilon - nr\right| \le K_\varepsilon \sqrt{n}\, \ell_n^{3/2}. \tag{1.6}$$

3. *There are finite $K, c > 0$ such that every $n \ge 1$, every exact minimizing pair $D(\rho\|\sigma) = D_n$, and every $0 \le t \le c/n$ satisfy*

$$\mathbb{E}\big[(Z-nr)^2 e^{t|Z-nr|}\big] \le K n \ell_n^3. \tag{1.7}$$

The constants K, c are chosen before the length, the minimizing pair, and t.

2 Quantum entropy inputs and pair repetition

For clarity, we record the quantum entropy inputs used in the proof. For $\alpha \ne 1$ put

$$\begin{aligned} D_{\mathrm{P},\alpha}(\rho\|\sigma) &= \frac{1}{\alpha-1}\log_2 \operatorname{Tr}\rho^\alpha\sigma^{1-\alpha}, \\ D_{\mathrm{S},\alpha}(\rho\|\sigma) &= \frac{1}{\alpha-1}\log_2 \operatorname{Tr}\left(\sigma^{(1-\alpha)/(2\alpha)}\rho\sigma^{(1-\alpha)/(2\alpha)}\right)^\alpha. \end{aligned}$$

Negative powers are evaluated on the support of σ, and above-one divergences are $+\infty$ for unsupported pairs. Let $D_{\mathrm{M},\alpha}$ denote the supremum of the classical Rényi divergence over finite POVMs. At order one these are Umegaki and measured relative entropy. For each divergence, its set version is the infimum over both matrix arguments. Write $M_n(\alpha)$ and $S_n(\alpha)$ for the measured and sandwiched set versions.

The tensor, pinching, and finite-estimation results of Fang–Fawzi–Fawzi [1, Lemmas 21, 23, 26, 28, 29] give, for $1/2 \le \alpha < 1$,

$$M_{m+n}(\alpha) \ge M_m(\alpha) + M_n(\alpha), \tag{2.1}$$

$$M_n(\alpha) \ge S_n(\alpha) - g(n), \tag{2.2}$$

$$nr \le D_n \le nr + g(n), \qquad g(n) = 2(d^2+d)\log_2(n+d). \tag{2.3}$$

Umegaki set entropy is subadditive. These results have exactly the compactness, convexity, permutation, primal, and positive-polar hypotheses stated above; their positive alternatives are allowed to be unnormalized. Choose a fixed G such that $g(n) \le G\ell_n$ for all $n \ge 1$.

The state comparisons used below are

$$D_{\mathrm{M},\alpha} \le D_{\mathrm{P},\alpha} \le D \quad (1/2 \le \alpha < 1), \tag{2.4}$$

$$D_{\mathrm{S},1+\delta} \le D_{\mathrm{P},1+\delta}, \tag{2.5}$$

$$D_{\mathrm{S},1-\delta}(\rho\|\sigma) \ge (1-\delta)D_{\mathrm{P},1-\delta}(\rho\|\sigma) - \delta\log_2 \operatorname{Tr}\sigma. \tag{2.6}$$

The last trace term must be retained. The comparisons and the one-shot inequality used later appear in [1, Lemma 12 and Eq. (269)] and follow from the usual Petz and sandwiched trace comparisons on supports.

For completeness, finiteness and the rate limit can already be settled. Write $\tau = \sigma/\operatorname{Tr}\sigma$. Nonnegativity of $D(\rho\|\tau)$ and (1.3) give $D(\rho\|\sigma) \ge -Cn$, while monotonicity of the logarithm gives $D(\rho\|\sigma) \le Cn$. Compactness and lower semicontinuity give an exact minimizer. Tensor additivity of D and (1.1) imply $D_{m+n} \le D_m + D_n$. Fekete's lemma therefore yields the full-sequence limit $D_n/n \to r$, with $|r| \le C$. The upper estimate in (2.3) follows by combining order-one measured superadditivity with the order-one pinching estimate. These observations establish part 1 of Theorem 1.1.

For $0 < \delta \le 1/2$ define the actual measured rate

$$R(\delta) = \lim_{k\to\infty} \frac{M_k(1-\delta)}{k} = \sup_{m\ge 1} \frac{M_m(1-\delta)}{m}. \tag{2.7}$$

The limit exists by (2.1). A one-outcome measurement bounds the sequence below by $-Ck$, and (2.4) bounds it above by $D_k \le Ck$. In particular $R(\delta) \le r$.

Lemma 2.1 (Repetition transfer)**.** *Every admissible pair at length n obeys*

$$D_{\mathrm{P},1-\delta}(\rho\|\sigma) \ge nR(\delta). \tag{2.8}$$

Proof. Primal closure admits $(\rho^{\otimes k}, \sigma^{\otimes k})$ at length kn. Thus

$$M_{kn}(1-\delta) \le D_{\mathrm{M},1-\delta}(\rho^{\otimes k}\|\sigma^{\otimes k}) \le kD_{\mathrm{P},1-\delta}(\rho\|\sigma).$$

Divide by kn and take the full rate limit along the multiples of n. The conclusion applies to every original pair; no tensor factorization of a minimizer is needed. □

3 Likelihood curvature and a fixed finite baseline

For $u > -1$, the spectral identities and half-order bounds of (1.5) are

$$\mathbb{E}Z = D(\rho\|\sigma), \qquad \mathbb{E}2^{uZ} = \operatorname{Tr}\rho^{1+u}\sigma^{-u}, \qquad \mathbb{E}2^{\pm Z/2} \le 2^{Cn/2}. \tag{3.1}$$

The positive half-order bound follows from $\sigma \ge 2^{-Cn}\rho$ and inverse-square-root antitonicity. One may apply it to strictly positive regularizations preserving this domination and pass to the supported limit. The negative half-order bound is $\operatorname{Tr}\sqrt{\rho}\sqrt{\sigma} \le \sqrt{\operatorname{Tr}\rho\ \operatorname{Tr}\sigma}$. Hence, for

$$X = Z - nr, \qquad s = \mathbb{E}X = D(\rho\|\sigma) - nr \ge 0,$$

Markov's inequality gives

$$\mathbb{P}\{X > x\} \le 2^{Cn - x/2}, \qquad \mathbb{P}\{-X > x\} \le 2^{Cn-x/2} \quad (x \ge 0). \tag{3.2}$$

All bounds are uniform over admissible pairs, including singular pairs.

We use the exact remainder

$$R_0(y) = e^{-y} - 1 + y = y^2 \int_0^1 (1-v)e^{-vy}\,dv \ge 0. \tag{3.3}$$

For $J = 4(C+1)n$ and $t \le a/n$, with fixed sufficiently small $a > 0$, integrating (3.2) outside $[-J, J]$ and bounding the integrand inside that interval yields

$$\mathbb{E}[X^2 e^{t|X|}] \le K_0 n^2. \tag{3.4}$$

Indeed, the inner contribution is at most $J^2 e^{aJ/n}$. The two outer contributions are bounded by the boundary value plus the integrals of $(2x + tx^2)e^{tx}2^{Cn-x/2}$, each of which is $O(n^2 2^{-n})$ after shrinking a. Taylor's estimate

$$e^{\pm tX} \le 1 \pm tX + \tfrac{1}{2}t^2X^2e^{t|X|} \tag{3.5}$$

then gives a below-one Petz loss $O(\delta n^2)$ uniformly over all pairs, where $t = \delta \ln 2$. For an exact minimizing pair, $s \le g(n)$ and the positive version gives the corresponding above-one upper bound. Applying (2.2)–(2.6) produces finite constants $A, B, c_0, N_0 > 0$ with

$$M_n(1-\delta) \ge nr - A\ell_n - B\delta n^2 \quad (n \ge N_0,\ 0 < \delta \le c_0/n). \tag{3.6}$$

We next improve this baseline a fixed number of times. Suppose for one fixed $q \in (0, 1]$ there are fixed A, B, c, N such that

$$M_m(1-\delta) \ge mr - A\ell_m - B\delta m^{1+q}\ell_m^{1-q} \quad (m \ge N,\ 0 < \delta \le c/m). \tag{3.7}$$

Put $\gamma = 1/(1+q) \in [1/2, 1)$. From (2.7),

$$r - R(\delta) \le A\ell_m/m + B\delta m^q \ell_m^{1-q}.$$

Choose $m = \lceil \delta^{-\gamma} L^{1-\gamma} \rceil$, where $L = \log_2(1/\delta)$. As $\delta \downarrow 0$, the block window holds because $\delta m \to 0$, and ℓ_m is comparable to L. The two terms have the same order, so

$$0 \le r - R(\delta) \le K\delta^\gamma L^\gamma. \tag{3.8}$$

Fix $t_0 = \zeta/n$, with sufficiently small constant $\zeta > 0$, and put $\delta_n = t_0/\ln 2$. Lemma 2.1 and (3.8) give

$$\ln \mathbb{E}e^{-t_0 X} = t_0\{nr - D_{\mathrm{P},1-\delta_n}(\rho\|\sigma)\} \le K_1 n^{-\gamma}\ell_n^\gamma =: a_n. \tag{3.9}$$

For large n, $e^{a_n}-1\le 2a_n$. On the negative half-line, (3.3) gives $R_0(-u)\ge u^2e^{u/2}/16$ by restricting the integral to $[1/2,3/4]$. On $0\le y\le t_0J$, it gives $R_0(y)\ge e^{-t_0J}y^2/2$. Consequently, for $0\le t\le t_0/2$,

$$\mathbb{E}[X^2e^{t|X|};X\le J]\le k_1t_0^{-2}\mathbb{E}R_0(t_0X)\le k_1t_0^{-2}(2a_n+t_0s). \tag{3.10}$$

The positive extreme tail beyond J is $O(n^22^{-n})$ by (3.2). Thus

$$\mathbb{E}[X^2e^{t|X|}]\le K_2\{ns+n^{2-\gamma}\ell_n^\gamma\}\quad(0\le t\le c_1/n). \tag{3.11}$$

Retaining s is essential. In the negative Taylor estimate its coefficient is $-t+(K_2/2)t^2n$, which is nonpositive for $t\le 1/(K_2n)$. Dropping that term, applying $\ln(1+z)\le z$, and then (2.2) and (2.6), proves (3.7) with

$$q'=1-\gamma=\frac{q}{1+q}.$$

The scale term costs at most $2Cn\delta$ and is absorbed because $n^{2-\gamma}\ell_n^\gamma\ge n$. Starting at (3.6), three applications give $q=1/4$. We therefore fix, once and for all, constants $G,B,c_*,N_*>0$ with $G\ell_n\ge g(n)$ and

$$M_n(1-\delta)\ge nr-G\ell_n-B\delta n^{5/4}\ell_n^{3/4}\quad(n\ge N_*,\ 0<\delta\le c_*/n). \tag{3.12}$$

These constants are fixed before the coefficient chosen in the next section.

4 A single-coefficient logarithmic induction

We prove that there are one finite $M>0$ and one fixed $c>0$ such that

$$M_n(1-\delta)\ge nr-G\ell_n-M\delta n\ell_n^3\quad(n\ge N_*,\ 0<\delta\le c/(n\ell_n)). \tag{4.1}$$

Choose a positive ζ satisfying

$$\zeta\le\min\left\{\frac{\ln 2}{8},\ \frac{1}{100(C+1)},\ 8\ln 2\,c_*\right\},\qquad c=\frac{\zeta}{8\ln 2}. \tag{4.2}$$

At a large current length n set

$$m=\lfloor n/16\rfloor,\qquad t_0=\frac{\zeta}{n\ell_n},\qquad \delta_n=t_0/\ln 2.$$

For all sufficiently large n,

$$n/32\le m<n,\qquad \ell_m\le\ell_n-2,\qquad \delta_nm\ell_m\le\frac{\zeta}{16\ln 2}<c. \tag{4.3}$$

Thus the induction hypothesis at m is applicable at the actual order δ_n.

By (2.7), the hypothesis at m and Lemma 2.1 yield, for every pair at the current length,

$$\ln\mathbb{E}e^{-t_0X}\le t_0n\{G\ell_m/m+M\delta_n\ell_m^3\}=:a_n, \tag{4.4}$$

$$a_n\le\frac{32\zeta G}{n}+\frac{M\zeta^2}{\ln 2}\frac{\ell_n}{n}. \tag{4.5}$$

Only the estimate at the strictly smaller length has been used.

To preserve the leading coefficient, use a sharper scalar remainder bound. In (3.3), the probability density $2(1-v)$ on $[0,1]$ has mean $1/3$. Jensen's inequality therefore gives, for $u \geq 0$,

$$R_0(-u) \geq \tfrac{1}{2}u^2 e^{u/3}, \qquad R_0(u) \geq \tfrac{1}{2}u^2 e^{-u/3}. \tag{4.6}$$

Let $J = 4(C+1)n$ and $T = t_0 J = 4\zeta(C+1)/\ell_n$. If $0 \leq t \leq t_0/3$, these inequalities imply

$$\mathbb{E}[X^2 e^{t|X|}; X \leq J] \leq \frac{2e^{2T/3}}{t_0^2}\mathbb{E}R_0(t_0 X). \tag{4.7}$$

For $X \leq 0$ the factor $2/t_0^2$ is sufficient; for $0 \leq X \leq J$ the remaining factor is at most $e^{2T/3}$. The positive tail still obeys

$$\mathbb{E}[X^2 e^{tX}; X > J] \leq K_f n^2 2^{-n}. \tag{4.8}$$

If $0 \leq a_n \leq 1$, then $e^{a_n} - 1 \leq a_n(1+2a_n)$. Define

$$H_n = e^{2T/3}(1+2a_n), \qquad P = 32G/\zeta.$$

Substitution of (4.4) into (4.7), retaining ℓ_m^3 in its leading term, yields

$$\mathbb{E}[X^2 e^{t|X|}] \leq 2H_n \left\{ \frac{M}{\ln 2} n\ell_m^3 + Pn\ell_n^2 + \frac{n\ell_n}{\zeta}s \right\} + K_f n^2 2^{-n}. \tag{4.9}$$

Here $0 \leq t \leq t_0/3$.

For an order δ in (4.1), $t = \delta \ln 2 \leq t_0/8$. The coefficient of s in the negative Taylor bound is

$$-t + H_n t^2 n\ell_n/\zeta \leq -t(1 - H_n/8).$$

It is nonpositive whenever $H_n \leq 2$. The factor 2 in (4.9) cancels the factor $1/2$ in (3.5). The below-one Petz loss is consequently bounded by

$$\delta \left\{ H_n M n\ell_m^3 + H_n \ln 2\, Pn\ell_n^2 + \tfrac{1}{2}\ln 2\, K_f n^2 2^{-n} \right\}.$$

State comparison adds at most $2Cn\delta$ and set pinching adds $g(n) \leq G\ell_n$. It follows that a fixed K_0, independent of M, n and the pair, closes the current induction step provided

$$H_n M\ell_m^3 + K_0\ell_n^2 \leq M\ell_n^3. \tag{4.10}$$

We now choose M without a circular dependence. Put $N = \lceil M^2 \rceil$. For $N_* \leq n \leq N$, the quarter-stage estimate (3.12) proves (4.1), because $c \leq c_*$ and

$$Bn^{1/4}\ell_n^{3/4} \leq B2^{1/4}M^{1/2}\ell_n^3 \leq M\ell_n^3$$

for sufficiently large $M \geq 1$. This base interval requires only a sublinear power $M^{1/2}$ of the coefficient.

Enlarge M so that $N \geq 32N_*$ and exceeds all fixed thresholds in (4.3). For $n > N$, eventual monotonicity of ℓ_n^2/n and (4.5) show uniformly that

$$a_n\ell_n = O\left(\frac{G\ell_N}{N} + \frac{M\ell_N^2}{N} \right) \longrightarrow 0 \quad \text{as } M \to \infty.$$

Thus we can arrange $a_n \leq 1/(100\ell_n)$ for all $n > N$. The choice (4.2) then implies

$$H_n \leq 1 + \frac{1}{4\ell_n} \leq 2. \tag{4.11}$$

For $\ell_n \geq 8$ and $\ell_m \leq \ell_n - 2$, expansion of the cubic gives

$$\left(1 + \frac{1}{4\ell_n}\right) \ell_m^3 \leq \ell_n^3 - 2\ell_n^2. \tag{4.12}$$

Finally take $M \geq K_0/2$. Equations (4.11) and (4.12) imply (4.10). Increasing M satisfies all these conditions while preserving the finite-base inequality. Strong induction on n, using $N_* \leq m < n$ beyond the base, proves (4.1) with this one fixed coefficient.

5 The global modulus and the wider moment window

For sufficiently small $\delta > 0$, put $L = \log_2(1/\delta)$ and choose

$$m = \left\lfloor \frac{a}{\delta L} \right\rfloor$$

with a sufficiently small fixed $a > 0$. Then $m \geq N_*$, $\ell_m = O(L)$, and $\delta m \ell_m \leq c$. The supremum formula (2.7) and (4.1) give

$$0 \leq r - R(\delta) \leq G\ell_m/m + M\delta\ell_m^3 \leq K\delta \log_2(1/\delta)^3. \tag{5.1}$$

In particular, its order window is now a fixed interval $(0, \delta_0]$ independent of the current length.

Take $t_0 = \zeta'/n$ for a fixed sufficiently small $\zeta' > 0$, and let $\delta_n = t_0/\ln 2$. Repetition transfer and (5.1) imply

$$\ln \mathbb{E} e^{-t_0 X} \leq t_0 n\{r - R(\delta_n)\} \leq K' \frac{\ell_n^3}{n}.$$

The right side tends to zero. Applying the fixed-scale curvature argument (3.10), with $t_0 J$ now constant, and the positive extreme tail gives

$$\mathbb{E}[X^2 e^{t|X|}] \leq K''\{ns + n\ell_n^3\} \quad (0 \leq t \leq c'/n) \tag{5.2}$$

for all large n and every admissible pair. For every exact minimizer, (2.3) gives $0 \leq s \leq g(n) \leq G\ell_n$, so (5.2) proves (1.7) at large lengths. The uniform half-order tails (3.2) bound the same exponential moments over the finitely many remaining lengths, after shrinking c' if necessary and enlarging K. This proves part 3, with a single window c/n and constants uniform over all exact minimizers.

6 Both testing bounds at the same rate

Fix $\varepsilon \in (0, 1)$ and set $f_n = \sqrt{n}\,\ell_n^{3/2}$. The Petz one-shot lower bound [1, Lemma 12] is

$$D_{\mathrm{H}}^{\varepsilon}(\rho\|\sigma) \geq D_{\mathrm{P},1-\delta}(\rho\|\sigma) - \frac{1-\delta}{\delta} \log_2(1/\varepsilon). \tag{6.1}$$

Choose $\delta = 1/f_n$ for sufficiently large n. This belongs to the fixed modulus window, and $\log_2(1/\delta) = O(\ell_n)$. Lemma 2.1 and (5.1) yield, uniformly before taking either family infimum,

$$D_{\mathrm{H}}^{\varepsilon}(\rho\|\sigma) \geq nr - Kn\delta \log_2(1/\delta)^3 - \delta^{-1} \log_2(1/\varepsilon) \geq nr - K_\varepsilon f_n.$$

The same lower bound therefore holds for H_n^{ε}.

For the upper bound we give a physical converse that uses only the second moment in part 3. Let $0 \le Q \le I$, $\operatorname{Tr}\rho Q \ge 1-\varepsilon$, and $\beta = \operatorname{Tr}\sigma Q$. For any $a \in \mathbb{R}$ and any $b > 0$,

$$\mathbb{P}\{Z \le a\} \le (1+b)\varepsilon + (1+b^{-1})2^a\beta. \tag{6.2}$$

To see this, split each overlap as

$$\langle u_i, v_j\rangle = \langle u_i, (I-Q)v_j\rangle + \langle u_i, Qv_j\rangle.$$

Use $|z+w|^2 \le (1+b)|z|^2 + (1+b^{-1})|w|^2$, multiply by p_i, and sum over the event $p_i \le 2^a q_j$. Enlarging each nonnegative sum to all spectral indices gives

$$(1+b)\operatorname{Tr}\rho(I-Q)^2 + (1+b^{-1})2^a \operatorname{Tr}\sigma Q^2.$$

Because $(I-Q)^2 \le I-Q$ and $Q^2 \le Q$, this is at most the right side of (6.2). Kernel columns have weight zero and do not affect the argument.

Choose an exact minimizing pair at length n. Set

$$\eta = (1-\varepsilon)/4, \quad b = (1-\varepsilon)/2, \quad h = \sqrt{K/\eta}\, f_n, \quad a = nr + h,$$

where K is the constant in (1.7). Chebyshev's inequality at $t = 0$ gives $\mathbb{P}\{Z \le a\} \ge 1-\eta$, and $1-\eta-(1+b)\varepsilon > 0$. Thus every accepting effect obeys

$$\beta \ge 2^{-a}\frac{1-\eta-(1+b)\varepsilon}{1+b^{-1}}.$$

Taking the minimum over effects and then the negative logarithm yields

$$D_{\mathrm{H}}^{\varepsilon}(\rho\|\sigma) \le nr + \sqrt{K/\eta}\, f_n - \log_2 \frac{1-\eta-(1+b)\varepsilon}{1+b^{-1}}. \tag{6.3}$$

The original pair-infimum H_n^{ε} is no larger than the value of this admissible pair. This proves its upper bound about the same nr.

Finally (1.3) gives, for every pair and every positive length,

$$2^{-Cn}(1-\varepsilon) \le \beta_{\varepsilon}(\rho\|\sigma) \le \operatorname{Tr}\sigma \le 2^{Cn}.$$

Consequently both pairwise and set testing values are finite. These inequalities absorb the finitely many lengths excluded when choosing $\delta = 1/f_n$. Enlarging K_{ε} proves (1.6) for every $n \ge 1$ and completes Theorem 1.1.

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Low-noise and boundary asymptotics for depolarizing-channel squashed entanglement

Abstract

For $\Lambda_p(X) = (1-p)X + p\ \mathrm{Tr}(X)I_2/2$, squashed entanglement with arbitrary finite quantum squashing outputs satisfies

$$E_{\mathrm{sq}}(\Lambda_p) = 1 - \frac{3p}{8}\log_2\frac{1}{p} + O(p) \quad (p \downarrow 0).$$

At $p = 2/3 - 4\delta/3$, its value is bounded above and below by positive constant multiples of δ^2. An explicit map from the canonical four-dimensional environment to a sixteen-dimensional output works for every input; its exact maximum is attained at the Bell input. Its cost is $J(r(\delta))$, with

$$J(r) = \frac{\kappa r^2}{2\ln 2} + O(r^3), \qquad \kappa = \frac{2059}{5600}\ln 29 - \frac{2171}{5800} > 0.$$

The remainders are uniform on the stated physical intervals.

Takeoka, Guha, and Wilde introduced channel squashed entanglement as a bound on two-way assisted quantum and private communication [5]. Discarding the squashing environment gives the standard upper bound of half the entanglement-assisted classical capacity. For the depolarizing channel this is

$$E_{\mathrm{sq}}(\Lambda_p) \le 1 - \tfrac{1}{2}H(1-3p/4, p/4, p/4, p/4),$$

where H is the base-two Shannon entropy. Its low-noise expansion already has the coefficient $3/8$ in the abstract. The new low-noise claim is the matching lower bound for arbitrary finite quantum squashing outputs. The boundary estimates and the explicit squashing map are stated separately below.

1 Definitions and the endpoint theorems

Every state and channel below acts on finite-dimensional complex Hilbert spaces. Density matrices may be singular. Use natural entropy $S_{\mathrm{nat}}(\rho) = -\operatorname{Tr}\rho\ln\rho$, with $0\ln 0 = 0$, and bit entropy $S_2 = S_{\mathrm{nat}}/\ln 2$. Conditional mutual information is

$$I_2(R:B\mid E) = S_2(RE) + S_2(BE) - S_2(E) - S_2(RBE), \qquad I_{\mathrm{nat}} = (\ln 2)I_2.$$

For a state,

$$E_{\mathrm{sq}}(\rho_{RB}) = \inf_{\substack{\omega_{RBE}\ \text{density} \\ \dim E<\infty,\ \mathrm{Tr}_E\,\omega=\rho_{RB}}} \frac{1}{2}I_2(R:B\mid E)_\omega.$$

Let $P_0 = I, P_1 = X, P_2 = Y, P_3 = Z$, and put

$$q_0 = 1 - \frac{3p}{4}, \quad q_1 = q_2 = q_3 = \frac{p}{4}, \qquad U_p|\psi\rangle = \sum_{i=0}^{3}\sqrt{q_i}\,P_i|\psi\rangle_B|i\rangle_{\mathsf{E}}.$$

For $0 \le p \le 1$ this is an isometry implementing Λ_p. The canonical environment E has dimension four. The channel value has the following order of quantifiers:

$$E_{\mathrm{sq}}(\Lambda_p) = \sup_{\substack{\dim R<\infty \\ |\psi\rangle_{RA}\ \text{unit}}} \frac{1}{2}\inf_{\substack{\dim E<\infty \\ \mathcal{S}:\mathsf{E}\to E\ \mathrm{CPTP}}} I_2(R:B\mid E)_{(\mathrm{id}_{RB}\otimes\mathcal{S})(\mathrm{id}_R\otimes U_p)|\psi\rangle\langle\psi|(\mathrm{id}_R\otimes U_p^\dagger)}. \tag{1.1}$$

All finite squashing output dimensions occur in the inner infimum. Write $|\Phi\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, $P = |\Phi\rangle\langle\Phi|$, and

$$\rho_\delta = \left(\frac{1}{2} + \delta\right) P + \left(\frac{1}{2} - \delta\right) \frac{I - P}{3}, \qquad p(\delta) = \frac{2}{3} - \frac{4\delta}{3}.$$

The normalized Choi state of $\Lambda_{p(\delta)}$ is ρ_δ.

Theorem 1.1. *There is $C_0 > 0$ such that, for every $0 < p \le 1/2$,*

$$\left| E_{\mathrm{sq}}(\Lambda_p) - 1 + \frac{3p}{8} \log_2 \frac{1}{p} \right| \le C_0 p. \tag{1.2}$$

There is $C_1 > 0$ such that, for every $0 \le \delta \le 21/512$,

$$\frac{\delta^2}{9 \ln 2} \le E_{\mathrm{sq}}(\Lambda_{p(\delta)}) \le C_1 \delta^2. \tag{1.3}$$

In particular $E_{\mathrm{sq}}(\Lambda_{p(\delta)}) = \Theta(\delta^2)$ as $\delta \downarrow 0$.

Theorem 1.2. *For every $0 \le \delta \le 21/512$ there is an explicitly defined CPTP map $\mathcal{S}_\delta : \mathsf{E} \to \mathbb{C}^{16}$ such that its half-CMI cost over all normalized pure reference/input states has a maximum attained at the Bell input, equal to*

$$J(r) = \frac{H(0,r) + H(r,0) - H(r,r) - H(0,0)}{2 \ln 2}, \qquad r = r(\delta) = \frac{320\delta}{\sqrt{324 + 21120\delta} + 18}, \tag{1.4}$$

$$H(x,y) = S_{\mathrm{nat}}((D_x \otimes D_y)\xi), \qquad \xi = \frac{7}{8}|v\rangle\langle v| + \frac{I_4}{32},$$

$$|v\rangle = \frac{2|00\rangle + |11\rangle}{\sqrt{5}}.$$

Here D_x is amplitude damping with Kraus operators $\mathrm{diag}(1, \sqrt{1-x})$ *and $\sqrt{x}|0\rangle\langle 1|$. There are finite constants $M_r, M_\delta \ge 0$ such that*

$$\left| J(r) - \frac{\kappa}{2 \ln 2} r^2 \right| \le M_r r^3 \qquad (0 \le r \le 1/4), \tag{1.5}$$

$$\left| J(r(\delta)) - \frac{3200\kappa}{81 \ln 2} \delta^2 \right| \le M_\delta \delta^3 \qquad (0 \le \delta \le 21/512). \tag{1.6}$$

The coefficient κ is the cost coefficient of this displayed map.

2 Finite extensions and the Bell input

Every finite extension of the output associated with an input in (1.1) is obtained by a finite channel on its purifying environment. To see this, purify the extension in an additional finite system G. Two purifications of the same output marginal are related by an isometry on the support of the purifying marginal. Extend its associated channel off that support by mapping the unused trace to a fixed output density. Conversely a channel on the environment supplies an extension with the same output marginal. Thus the inner infimum is exactly the state squashed entanglement of that input's output.

For this Pauli channel,

$$E_{\mathrm{sq}}(\Lambda_p) = E_{\mathrm{sq}}((\mathrm{id} \otimes \Lambda_p)P). \tag{2.1}$$

Here is the finite channel argument. Standard Bell measurement and Pauli correction simulate Λ_p from its Choi state: for the resource $(I \otimes P_i)P(I \otimes P_i)^\dagger$, teleportation implements $X \mapsto P_i X P_i$; their mixture with weights q_i implements Λ_p. This is a finite LOCC protocol [2, Appendix B, Proposition 26]. The pure input and its reference start locally with the sender. Adding them does not change the resource's squashed entanglement, and LOCC monotonicity [1] bounds the output value by that resource's value. Bell input gives the reverse inequality. The extension argument above identifies the resource infimum with the canonical-environment infimum. This proves (2.1) with the order in (1.1).

3 A uniform binary-square entropy bound

For positive, possibly subnormalized X, set $S_{\mathrm{sub}}(X) = -\operatorname{Tr} X \ln X$. For a binary orthogonal pinching Δ, define its natural entropy cost $C(X) = S_{\mathrm{sub}}(\Delta X) - S_{\mathrm{sub}}(X)$. Pinching preserves trace, so $C(\alpha X) = \alpha C(X)$.

Lemma 3.1. *Let H be any finite Hermitian binary-block matrix. Put*

$$t = \operatorname{Tr} H^2, \qquad u = \|PH(I-P)\|_{\mathrm{HS}}^2.$$

If $0 < u \le t/2$, then

$$C(H^2) \le u\,[\mathcal{F}(t/u) + 4], \qquad \mathcal{F}(m) = \operatorname{arcosh}(m/2)\sqrt{\frac{m+2}{m-2}}, \quad \mathcal{F}(2) = 2. \tag{3.1}$$

At $u = 0$, $C(H^2) = 0$. For $m \ge 2$,

$$\mathcal{F}(m) \le \ln m + \frac{2\ln m}{m-1} \le \ln m + 2. \tag{3.2}$$

There is no assumption on the signs of the diagonal blocks or on rank.

Proof. Write

$$H = \begin{pmatrix} A & Z^\dagger \\ Z & B \end{pmatrix}, \quad \rho = H^2, \quad D = \Delta\rho = D_0 \oplus D_1,$$

$$D_0 = A^2 + Z^\dagger Z, \qquad D_1 = B^2 + ZZ^\dagger, \qquad Y = ZA + BZ.$$

For $x > 0$ let

$$S_x = (D_1 + xI)^{-1/2} Y (D_0 + xI)^{-1/2}, \quad Q_x = S_x^\dagger S_x < I.$$

The strict inequality is the Schur complement of $\rho + xI > 0$. The determinant formula gives

$$\ln\det(D + xI) - \ln\det(\rho + xI) = -\operatorname{Tr}\ln(I - Q_x).$$

Integrate the scalar primitive $(x+\lambda)\ln(x+\lambda) - (x+\lambda)$ over the two spectra. Equal dimension and trace cancel the infinite endpoint. The finite endpoint is $\operatorname{Tr}\rho\ln\rho - \operatorname{Tr} D\ln D$. Zero eigenvalues produce integrable $\ln x$ terms, and the infinite tail is $O(x^{-2})$. Thus

$$C(\rho) = \mathcal{J}(Y) + N, \tag{3.3}$$

where

$$\mathcal{J}(V) = \int_0^\infty \operatorname{Tr} V^\dagger (D_1 + xI)^{-1} V (D_0 + xI)^{-1}\, dx,$$
$$N = \int_0^\infty \operatorname{Tr}[-\ln(I - Q_x) - Q_x]\, dx.$$

If $z \in \ker D_0$, then $Az = Zz = 0$; if $z \in \ker D_1$, then $Bz = Z^\dagger z = 0$. All subsequent inverses and logarithms can be restricted to these positive ranges without changing the integrals.

We first bound the nonlinear term. Let $H_0 = A \oplus B$ and $W = H - H_0$. For every $\varepsilon > 0$, expansion of a positive Hermitian square gives

$$D = H_0^2 + W^2 \le (1+\varepsilon)\rho + (2 + \varepsilon^{-1}) W^2. \tag{3.4}$$

Let $w_k = \sigma_k(Z)^2$ be in decreasing order. Split $W^2 = F_k + R_k$, keeping its largest $k-1$ singular modes in each sector. Then $\operatorname{rank} F_k \le 2(k-1)$ and $0 \le R_k \le w_k I$. For $w_k > 0$, put $r = x/w_k$, $\varepsilon = (\sqrt{1+r} - 1)^{-1}$, and $c = 1 - (1+r)^{-1/2}$. These satisfy $c(1+\varepsilon) = 1$ and $(1-c)x = c(2+\varepsilon^{-1})w_k$. Equation (3.4) implies

$$\rho + xI - c(D + xI) \ge -c(2 + \varepsilon^{-1}) F_k.$$

After congruence by $(D + xI)^{-1/2}$, the matrix $\left(\begin{smallmatrix} I & S_x^\dagger \\ S_x & I \end{smallmatrix}\right)$ is at least cI minus a positive matrix of rank at most $2(k-1)$. Its $(2k-1)$st smallest eigenvalue is therefore at least c. Its small eigenvalues are $1 - \sigma_j(S_x)$, with extra ones for unequal sectors. Consequently

$$q_{2k-1}(x), q_{2k}(x) \le \frac{w_k}{x + w_k}, \quad q_j(x) = \sigma_j(S_x)^2.$$

Extend the lists by zero; a pair with only one index decreases the sum. For $f_0(q) = -\ln(1-q) - q$, which is increasing and nonnegative,

$$\int_0^\infty f_0\!\left(\frac{w}{x+w}\right) dx = w \int_0^1 \frac{f_0(q)}{q^2}\, dq = w.$$

An antiderivative in the last integral is $-\ln(1-q) - f_0(q)/q$, with endpoint values 0 and 1. It follows that

$$N \le 2 \sum_k w_k = 2u. \tag{3.5}$$

Next let $P_0 = \sqrt{D_0}$, $P_1 = \sqrt{D_1}$, and $R_i = (D_i + xI)^{-1}$. The diagonal blocks A, B may be indefinite. Nevertheless inverse order and congruence give

$$AR_0A \le A(A^2 + xI)^{-1} A = A^2 (A^2 + xI)^{-1} \le D_0 (D_0 + xI)^{-1} = P_0 R_0 P_0.$$

The analogous inequality holds for B. Taking positive traces yields

$$\mathcal{J}(ZA) + \mathcal{J}(BZ) \le \mathcal{J}(ZP_0) + \mathcal{J}(P_1 Z).$$

The mixed form is the integral of the Hilbert–Schmidt inner product of BR_1Z and ZR_0A. Each of their integrated squared norms is at most u, because $B^2 \le D_1, A^2 \le D_0$ and $\int_0^\infty d/(d+x)^2\, dx = 1$ on positive support. Two applications of Cauchy–Schwarz give

$$\mathcal{J}(Y) \le \mathcal{J}(ZP_0) + \mathcal{J}(P_1 Z) + 2u. \tag{3.6}$$

In separate eigenbases of P_0, P_1, the self-term is

$$\sum_{b,a} |Z_{ba}|^2 (p_a^2 + q_b^2) \frac{\ln p_a^2 - \ln q_b^2}{p_a^2 - q_b^2}.$$

The equal-eigenvalue formula is its continuous limit. On Hilbert–Schmidt matrices let $X = \ln L_{D_1} - \ln R_{D_0}$. The commuting left and right multiplication operators give a spectral probability measure associated with $Z/\sqrt{u}$. For $y = |X|$, the last display is $u\,\mathbb{E}[y \coth(y/2)]$, taking value 2 at $y = 0$. The inverse contractions

$$Z D_0^{-1} Z^\dagger \le I, \qquad Z^\dagger D_1^{-1} Z \le I$$

imply $\langle Z, e^X Z\rangle \le \operatorname{Tr} D_1$ and $\langle Z, e^{-X} Z\rangle \le \operatorname{Tr} D_0$. Thus

$$\mathbb{E} \cosh y \le \frac{t}{2u}.$$

With $r = \sinh^2(y/2)$, direct integration gives

$$y \coth(y/2) = 2 \int_0^1 \sqrt{\frac{1+r}{1+v^2 r}}\, dv.$$

For $\beta \in [0,1]$, $g_\beta(r) = \sqrt{(1+r)/(1+\beta r)}$ is increasing and concave:

$$g_\beta' = \frac{(1-\beta) g_\beta}{2(1+r)(1+\beta r)}, \qquad g_\beta'' = -\frac{(1-\beta)(1+3\beta+4\beta r) g_\beta}{4(1+r)^2(1+\beta r)^2}.$$

Since $\mathbb{E} r \le (t/u - 2)/4$, Jensen gives

$$\mathcal{J}(ZP_0) + \mathcal{J}(P_1 Z) \le u \mathcal{F}(t/u).$$

Combine this with (3.3), (3.5), and (3.6) to obtain (3.1).

Finally $y \coth(y/2)$ is increasing, since its derivative multiplied by $\sinh^2(y/2)$ is $(\sinh y - y)/2 \ge 0$. Use $\operatorname{arcosh}(m/2) \le \ln m$ and $\coth(\ln m/2) = 1 + 2/(m-1)$ to obtain the first bound in (3.2). The second follows from $\ln m \le m - 1$. □

For later use, if T is a rectangular amplitude whose input and output each have a binary decomposition, let

$$t_T = \|T\|_{\mathrm{HS}}^2, \qquad u_T = \|T_{01}\|_{\mathrm{HS}}^2 + \|T_{10}\|_{\mathrm{HS}}^2.$$

The Hermitian dilation $\left(\begin{smallmatrix} 0 & T \\ T^\dagger & 0 \end{smallmatrix}\right)$, grouped by the two binary labels, has budgets $t = 2t_T, u = u_T$. Its square has opposite Grams. When $u_T > 0$, Lemma 3.1 gives

$$C_{\mathrm{out}}(TT^\dagger) + C_{\mathrm{in}}(T^\dagger T) \le u_T[\mathcal{F}(2t_T/u_T) + 4]. \tag{3.7}$$

At $u_T = 0$, both costs are zero. The sectors may have unequal dimensions.

4 Four-image entropy sharing at low noise

Put $s = 3p/4$, $a = p/4 = s/3$, $b = 1 - s$, and $w = b + a = 1 - 2a$. A finite CPTP squashing map has a finite Stinespring isometry $\mathsf{E} \to E \otimes G$. Its four images $v_i = \mathrm{vec}(M_i)$ are arbitrary orthonormal vectors:

$$M_i \in \mathbb{C}^{d_E \times d_G}, \qquad \mathrm{Tr}\, M_i^\dagger M_j = \delta_{ij}.$$

The pure squashed Bell extension is

$$|\Psi\rangle = \sqrt{b}\,|\Phi\rangle v_0 + \sqrt{a}\sum_{j=1}^{3}(I \otimes P_j)|\Phi\rangle\, v_j.$$

Its AE/BG rectangular amplitude, and its AG/BE amplitude, are

$$T_E = \frac{1}{\sqrt{2}}\left[\sqrt{b}\, I \otimes M_0 + \sqrt{a}\sum_{j=1}^{3} P_j^{\mathsf{T}} \otimes M_j\right], \tag{4.1}$$
$$T_G = \frac{1}{\sqrt{2}}\left[\sqrt{b}\, I \otimes M_0^{\mathsf{T}} + \sqrt{a}\sum_{j=1}^{3} P_j^{\mathsf{T}} \otimes M_j^{\mathsf{T}}\right].$$

Each has Hilbert–Schmidt norm one. Its left Gram is the named first marginal; its physical right marginal is the total transpose of the right Gram. Entropy and the pinching by real computational projectors are invariant under that transpose. The environment marginals are exactly

$$\rho_E = bM_0M_0^\dagger + a\sum_j M_jM_j^\dagger, \qquad \rho_G = \left(bM_0^\dagger M_0 + a\sum_j M_j^\dagger M_j\right)^{\mathsf{T}}.$$

Hence the half-CMI bit cost U satisfies

$$\begin{aligned} U &= \frac{1}{2}[S_2(AE) - S_2(E) + S_2(AG) - S_2(G)] \\ &= 1 - \frac{1}{4}J_4, \qquad J_4 = I_2(A:E) + I_2(A:G) + I_2(B:E) + I_2(B:G). \end{aligned} \tag{4.2}$$

The four-MI identity follows from purity and maximally mixed A, B; it uses both environment entropy subtractions.

For a spin/environment marginal, computational pinching gives the exact decomposition

$$I_2(A:E) = I_2(Z_A:E) + C_A(\rho_{AE})/\ln 2.$$

There are four such decompositions. In (4.1) the two off-spin blocks are $\sqrt{a/2}(M_1 + iM_2)$ and $\sqrt{a/2}(M_1 - iM_2)$, each of squared norm a. Each of the two rectangular dilations therefore has $t = 2, u = 2a$. Equation (3.7) bounds the sum of four coherence costs, in bits, by

$$\frac{4a}{\ln 2}[\mathcal{F}(1/a) + 4]. \tag{4.3}$$

Let $A_0 = \sqrt{b}M_0$, $V = \sqrt{a}M_3$. The two conditional E states after A pinching are

$$(A_0 \pm V)(A_0 \pm V)^\dagger + a(M_1 \pm iM_2)(M_1 \pm iM_2)^\dagger.$$

Each has trace $w + 2a = 1$. The coherent branch of weight w is the equal-prior ensemble

$$v_{\pm} = \frac{\sqrt{b}\, v_0 \pm \sqrt{a}\, v_3}{\sqrt{w}}.$$

Write χ_E, χ_G for its two reduced Holevo quantities. The XY branch has weight $2a$ and equal priors. For B pinching, the same coherent branch appears, while the XY states exchange. Convexity of mutual information at fixed classical prior and the binary information bound 1 give

$$\sum_{\text{four marginals}} I_2(Z : \text{environment}) \le 2w(\chi_E + \chi_G) + 8a. \tag{4.4}$$

The equality of priors in both branches justifies this convexity step.

Form a second rectangular matrix

$$L = \frac{1}{\sqrt{2}} \begin{pmatrix} A_0 & V \\ V & A_0 \end{pmatrix}.$$

It has squared norm w. A Hadamard on each binary copy label turns it into the blocks $(A_0 + V)/\sqrt{2}, (A_0 - V)/\sqrt{2}$, each of squared norm $w/2$. The block entropy formula gives

$$C(LL^\dagger) + C(L^\dagger L) = (\ln 2) w(\chi_E + \chi_G).$$

Its Hermitian dilation has budgets $t = 2w, u = a$. Therefore

$$(\ln 2) w(\chi_E + \chi_G) \le a[\mathcal{F}(2w/a) + 4].$$

Combine this with (4.2), (4.3), and (4.4). For every finite squashing map,

$$U \ge 1 - \frac{a}{\ln 2}\mathcal{F}(1/a) - \frac{a}{2\ln 2}\mathcal{F}(2w/a) - \frac{6a}{\ln 2} - 2a. \tag{4.5}$$

This bound holds throughout $0 < p \le 1$; both ratios are at least 4.

In the original p convention the natural loss in (4.5) is

$$\mathcal{L}(p) = \frac{p}{4}\mathcal{F}(4/p) + \frac{p}{8}\mathcal{F}((8-4p)/p) + (6 + 2\ln 2)\frac{p}{4}.$$

Using (3.2) and $\ln((8-4p)/p) \le \ln(8/p)$ gives, for $0 < p \le 1/2$,

$$\mathcal{L}(p) \le \frac{3p}{8}\ln\frac{1}{p} + \frac{18 + 11\ln 2}{8}p. \tag{4.6}$$

Thus every finite map obeys

$$U \ge 1 - \frac{3p}{8}\log_2\frac{1}{p} - \frac{18 + 11\ln 2}{8\ln 2}p.$$

The constant does not depend on the images, spectra, or either environment dimension. Infimizing and admitting Bell input to (1.1) preserves the inequality.

5 An admissible splitter and the matching low-noise upper

Let $E = G = \mathbb{C}^4$ and choose the four orthonormal images

$$v_0 = |0,0\rangle, \qquad v_j = \frac{|j,0\rangle + |0,j\rangle}{\sqrt{2}} \quad (1 \le j \le 3).$$

They define an isometry on E, and tracing out G defines a legal squashing map. Direct calculation gives the environment eigenvalues

$$1 - \frac{s}{2}, \quad \frac{s}{6}, \quad \frac{s}{6}, \quad \frac{s}{6}.$$

To compute the active AE spectrum, write its amplitude as

$$T = \begin{pmatrix} \alpha I_2 & \gamma X^{\mathsf{T}} & \gamma Y^{\mathsf{T}} & \gamma Z^{\mathsf{T}} \\ \gamma X^{\mathsf{T}} & 0 & 0 & 0 \\ \gamma Y^{\mathsf{T}} & 0 & 0 & 0 \\ \gamma Z^{\mathsf{T}} & 0 & 0 & 0 \end{pmatrix}, \qquad \alpha = \sqrt{(1-s)/2}, \quad \gamma = \sqrt{s/12}.$$

The map from the vacuum spin space to the three other sectors has Gram $3\gamma^2 I_2$. On its image and the vacuum spin space, T reduces to $\left(\begin{smallmatrix} \alpha & \sqrt{3}\gamma \\ \sqrt{3}\gamma & 0 \end{smallmatrix}\right) \otimes I_2$; it is zero on the orthogonal complement. Its square therefore has eigenvalues

$$\frac{1-q}{2}, \quad \frac{q}{2}, \quad \text{each twice}, \qquad q = \frac{1-\sqrt{1-s^2}}{2}.$$

Both environment and AE/AG marginals are exchange symmetric. The exact half-CMI bit cost is consequently

$$U_{\text{split}} = 1 - h_2(s/2) + h_2(q) - \frac{s}{2}\log_2 3. \tag{5.1}$$

Since $q = O(s^2)$, $h_2(q) = O(s^2 \ln(1/s))$, and $h_2(s/2) = (s/2)\log_2(1/s) + O(s)$. Equation (2.1) and the legal extension give

$$E_{\text{sq}}(\Lambda_p) \le U_{\text{split}} = 1 - \frac{3p}{8}\log_2\frac{1}{p} + O(p).$$

Together with (4.6), this proves the low-noise part of Theorem 1.1. The finite formula on $0 < p \le 1/2$, its continuous small-p expansion, and compactness away from zero give a single finite remainder constant C_0 on that full interval.

6 A faithful core and weak local isometries

Let L, R be qubits and take the core in Theorem 1.2:

$$\xi = \frac{1}{160}\begin{pmatrix} 117 & 0 & 0 & 56 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 56 & 0 & 0 & 33 \end{pmatrix}. \tag{6.1}$$

Its eigenvalues are $29/32, 1/32, 1/32, 1/32$. It is normalized, satisfies $\xi \ge I/32$, and is entangled: the odd block of its partial transpose has eigenvalue $(5-56)/160 < 0$. Also $\mathrm{Tr}(P\xi) = 131/160 > 1/2$. For an explicit purification in $G = \mathbb{C}^4$, let $v_\perp = (|00\rangle - 2|11\rangle)/\sqrt{5}$ and set

$$|\zeta\rangle = \sqrt{\frac{29}{32}}|v\rangle|0\rangle_G + \sqrt{\frac{1}{32}}(|v_\perp\rangle|1\rangle_G + |01\rangle|2\rangle_G + |10\rangle|3\rangle_G). \tag{6.2}$$

The four core vectors are orthonormal, so this is normalized and has marginal ξ.

For $0 \le r \le 1/4$, define $V_r : L \to A \otimes L$ by

$$V_r|0\rangle = |0_A 0_L\rangle, \qquad V_r|1\rangle = \sqrt{1-r}|0_A 1_L\rangle + \sqrt{r}|1_A 0_L\rangle. \tag{6.3}$$

Its columns are orthonormal. Use the same isometry on $R \to B \otimes R$ and define

$$|\psi_r\rangle = (V_r \otimes V_r \otimes I_G)|\zeta\rangle, \qquad E = LR.$$

It is a normalized pure $ABEG$ state with E, G both of dimension four. Tracing out the retained qubit in (6.3) gives D_r. Tracing out its environment gives a complementary qubit channel T_r whose excited-state survival is r and whose off-diagonal multiplier is $\sqrt{r}$.

The AB marginal is $(T_r \otimes T_r)\xi$. Its 00/11 off-diagonal entry is $7r/20$, and its populations are

$$p_{11} = \frac{33r^2}{160}, \qquad p_{01} = p_{10} = \frac{r(38-33r)}{160}, \qquad p_{00} = 1 - \frac{76r - 33r^2}{160}.$$

Hence

$$\mathrm{Tr}(P\rho_{AB}(r)) = \frac{1}{2} + \delta(r), \qquad \delta(r) = \frac{18r + 33r^2}{160}. \tag{6.4}$$

This is strictly increasing on $r \ge 0$, with $\delta(1/4) = 21/512$. Solving its quadratic gives exactly

$$r(\delta) = \frac{\sqrt{324 + 21120\delta} - 18}{66} = \frac{320\delta}{\sqrt{324 + 21120\delta} + 18}, \qquad 0 \le r(\delta) \le \frac{80}{9}\delta.$$

Thus it covers every δ in the claimed interval, including zero.

7 Entropy support, the Hessian, and uniform cubic remainders

Let $H(x, y)$ be as in (1.4). The four actual entropies of the seed are

$$S_{\mathrm{nat}}(AE) = H(0, r), \qquad S_{\mathrm{nat}}(BE) = H(r, 0),$$
$$S_{\mathrm{nat}}(E) = H(r, r), \qquad S_{\mathrm{nat}}(ABE) = S_{\mathrm{nat}}(G) = H(0, 0).$$

The first two follow by tracing the other retained qubit and removing the remaining isometry; the last follows by purity. Thus the seed's half-CMI bit cost is exactly $J(r)$. For $0 \le x, y \le 1/4$,

$$(D_x \otimes D_y)\xi \ge \frac{1}{32} D_x(I) \otimes D_y(I) \ge \frac{9}{512} I. \tag{7.1}$$

The four nonzero spectra on the isometric AE, BE supports have the same lower bound.

For completeness the matrix inside H has X-state entries

$$a' = \frac{117 + 5(x+y) + 33xy}{160}, \qquad b' = \frac{(1-y)(5+33x)}{160},$$
$$c'_{\mathrm{odd}} = \frac{(1-x)(5+33y)}{160}, \qquad d' = \frac{33(1-x)(1-y)}{160},$$
$$z' = \frac{56}{160}\sqrt{(1-x)(1-y)}.$$

Its eigenvalues are b', c'_{odd}, and

$$\lambda_\pm = \frac{a' + d' \pm \sqrt{(a'-d')^2 + 4(z')^2}}{2}.$$

On the whole square $[0,1/4]^2$, (7.1) keeps them positive, and $z' \neq 0$ keeps the discriminant positive. They are smooth on a neighborhood of this compact square. Their entropy derivatives through order three are bounded. Taylor's formula for $g(r) = H(0,r) + H(r,0) - H(r,r) - H(0,0)$ therefore gives a uniform remainder on $0 \le r \le 1/4$. Its constant and linear terms cancel.

We compute its quadratic coefficient exactly. The derivative of damping at zero is

$$\mathcal{G}(X) = |0\rangle\langle 1|\, X\, |1\rangle\langle 0| - \frac{1}{2}\{|1\rangle\langle 1|, X\}.$$

Let $A = (\mathcal{G} \otimes \mathrm{id})\xi$, $B = (\mathrm{id} \otimes \mathcal{G})\xi$, and $C = (\mathcal{G} \otimes \mathcal{G})\xi$. In units $1/160$, their diagonals are

$$A : (5, 33, -5, -33), \quad B : (5, -5, 33, -33), \quad C : (33, -33, -33, 33);$$

their $00/11$ off-diagonals are respectively $-28, -28, 14$. The differential of natural entropy gives

$$g(r) = \kappa r^2 + O(r^3), \qquad \kappa = \operatorname{Tr} C \ln \xi + \operatorname{Tr} A(D \ln_\xi)(B). \tag{7.2}$$

Indeed $\kappa = -\partial_x \partial_y H(0,0)$; the trace of C is zero.

Put $Q = I - |v\rangle\langle v|$. Since $\ln \xi = \ln(1/32) I + \ln 29\, |v\rangle\langle v|$, the required matrix elements are

$$\langle v, Cv\rangle = \frac{221}{800}, \quad \langle v, Av\rangle = \langle v, Bv\rangle = -\frac{5}{32}, \quad \langle v_\perp, Av\rangle = \langle v_\perp, Bv\rangle = \frac{1}{5},$$

$$\operatorname{Tr}(QAQBQ) = -\frac{321}{25600}.$$

In an eigenbasis of ξ, $D \ln_\xi$ multiplies entries by the logarithmic divided difference $(\ln \lambda_i - \ln \lambda_j)/(\lambda_i - \lambda_j)$, with diagonal value $1/\lambda_i$. Between $29/32$ and $1/32$ it is $(8/7)\ln 29$. Therefore the second trace in (7.2) is

$$\frac{25}{928} - \frac{321}{800} + \frac{16}{175} \ln 29.$$

Together with the first trace this gives

$$\kappa = \frac{2059}{5600} \ln 29 - \frac{2171}{5800}.$$

It is positive: $e < 3$ implies $\ln 29 > 2$, and the resulting rational lower bound is positive.

Taylor's formula and the bounded third derivative prove (1.5) on the entire interval. The inverse parameter is smooth there and satisfies

$$r(\delta) = \frac{80}{9}\delta + O(\delta^2).$$

Its derivatives are bounded on $[0, 21/512]$. Thus $r(\delta)^2 = (6400/81)\delta^2 + O(\delta^3)$ with a uniform constant, and $r(\delta)^3 \le (80/9)^3 \delta^3$. Substitution proves (1.6). Strong subadditivity gives $J(r) \ge 0$. Enlarging the finite remainder constant yields $J(r(\delta)) \le C_1 \delta^2$ on the full interval.

8 Four tetrahedral branches and the original-environment map

Let the four Bloch directions be

$$n_1 = \frac{(1,1,1)}{\sqrt{3}}, \quad n_2 = \frac{(1,-1,-1)}{\sqrt{3}}, \quad n_3 = \frac{(-1,1,-1)}{\sqrt{3}}, \quad n_4 = \frac{(-1,-1,1)}{\sqrt{3}}.$$

They satisfy $\frac{1}{4}\sum_j n_j = 0$ and $\frac{1}{4}\sum_j n_j n_j^{\mathsf{T}} = I_3/3$. For $n_j = (n_{jx}, n_{jy}, n_{jz})$ an explicit choice is

$$U_j = \begin{pmatrix} \alpha_j & -\overline{\beta_j} \\ \beta_j & \alpha_j \end{pmatrix}, \quad \alpha_j = \sqrt{\frac{1+n_{jz}}{2}}, \quad \beta_j = \frac{n_{jx}+in_{jy}}{\sqrt{2(1+n_{jz})}}.$$

It is unitary and satisfies $U_j Z U_j^\dagger = n_j \cdot \sigma$. Let $W_j = \overline{U_j} \otimes U_j$, which fixes Φ. Use independent four-dimensional flag spaces J, K and define

$$|\Omega_r\rangle = \frac{1}{2}\sum_{j=1}^{4} (W_j \otimes I_{EG})|\psi_r\rangle|j\rangle_J|j\rangle_K, \qquad E' = J \otimes LR, \quad G' = K \otimes G. \tag{8.1}$$

This is normalized and pure, with $\dim E' = \dim G' = 16$. Tracing out K kills cross-branch terms. Each branch is a local unitary of the seed, and the common J flag entropies cancel in CMI:

$$\frac{1}{2} I_2(A : B \mid E')_{\Omega_r} = J(r). \tag{8.2}$$

The seed's AB marginal has Bloch form

$$\frac{1}{4}[I + z(Z \otimes I + I \otimes Z) + s_0(X \otimes X - Y \otimes Y) + t_0 Z \otimes Z],$$

where

$$z = 1 - \frac{19r}{40}, \qquad s_0 = \frac{7r}{10}, \qquad t_0 = 1 - \frac{19r}{20} + \frac{33r^2}{40}.$$

Partial transpose on A changes the correlation tensor to $s_0 I_3 + (t_0 - s_0) e_z e_z^{\mathsf{T}}$. Under W_j, this transposed state transforms by $U_j \otimes U_j$. The tetrahedral first and second moments kill its local Bloch vectors and replace its correlation tensor by $(2s_0 + t_0)I_3/3$. Undoing partial transpose therefore gives exactly

$$\mathrm{Tr}_{E'G'} |\Omega_r\rangle\langle\Omega_r| = \rho_{\delta(r)}. \tag{8.3}$$

This also follows by comparing $(2s_0 + t_0)/3 = (1 + 4\delta(r))/3$.

For the canonical Bell basis $b_i = (I \otimes P_i)\Phi$, let $q_0 = 1/2 + \delta$, $q_i = (1/2 - \delta)/3$ for $i > 0$. All are positive on the full interval under consideration. At $r = r(\delta)$, define

$$w_i = q_i^{-1/2}(\langle b_i| \otimes I_{E'G'})|\Omega_{r(\delta)}\rangle, \quad V_\delta|i\rangle = w_i, \quad \mathcal{S}_\delta(X) = \mathrm{Tr}_{G'}\, V_\delta X V_\delta^\dagger. \tag{8.4}$$

Equation (8.3) gives $\langle w_i, w_j\rangle = \delta_{ij}$. Thus V_δ is an actual isometry from the original environment E, and $\mathcal{S}_\delta$ is CPTP with output dimension sixteen. It maps the canonical Bell purification precisely to $\Omega_{r(\delta)}$.

9 Purifier phases, coherent Pauli covariance, and every input

We verify the covariance of the pure extension, including its purifier. For a diagonal unitary $D = \mathrm{diag}(d_0, d_1)$, let

$$E_D = \overline{D}_L \otimes D_R, \quad \alpha = \overline{d_0} d_1, \quad \beta = \overline{d_1} d_0, \quad G_D = \mathrm{diag}(1, 1, \alpha^{-1}, \beta^{-1}).$$

Both $v, v_\perp$ have phase 1 under E_D; the odd core vectors have phases α, β. The explicit purifier (6.2) therefore obeys $(E_D \otimes G_D)\zeta = \zeta$. Directly on the two input basis vectors,

$$(D_{\text{retained}} \otimes D_{\text{environment}})V_r = d_0 V_r D_{\text{input}}.$$

Use its conjugate on L and ordinary version on R. The scalar factors cancel, and

$$(\overline{D}_A \otimes D_B \otimes E_D \otimes G_D)\psi_r = \psi_r. \tag{9.1}$$

For $P_i = X, Y, Z$, its Bloch rotation permutes the tetrahedral directions; write the permutation as π_i. Since the two columns of $P_i U_j$ have the eigenrays of $U_{\pi_i(j)} Z U_{\pi_i(j)}^\dagger$, there is a diagonal unitary D_{ij} such that $P_i U_j = U_{\pi_i(j)} D_{ij}$. Then

$$(P_i^{\mathsf{T}} \otimes P_i) W_j = W_{\pi_i(j)} (\overline{D_{ij}} \otimes D_{ij}).$$

Here $P_i^{\mathsf{T}} = \overline{P_i}$, including the sign for Y. Define $L_E(P_i)$ to permute branch j to $\pi_i(j)$ and apply $E_{D_{ij}}$ inside it. Define $L_G(P_i)$ similarly with $G_{D_{ij}}$. These are unitary block permutations. Applying (9.1) branch by branch in (8.1) proves

$$(P_i^{\mathsf{T}} \otimes P_i \otimes L_E(P_i) \otimes L_G(P_i))\Omega_r = \Omega_r. \tag{9.2}$$

Let $T_\delta = (I_B \otimes V_\delta) U_{p(\delta)}$. It is the joint isometry for one and the same squashing map (8.4) on every input. Since $(\mathrm{id} \otimes T_\delta)\Phi = \Omega_{r(\delta)}$, (9.2) is equivalent to

$$(P_i \otimes L_E(P_i) \otimes L_G(P_i)) T_\delta P_i = T_\delta.$$

For an input density τ, the half-CMI for any pure reference purification of τ is

$$f(\tau) = \frac{1}{2}\left[S_2(B \mid E')_{T_\delta \tau T_\delta^\dagger} + S_2(B \mid G')_{T_\delta \tau T_\delta^\dagger} \right].$$

This identity is purity's entropy duality on the reference and the three joint output systems. Conditional entropy is concave, so f is concave. The proved local covariance gives $f(P_i \tau P_i) = f(\tau)$. The qubit Pauli average is $I_2/2$, whence

$$f(\tau) \le f\left(\frac{1}{4} \sum_{i=0}^{3} P_i \tau P_i \right) = f(I_2/2) = J(r(\delta)).$$

Bell input attains equality. This proves the exact same-map maximum in Theorem 1.2. Using that single map in the inner infimum of (1.1), then taking the outer supremum, proves the boundary upper in Theorem 1.1. Together with the previous entropy section it completes all assertions of Theorem 1.2.

10 The unrestricted quadratic lower

For a Hermitian bipartite operator X, let $\|X\|_\to$ be the supremum of the classical ℓ^1 norm of the measurement outcome vector over all finite Alice-to-Bob one-way LOCC measurements. Put

$$d_\to(\rho, \mathrm{SEP}) = \inf_{\sigma \in \mathrm{SEP}} \|\rho - \sigma\|_\to.$$

We use the finite-dimensional faithfulness theorem [3, Theorem, equation (9), and Corollary 1]: for every density ω_{ABE},

$$I_{\mathrm{nat}}(A : B \mid E)_\omega \ge \frac{1}{8} d_\to(\omega_{AB}, \mathrm{SEP})^2. \tag{10.1}$$

Equivalently its half-CMI bit cost is at least $d_\to^2/(16 \ln 2)$. The one-way classical communication is from Alice to Bob. The entanglement-testing identity used in its proof is the generalized Stein equality with normalized separable alternatives [4]; it applies to singular null states as well.

The numerical witness for ρ_δ is explicit. Alice chooses one of the three Pauli axes uniformly, measures the transposed observable on her qubit, and sends the axis and outcome to Bob. Bob measures the ordinary observable and accepts equal outcomes. The binary acceptance effect is

$$M = \frac{1}{3} \sum_{i=X,Y,Z} \frac{I + P_i^{\mathsf{T}} \otimes P_i}{2} = \frac{I + 2P}{3}. \tag{10.2}$$

For any product density $\alpha \otimes \beta$, $\mathrm{Tr}(P(\alpha \otimes \beta)) = \mathrm{Tr}(\alpha^{\mathsf{T}}\beta)/2 \leq 1/2$. Convexity gives the same inequality for every separable density. Thus all separable densities have acceptance at most $2/3$, whereas

$$\mathrm{Tr}(M\rho_\delta) = \frac{2}{3} + \frac{2\delta}{3}.$$

For a binary measurement the outcome ℓ^1 distance is twice the acceptance difference. Hence

$$d_{\to}(\rho_\delta, \mathrm{SEP}) \geq \frac{4\delta}{3}.$$

Every finite squashing map on the canonical Bell purification leaves this AB marginal exactly ρ_δ. Apply (10.1) to each of these actual extensions:

$$\frac{1}{2} I_2(A : B \mid E) \geq \frac{(4\delta/3)^2}{16 \ln 2} = \frac{\delta^2}{9 \ln 2}. \tag{10.3}$$

The inequality has no conditioning-dimension restriction. Take the infimum over all finite outputs and use Bell input in the outer supremum. This proves the lower in (1.3). The already proved upper and the positive lower coefficient give its right-neighborhood $\Theta(\delta^2)$ statement.

For the core ξ, the same actual one-way effect gives $\mathrm{Tr}(M\xi) = 211/240 > 2/3$, also verifying its nonseparability directly from the original convex-hull definition of SEP.

11 A capped pressure consequence

For any positive integer D, define the natural-unit pressure

$$K_D(\ell) = \sup_{\substack{1 \leq \dim E \leq D \\ \omega_{ABE} \text{ density}}} \left\{ \ell \left(\mathrm{Tr}(P\omega_{AB}) - \frac{1}{2} \right) - I_{\mathrm{nat}}(A : B \mid E)_\omega \right\}. \tag{11.1}$$

This domain contains all normalized densities in the admitted dimensions. For $\ell \geq 0$, the Bell effect bound and strong subadditivity give every payoff at most $\ell/2$, so the supremum is finite.

Proposition 11.1. *There exist $c, \ell_0 > 0$ such that, for every $D \geq 16$ and every $0 < \ell \leq \ell_0$,*

$$K_D(\ell) \geq c\ell^2.$$

Proof. The actual E16 extension from (8.1) and (8.4) has Bell fidelity $1/2 + \delta$. The uniform entropy bounds supply a constant $C > 0$, common to $0 \leq \delta \leq 21/512$, for which its natural CMI is at most $C\delta^2$. For every $D \geq 16$ this state is admitted in (11.1); therefore $K_D(\ell) \geq \ell\delta - C\delta^2$. Choose

$$\delta = \frac{\ell}{2C}, \qquad \ell_0 = 2C\frac{21}{512}, \qquad c = \frac{1}{4C}.$$

The parameter is physical for $0 < \ell \leq \ell_0$, and substitution gives $K_D(\ell) \geq \ell^2/(4C)$. □

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Squashed entanglement at the Bell edge and a fidelity–Holevo deficit bound

Abstract

For the two-qubit state $\rho_c = (|00\rangle\langle 00| + |11\rangle\langle 11| + c|00\rangle\langle 11| + c|11\rangle\langle 00|)/2$, the infimum of half the conditional mutual information over all finite quantum conditioning systems satisfies

$$\frac{c^2}{2\ln 2}\ln\frac{1}{4c} \le E_{\mathrm{sq}}(\rho_c) \le h_2\left(\frac{1+\sqrt{1-c^2}}{2}\right) \quad (0 < c \le \tfrac{1}{4}).$$

Thus $E_{\mathrm{sq}}(\rho_c) = (c^2/2)\log_2(1/c) + O(c^2)$. The proof uses an exact reduction to finite density pairs, a jointly convex column entropy deficit, and a common-channel inequality bounding the square of the increase in root fidelity by the binary Holevo loss. All matrix statements include singular and noncommuting densities.

The upper bound is the entanglement-of-formation formula of Wootters [3], combined with $E_{\mathrm{sq}} \le E_F$ [4]. The contribution here is the lower bound with the same leading behavior as $c \downarrow 0$. For comparison, the hashing inequality [5] and $E_{\mathrm{sq}} \ge E_D$ [4] also give

$$E_{\mathrm{sq}}(\rho_c) \ge 1 - h_2\left(\frac{1+c}{2}\right).$$

Indeed ρ_c has maximally mixed marginals and nonzero eigenvalues $(1 \pm c)/2$. This bound can be stronger away from the asymptotic regime (for example at $c = 0.1$), so the new lower bound is not claimed to dominate it throughout the interval.

1 Definitions and statements

All Hilbert spaces in this paper are finite dimensional and complex. A density matrix is positive semidefinite with trace one. A channel is a completely positive trace-preserving linear map. Write

$$S(\rho) = -\operatorname{Tr}\rho\log_2\rho, \qquad I(A:B\mid E)_\omega = S(AE) + S(BE) - S(E) - S(ABE),$$

where $0\log 0 = 0$. Squashed entanglement means

$$E_{\mathrm{sq}}(\rho_{AB}) = \inf_{\substack{\dim E<\infty,\ \omega_{ABE}\ \text{density} \\ \operatorname{Tr}_E\omega=\rho_{AB}}} \frac{1}{2} I(A:B\mid E)_\omega. \tag{1.1}$$

The conditioning dimension is included in the infimum. For two densities on a common space define their affinity, root fidelity, equal-prior Holevo information, and cost by

$$a(\rho,\sigma) = \operatorname{Tr}\sqrt{\rho}\sqrt{\sigma}, \qquad F(\rho,\sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1, \tag{1.2}$$

$$\chi(\rho,\sigma) = S\left(\frac{\rho+\sigma}{2}\right) - \frac{S(\rho)+S(\sigma)}{2}, \qquad V(\rho,\sigma) = 1 - \chi(\rho,\sigma). \tag{1.3}$$

The trace in the affinity is real and nonnegative. Cauchy–Schwarz and the trace norm give $0 \le a \le F \le 1$. Let

$$h_2(t) = -t\log_2 t - (1-t)\log_2(1-t), \qquad \phi(v) = h_2\left(\frac{1+\sqrt{1-v}}{2}\right), \quad 0 \le v \le 1.$$

Theorem 1.1. *For every* $0 < c \le 1/4$,

$$\frac{c^2}{2\ln 2}\ln\frac{1}{4c} \le E_{\mathrm{sq}}(\rho_c) \le \phi(c^2). \tag{1.4}$$

There exist constants $C, c_0 > 0$ *such that, for* $0 < c \le c_0$,

$$\left| E_{\mathrm{sq}}(\rho_c) - \frac{c^2}{2}\log_2\frac{1}{c} \right| \le Cc^2, \qquad 0 \le \phi(c^2) - E_{\mathrm{sq}}(\rho_c) \le Cc^2.$$

Theorem 1.2. *For any finite density pair* η_0, η_1 *and any common channel* $\mathcal{N}$*, including singular inputs,*

$$\left[F(\mathcal{N}\eta_0, \mathcal{N}\eta_1) - F(\eta_0, \eta_1)\right]^2 \le 4\ln 2\,\left[\chi(\eta_0, \eta_1) - \chi(\mathcal{N}\eta_0, \mathcal{N}\eta_1)\right]. \tag{1.5}$$

Theorem 1.3. *For every finite density pair* ρ, σ *of affinity* $c > 0$*, with* $F = F(\rho, \sigma)$,

$$1 - \chi(\rho, \sigma) \ge \phi(F^2) - \phi(F^2 - c^2) + \frac{(F - \sqrt{Fc})^2}{4\ln 2}. \tag{1.6}$$

2 Exact reduction to density pairs

Proposition 2.1. *For* $0 \le c \le 1$,

$$E_{\mathrm{sq}}(\rho_c) = \inf_{\substack{\rho,\sigma \text{ finite densities} \\ a(\rho,\sigma)=c}} V(\rho, \sigma). \tag{2.1}$$

Both sides range over every finite dimension.

Proof. Given a finite pair with affinity c, place the following block matrix on $\mathrm{span}\{|00\rangle, |11\rangle\} \otimes E$:

$$\Gamma = \frac{1}{2}\begin{pmatrix} \rho & \sqrt{\rho}\sqrt{\sigma} \\ \sqrt{\sigma}\sqrt{\rho} & \sigma \end{pmatrix}. \tag{2.2}$$

It is a column Gram, has trace one, and has AB marginal ρ_c. Both AE and BE marginals are the block diagonal state $\frac{1}{2}(\rho \oplus \sigma)$. The E marginal is $(\rho+\sigma)/2$. If $M = (\sqrt{\rho}, \sqrt{\sigma})^{\mathsf{T}}/\sqrt{2}$, then $\Gamma = MM^\dagger$ and $M^\dagger M = (\rho + \sigma)/2$. The opposite rectangular Grams have the same nonzero eigenvalues. Consequently

$$\frac{1}{2}I(A : B \mid E)_\Gamma = 1 + \frac{S(\rho) + S(\sigma)}{2} - S\left(\frac{\rho+\sigma}{2}\right) = V(\rho, \sigma).$$

This proves the inequality from the left side of (2.1) to the right.

Conversely let ω_{ABE} be any extension in (1.1). Positivity confines its support to $\mathrm{span}\{|00\rangle, |11\rangle\} \otimes E$. Indeed an outside-support diagonal entry of its AB marginal is zero; the nonnegative diagonal entries over E must all vanish, and a zero diagonal of a positive matrix forces its corresponding row and column to vanish. A finite purification therefore has the form

$$|\Psi\rangle = \frac{|00\rangle|w_0\rangle + |11\rangle|w_1\rangle}{\sqrt{2}}, \qquad \|w_i\| = 1, \quad \langle w_1, w_0\rangle = c,$$

with $w_i \in E \otimes G$. Let ρ_i and τ_i be the E and G marginals of $|w_i\rangle$, and let their Holevo quantities be χ_E, χ_G. The conditional marginal spectra of a pure vector agree. The block entropy formula and purity of Ψ give

$$J := \frac{1}{2}I(A : B \mid E)_\omega = 1 - \frac{\chi_E + \chi_G}{2} \ge 0. \tag{2.3}$$

Reshape w_i into rectangular matrices M_i of Hilbert–Schmidt norm one. For any Hermitian X, Y, spectral decomposition gives

$$\begin{aligned}\||X| - |Y|\|_{\mathrm{HS}}^2 &= \sum_{j,k} (|\lambda_j| - |\mu_k|)^2 |\langle j,k\rangle|^2 \\ &\le \sum_{j,k} (\lambda_j - \mu_k)^2 |\langle j,k\rangle|^2 = \|X - Y\|_{\mathrm{HS}}^2.\end{aligned}$$

Apply this to $X_i = \left(\begin{smallmatrix} 0 & M_i \\ M_i^\dagger & 0 \end{smallmatrix}\right)$. Its absolute value has blocks $\sqrt{M_i M_i^\dagger}$ and $\sqrt{M_i^\dagger M_i}$. If a_E, a_G are the affinities of the two marginal pairs, this yields

$$4 - 2(a_E + a_G) \le 4 - 4c, \qquad a := \frac{a_E + a_G}{2} \ge c.$$

The possible transpose in the vectorized G marginal preserves this affinity. The equal direct sum $\widehat{\rho}_i = (\rho_i \oplus \tau_i)/2$ has affinity a and cost J; the common flag entropy cancels. If $c > 0$, mix this pair in an orthogonal block, with weight c/a, with two mutually orthogonal pure states. The latter pair has affinity zero and cost zero. The new finite pair has affinity exactly c and cost $(c/a)J \le J$. Thus every finite extension produces a finite admissible pair of no larger cost. Taking infima proves the reverse inequality. At $c = 0$, ρ_0 has a labelled product extension of zero CMI, and the orthogonal pure pair has zero cost, so both infima are zero. □

3 Common recovery and the fidelity–Holevo inequality

For densities α, β write

$$D_2(\alpha\|\beta) = \operatorname{Tr} \alpha(\log_2 \alpha - \log_2 \beta)$$

with value $+\infty$ when the support inclusion fails. $D_{M,2}$ is the supremum of classical relative entropy in bits over all finite POVMs. We use the common-reference recovery theorem [1, Corollary 4.2, equations (60)–(61)]: if α is supported in a density ζ, then

$$D_2(\alpha\|\zeta) - D_2(\mathcal{N}\alpha\|\mathcal{N}\zeta) \ge D_{M,2}(\alpha\|\mathcal{R}_{\zeta,\mathcal{N}}\mathcal{N}\alpha). \tag{3.1}$$

The completely positive recovery map depends only on $\zeta, \mathcal{N}$. With $\mu = \mathcal{N}\zeta$, it is the average of

$$\mathcal{R}^{[t]}(Z) = \zeta^{(1+it)/2}\mathcal{N}^\dagger\big(\mu^{-(1+it)/2} Z \mu^{-(1-it)/2}\big)\zeta^{(1-it)/2}$$

against $\beta_0(t) = \pi/[2(\cosh(\pi t) + 1)]$. Inverse powers act on the respective supports. Cyclicity and the adjoint identity give $\operatorname{Tr} \mathcal{R}^{[t]}(Z) = \operatorname{Tr} Z\Pi_\mu$. The map has a CPTP extension to the whole output space:

$$Z \longmapsto \mathcal{R}_{\zeta,\mathcal{N}}(Z) + \operatorname{Tr}[(I - \Pi_\mu)Z]\,\tau, \tag{3.2}$$

where τ is any fixed input density. The original recovery annihilates off-support blocks, and the additional term is completely positive. This extension is common to every target supported in ζ.

Measured relative entropy obeys

$$D_{M,2}(\alpha\|\beta) \ge -2\log_2 F(\alpha, \beta). \tag{3.3}$$

One may use a fidelity-minimizing Fuchs–Caves measurement [2] and classical Jensen to obtain this inequality. The factor two corresponds to root fidelity.

Proof of Theorem 1.2. Put $\zeta = (\eta_0+\eta_1)/2$, $\sigma_i = \mathcal{N}\eta_i$, and $\mu = (\sigma_0+\sigma_1)/2$. Each target is supported in its corresponding average. Define

$$g_i = D_2(\eta_i\|\zeta) - D_2(\sigma_i\|\mu) \geq 0, \qquad \lambda = \frac{g_0+g_1}{2} = \chi(\eta_0,\eta_1) - \chi(\sigma_0,\sigma_1).$$

Use (3.1) with the same $\zeta, \mathcal{N}$ for both i, and the common completion (3.2). Writing $f_i = F(\eta_i, \mathcal{R}\sigma_i)$,

$$g_i \geq -2\log_2 f_i, \qquad B_i^2 := 2(1-f_i) \leq (\ln 2) g_i.$$

The last inequality follows from $f_i \geq e^{-(\ln 2)g_i/2}$ and $1 - e^{-x} \leq x$. Root fidelity is monotone under a channel, so

$$F(\mathcal{R}\sigma_0, \mathcal{R}\sigma_1) \geq F(\sigma_0,\sigma_1).$$

For any densities a, a', b,

$$|F(a,b) - F(a',b)| \leq \sqrt{2-2F(a,a')}. \tag{3.4}$$

To verify this, choose unit purifications of a, b attaining their root fidelity, and choose a purification of a' aligned with that of a with real overlap $F(a,a')$. The polar-decomposition formula for fidelity permits these choices in a sufficiently large finite purifying space. The difference of the first and third vectors has norm $\sqrt{2-2F(a,a')}$; the inner-product triangle gives one inequality, and swapping a, a' gives the other. The same formula also proves fidelity monotonicity by applying a Stinespring isometry to maximizing purifications. Apply (3.4) in both arguments:

$$0 \leq F(\sigma_0,\sigma_1) - F(\eta_0,\eta_1) \leq B_0 + B_1.$$

Hence the square of the difference is at most

$$2(B_0^2 + B_1^2) \leq 2\ln 2 (g_0+g_1) = 4\ln 2\, \lambda,$$

as asserted. All support and purification arguments allow singular states. □

4 Affinity under a partial trace

Lemma 4.1. *For finite densities ρ_{AB}, σ_{AB},*

$$\operatorname{Tr}\sqrt{\rho_{AB}}\sqrt{\sigma_{AB}} \leq \operatorname{Tr}\sqrt{\rho_A}\sqrt{\sigma_A}. \tag{4.1}$$

Proof. First assume both matrices are strictly positive. On Hilbert–Schmidt matrix spaces let

$$\Delta_{AB}(Z) = \sigma_{AB} Z \rho_{AB}^{-1}, \qquad \Delta_A(Z) = \sigma_A Z \rho_A^{-1}, \qquad W(Z) = (Z\rho_A^{-1/2} \otimes I_B)\sqrt{\rho_{AB}}.$$

Partial trace gives $W^\dagger W = I$, and direct trace expansion gives

$$\langle W(X), \Delta_{AB} W(Y)\rangle_{\mathrm{HS}} = \operatorname{Tr} X^\dagger \sigma_A Y \rho_A^{-1}, \qquad W^\dagger \Delta_{AB} W = \Delta_A.$$

For a positive M, inverse compression satisfies $(W^\dagger M W)^{-1} \leq W^\dagger M^{-1} W$, by the block Schur complement after extending W to a unitary. Integrating this inequality for $M = \Delta_{AB} + sI$ in

$$\sqrt{x} = \frac{1}{\pi}\int_0^\infty s^{-1/2}\frac{x}{x+s}\,ds$$

gives $\sqrt{\Delta_A} \geq W^\dagger \sqrt{\Delta_{AB}} W$. The left and right multiplication operators commute and are positive. Thus $\sqrt{\Delta_{AB}}(Z) = \sqrt{\sigma_{AB}} Z \rho_{AB}^{-1/2}$. Evaluate the compression inequality on $\sqrt{\rho_A}$, whose image under W is $\sqrt{\rho_{AB}}$. The two quadratic forms are exactly the two sides of (4.1).

For semidefinite matrices, replace both by $(1-\varepsilon)\rho + \varepsilon I/d$ on the original fixed space. Their marginals converge to the original marginals. Continuity of square root and trace proves (4.1) in the limit. □

5 Joint convexity of the column deficit

Lemma 5.1. *The function*

$$T(b,u) = \phi(b^2) - \phi(b^2 - u^2)$$

is jointly convex on $\mathcal{D} = \{(b,u) : 0 \le b \le 1,\ |u| \le b\}$.

Proof. Multiply by $\ln 2$ and set

$$f(x) = -\frac{1+x}{2}\ln\frac{1+x}{2} - \frac{1-x}{2}\ln\frac{1-x}{2}, \quad A(q) = \frac{\operatorname{atanh}\sqrt{q}}{\sqrt{q}}, \quad K(q) = \frac{A(q)-1}{q}.$$

Continuous values at zero are $A(0) = 1, K(0) = 1/3$. In the interior put $q = 1 - b^2$, $Q = q + u^2$, and $\delta = u^2$. Then $T_{\mathrm{nat}} = f(\sqrt{q}) - f(\sqrt{Q})$ and $2qA'(q) = (1-q)^{-1} - A(q)$. Differentiation yields

$$\begin{aligned} T_{bb} &= K(q) - A(Q) + 2b^2A'(Q), & T_{bu} &= -2buA'(Q), \\ T_{uu} &= A(Q) + 2u^2A'(Q) > 0, & & \\ \det \nabla^2 T_{\mathrm{nat}} &= K(q)\big[A(Q) + 2\delta A'(Q)\big] - A(Q)K(Q). & & \end{aligned} \tag{5.1}$$

The integral

$$K(q) = \int_0^1 \frac{t^2}{1-qt^2}\,dt$$

and Cauchy–Schwarz give $KK'' \ge 2(K')^2$. Indeed use factors $t/(1-qt^2)^{1/2}$ and $t^3/(1-qt^2)^{3/2}$. Therefore $1/K$ is concave on $[0,1)$.

We also have $2A'K - AK' > 0$. To establish the sign explicitly, expand

$$A(q) = \sum_{n\ge0} \frac{q^n}{2n+1}, \qquad K(q) = \sum_{n\ge0} \frac{q^n}{2n+3}.$$

Let $S_n = \sum_{j=0}^{n} (2j+3)^{-1}$. The coefficient of q^n in $2A'K - AK'$ equals $E_n/[4(n+3)]$, where

$$E_n = 2(n+4)S_n - (2n+3) + \frac{3}{2n+5}, \qquad E_0 = \frac{4}{15}.$$

For verification, the coefficient in $2A'K$ is $(n+2)S_n/(n+3)$, while that in AK' is

$$\frac{1}{2}\sum_{i=0}^{n}\frac{1}{2i+1} - \frac{3}{4(n+3)}\left(\sum_{i=0}^{n}\frac{1}{2i+1} + \sum_{j=0}^{n}\frac{1}{2j+5}\right).$$

Their difference gives the displayed expression. Using $S_n \ge (n+1)/(2n+3)$,

$$E_{n+1} - E_n = 2S_n - \frac{2n}{2n+5} - \frac{6}{(2n+5)(2n+7)} \ge \frac{16n^2+64n+52}{(2n+3)(2n+5)(2n+7)} > 0.$$

Absolute convergence justifies all series products and derivatives.

The tangent inequality for the concave reciprocal, followed by this positive sign, gives for $\delta > 0$

$$\frac{1}{K(q)} - \frac{1}{K(Q)} \le \frac{\delta K'(Q)}{K(Q)^2} < \frac{2\delta A'(Q)}{A(Q)K(Q)}.$$

Multiplication by $K(q)K(Q)A(Q) > 0$ proves that the determinant in (5.1) is positive. At $u = 0$, $T_{bb} = T_{bu} = 0$ and $T_{uu} = A(q) > 0$. The Hessian is consequently positive semidefinite throughout the interior. Continuity extends convexity to the entire closed triangle, including $b = 0, b = 1, |u| = b$. Dividing by $\ln 2$ restores bits. □

6 Canonical flags and the additive pair bound

Proof of Theorem 1.3. First take strictly positive ρ, σ. Put $A = \sqrt{\rho}$, $B = \sqrt{\sigma}$, and

$$R = \rho^{-1/2}(\rho^{1/2}\sigma\rho^{1/2})^{1/2}\rho^{-1/2}.$$

This is the positive solution of $R\rho R = \sigma$. Choose an orthonormal eigenbasis $|j\rangle$ of R, with eigenvalues $t_j > 0$, and define

$$p_j = \|A|j\rangle\|^2, \quad q_j = \|B|j\rangle\|^2 = t_j^2 p_j, \quad |a_j\rangle = \frac{A|j\rangle}{\sqrt{p_j}}, \quad |b_j\rangle = \frac{B|j\rangle}{\sqrt{q_j}},$$

$$z_j = \langle a_j, b_j\rangle, \quad r_j = \frac{p_j + q_j}{2}, \quad \beta_j = \frac{2\sqrt{p_j q_j}}{p_j + q_j}.$$

The exact moment identities are

$$\sum_j r_j = 1, \qquad \sum_j r_j\beta_j = \operatorname{Tr} R\rho = F, \qquad \sum_j r_j\beta_j \operatorname{Re} z_j = \operatorname{Tr} AB = c. \tag{6.1}$$

Here $\operatorname{Tr} R\rho = \operatorname{Tr}\sqrt{\rho^{1/2}\sigma\rho^{1/2}} = F$. This spectral basis is the fidelity-optimal Fuchs–Caves basis [2]; the identities above specify exactly the part required.

Supply an actual orthogonal flag:

$$\eta_0 = \sum_j p_j |a_j\rangle\langle a_j| \otimes |j\rangle\langle j|, \qquad \eta_1 = \sum_j q_j |b_j\rangle\langle b_j| \otimes |j\rangle\langle j|.$$

Discarding it gives ρ, σ. The orthogonal-block formulas give

$$\widehat{F} = F(\eta_0, \eta_1) = \sum_j r_j\beta_j |z_j|, \qquad \widehat{c} = a(\eta_0, \eta_1) = \sum_j r_j\beta_j |z_j|^2.$$

Weighted Cauchy–Schwarz and Lemma 4.1 imply

$$\widehat{F}^2 \le F\widehat{c} \le Fc. \tag{6.2}$$

Let $\widehat{V} = 1 - \chi(\eta_0, \eta_1)$ and $\lambda = \chi(\eta_0, \eta_1) - \chi(\rho, \sigma)$. The entropy loss is additive:

$$V(\rho, \sigma) = \widehat{V} + \lambda. \tag{6.3}$$

With posterior $w_j = p_j/(p_j + q_j)$, the conditional pure-state mixture has determinant $w_j(1-w_j)(1-|z_j|^2)$. The classical flag entropy cancels, giving exactly

$$\widehat{V} = \sum_j r_j\big[\phi(\beta_j^2) - \phi(\beta_j^2(1 - |z_j|^2))\big].$$

The function ϕ is increasing. Thus, with $u_j = \beta_j \operatorname{Re} z_j$, each bracket is at least $T(\beta_j, u_j)$. Signed u_j satisfy $|u_j| \le \beta_j$; Lemma 5.1, Jensen, and (6.1) give

$$\widehat{V} \ge T(F, c) = \phi(F^2) - \phi(F^2 - c^2).$$

Theorem 1.2 applied to these flagged densities and the actual partial trace gives

$$\lambda \ge \frac{(F - \widehat{F})^2}{4\ln 2} \ge \frac{(F - \sqrt{Fc})^2}{4\ln 2}.$$

The second step uses $c \le F$ and (6.2). Combining it with (6.3) proves (1.6).

For a singular pair on dimension d, apply the faithful result to $\rho_\varepsilon = (1-\varepsilon)\rho + \varepsilon I/d$ and $\sigma_\varepsilon = (1-\varepsilon)\sigma + \varepsilon I/d$. For sufficiently small ε the affinity is positive. Entropy, root fidelity, and affinity are continuous at this fixed dimension. Taking $\varepsilon \downarrow 0$ proves the displayed bound. The flags and their bases need not converge because the final expression depends only on these continuous quantities. □

7 The Bell-edge lower bound and asymptotic

Proof of Theorem 1.1. For $0 < v < 1$, differentiation gives

$$\phi'(v) = \frac{\operatorname{atanh}\sqrt{1-v}}{2\ln 2\sqrt{1-v}}.$$

This is positive and decreasing: the series $\operatorname{atanh} s/s = \sum_{n\geq 0} s^{2n}/(2n+1)$ increases with s. Hence ϕ is concave, and for $0 < F \leq 1$

$$\phi(F^2) - \phi(F^2 - c^2) \geq c^2\phi'(F^2), \qquad \phi'(F^2) \geq \frac{\ln(1/F)}{2\ln 2}.$$

For the second inequality set $s = \sqrt{1-F^2}$ and use

$$\operatorname{atanh} s = \frac{1}{2}\ln\frac{(1+s)^2}{1-s^2} \geq \ln(1/F);$$

division by $s \leq 1$ preserves this lower bound. The endpoints follow by continuity.

For any finite pair of affinity $0 < c \leq 1/4$, let $L = \ln(1/c)$. If $F \leq 4c$, Theorem 1.3 gives

$$V \geq \frac{c^2}{2\ln 2}(L - \ln 4).$$

If $F \geq 4c$, then $\sqrt{Fc} \leq F/2$. Put $r = F/c \geq 4$. Keep both terms of the pair bound:

$$V \geq \frac{c^2}{2\ln 2}\left(L - \ln r + \frac{r^2}{8}\right) \geq \frac{c^2}{2\ln 2}L.$$

The final inequality follows because $r^2/8 - \ln r$ increases for $r \geq 4$ and equals $2 - \ln 4 > 0$ at 4. The dimension-independent lower bound passes through Proposition 2.1 to the original infimum.

For the upper bound set $p = (1 - \sqrt{1-c^2})/2$. The commuting pair $\rho = \operatorname{diag}(1-p, p)$, $\sigma = \operatorname{diag}(p, 1-p)$ has affinity $2\sqrt{p(1-p)} = c$, average $I/2$, and cost $h_2(p) = \phi(c^2)$. Its Gram (2.2) is an explicit feasible extension with conditioning dimension two.

Finally $p = c^2/4 + O(c^4)$ and $-p\ln p - (1-p)\ln(1-p) = p\ln(1/p) + p + O(p^2)$. Therefore

$$\phi(c^2) = \frac{c^2}{2\ln 2}L + \frac{c^2}{4\ln 2}(\ln 4 + 1) + O(c^4 L).$$

The lower in (1.4) differs from the leading term by exactly $-c^2$. Since $c^4 L = O(c^2)$, the two asserted remainder bounds follow. They hold for the all-finite infimum even when feasible conditioning dimensions grow with c. □

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Label symmetry is necessary for the Magniez–Zur collision tradeoff

Abstract

For a uniformly random function on N labels, we construct a quantum collision algorithm with average success probability at least $2/3$, total retained space $S = O(\log N)$, and query complexity $T = O(\sqrt{N \log\log N / \log N})$. Thus $T^2 S = O(N \log\log N)$, showing that the Magniez–Zur lower bound for label-symmetric algorithms does not extend to unrestricted value-oracle algorithms. The construction combines numerical fingerprints with quantum search.

For each integer N, choose a function $f : [N] \to [N]$ uniformly from all N^N functions. Its coherent value oracle acts by

$$O_f|x, y\rangle = |x, y \oplus \mathrm{enc}(f(x))\rangle.$$

The algorithm is fixed before f is chosen. Its success probability is averaged over that one function and its measurement outcomes. The resource T counts oracle queries; S counts all retained classical bits and qubits, including workspace and output registers.

Theorem. The bound $T^2 S = \Omega(N \log N)$ in the unrestricted random-function query/space model is false. For every sufficiently large N, there is an algorithm with unconditional average success at least $2/3$, a fixed number of coherent value-oracle queries, and

$$S = O(\log N), \qquad T = O\left(\sqrt{\frac{N \log\log N}{\log N}}\right).$$

Thus

$$T^2 S = O(N \log\log N) = o(N \log N).$$

All logarithms in parameter definitions are base two. Identify $[N]$ with $\{0, \ldots, N-1\}$ through fixed input/output encodings. Such a fixed relabelling changes no collision relation and adds only input-independent interface gates and $O(\log N)$ scratch if needed.

Magniez and Zur [MZ26, Theorem 4.19] prove $T^2 S = \Omega(N \log N)$ for label-symmetric algorithms and leave the unrestricted case open in their Section 1.4. The construction below uses numerical fingerprints of output labels and shows that the same bound does not extend to arbitrary value-oracle algorithms. The query and space resources have the same interpretation: intermediate operations are input-independent, and all retained data are charged. If a collision output is required to include its common value, retain that value from the final verification query; this adds at most n bits to the space bound and leaves the asymptotic separation unchanged.

The construction follows the sample-and-search approach of Brassard, Høyer and Tapp [BHT97], using Grover search [G96]. It stores numerical fingerprints in place of full sample values and checks matches using the oracle.

1 A parameterized algorithm

First let r be any integer with $r \geq 1024$ and $N \geq 64r^2$. Write

$$n = \lceil \log_2 N \rceil, \quad b = \lceil \log_2(64r^2) \rceil, \quad B = 2^b,$$
$$A = \{0, \ldots, r-1\}, \quad D = \{r, \ldots, N-1\}, \quad L = N - r.$$

Because $N \geq 64r^2$, $b \leq n$. Define the fixed numerical fingerprint

$$h(y) = y \bmod B.$$

For each fixed sample input $i \in A$, store only

$$a_i = h(f(i))$$

in a b-bit table slot. The corresponding original input index is its fixed slot; no full sample values or n-bit index array are retained. Each entry uses two queries: load $f(i)$ into an n-bit value register, copy its low b bits to the slot, and erase the full value by an inverse query. This uses $2r$ queries.

For an XOR value oracle the inverse is the oracle itself. For addition modulo N, the inverse uses one forward query conjugated by the fixed response negation $J|y\rangle = |-y \bmod N\rangle$. Fixed output encodings can likewise be converted before copying the tag and converted back before unloading the value.

More explicitly, an XOR query can be simulated using the addition-modulo-N oracle of [MZ26]. On a valid address x, load $f(x)$ into a zeroed $\lceil \log_2 N \rceil$-qubit temporary register, XOR its binary encoding into the response register, and erase the temporary register with an inverse addition query. The preceding negation identity implements that inverse with one forward query. Thus each XOR query costs two addition queries and $O(\log N)$ extra workspace; the asymptotic bounds are unchanged.

Compute a retained bit F indicating that all a_i are distinct. This requires no queries and can be done reversibly with small scratch: count equal-tag pairs in a counter of width $\lceil \log_2(\binom{r}{2} + 1) \rceil$, using one equality flag at a time; toggle F iff the count is zero; reverse the scan to erase the counter. Equality testing and conditional increments use $O(b + \log r)$ clean scratch. The table is preserved. F and all counter/controller bits are charged.

For a full value y, a reversible sequential table scan computes

$$J(a, y) = \bigoplus_{\{i : a_i = h(y)\}} i, \qquad C(a, y) = \#\{i : a_i = h(y)\} \bmod 2.$$

The empty XOR is zero. On $F = 1$, these are exactly the unique matching original sample index and its existence bit.

The scan consists of reversible gates on the stored table. For each fixed i, compute the equality flag $[a_i = h(y)]$ with $O(b)$ clean scratch, conditionally XOR the fixed binary representation of i into the output index and toggle the output bit, then erase the equality flag. Reverse the scan to erase the index and bit when desired. A live loop index costs $O(\log r)$ charged bits; an unrolled circuit has the same input-independent schedule. Every function-dependent table bit remains a data qubit.

Even when $F = 0$, the XOR index is valid:

$$0 \le J < 2^{\lceil \log_2 r \rceil} \le 2r < N.$$

Thus no bad-table or absent-tag branch requests an invalid oracle address.

Define, on D, the phase predicate

$$P_f(x) = [F = 1]\,[C(a, f(x)) = 1]\,[f(J(a, f(x))) = f(x)].$$

When $F = 1$, this is exactly membership of $f(x)$ in the r-element set $\{f(i) : i \in A\}$. When $F = 0$, it is identically zero.

A clean phase marker costs four queries:

1. Load $f(x)$ into Y.
2. Compute J, C by the table scan.

3. Load $f(J)$ into Z.
4. Apply the sign conditioned on F, C and the full n-bit equality $Y = Z$.
5. Erase Z with an inverse query, reverse the table scan, and erase Y with an inverse query.

All scratch is returned to zero. If there is no matching tag, $J = 0, C = 0$; querying the valid address zero and uncomputing it is harmless. The full-value comparison prevents a fingerprint false positive from being accepted.

The coherent value oracle acts through a single query interface, implemented by fixed swaps or by copying X or J into a charged temporary address Q, making the query, and erasing Q. If the physical response port is Y, place Z there by swapping Y, Z before and after each query intended for Z; this preserves the other full value and adds no register or query. These registers are included in the space bound below.

Prepare

$$|u_D\rangle = L^{-1/2} \sum_{x \in D} |x\rangle$$

and its reflection. These are input-independent unitaries of width $O(n)$, including scratch. An explicit preparation uses binary prefix rotations: for each prefix p, its weight $w(p)$ is the number of elements of the fixed interval D having that prefix, and the next-bit amplitudes are $\sqrt{w(p0)/w(p)}$ and $\sqrt{w(p1)/w(p)}$. Zero-weight prefixes use arbitrary fixed rotations. The rotations can be arranged on the address bits with $O(n)$ scratch; their gate count does not enter the query complexity. The reflection is the preparation, a zero-state reflection, and inverse preparation.

Run exactly

$$m = \left\lfloor \frac{3}{4}\sqrt{L/r} \right\rfloor = \left\lfloor \sqrt{\left\lfloor \frac{9L}{16r} \right\rfloor} \right\rfloor$$

Grover iterations. This integer is determined by N, r, never by the unknown marked count. Computing or retaining it and all control counters costs $O(n)$ bits.

After measuring x, make two further value queries to compare its full value with the selected sample value. On a marked result the measured pair (J, x) consists of distinct valid indices with equal full oracle values. The success event consists of distinct measured addresses with equal oracle values; all other outcomes count as failures.

The query count is identical on every function:

$$T = 2r + 4m + 2 \le 2r + 3\sqrt{L/r} + 2.$$

2 Unconditional success probability

A fingerprint bucket has at most $\lceil N/B \rceil$ values, even when N is not a multiple of B. For independent uniform $U, V \in [N]$, if $q_t = \Pr(h(U) = t)$,

$$\Pr(h(U) = h(V)) = \sum_t q_t^2 \le \max_t q_t \le 1/B + 1/N.$$

Let $E = \{F = 1\}$. Independence of the sample coordinates and the union bound give

$$\Pr(E) \ge 1 - \binom{r}{2}(1/B + 1/N) \ge 63/64.$$

Condition on any complete sample vector satisfying E. Its full values are distinct. The same fixed oracle's outside coordinates $f(x)$, $x \in D$, are still independent uniform values. If K counts marked search inputs, then

$$K \sim \mathrm{Bin}(L, r/N), \qquad \mu = r(1 - r/N), \qquad \mathrm{Var}\, K \le r.$$

This law also holds conditional only on E, since its parameter is the same for every such complete sample.

Let $G = \{|K - \mu| \le r/8\}$. Chebyshev yields

$$\Pr(G \mid E) \ge 1 - 64/r \ge 15/16.$$

On G, using $r^2/N \le 1/64$,

$$3r/4 \le K \le 5r/4.$$

For a fixed oracle, put

$$a = \sqrt{r/L} \le 1/10, \qquad \theta = \arcsin\sqrt{K/L}, \qquad q = \theta/a.$$

The lower count bound gives $q \ge \sqrt{3/4} > 5/6$. The elementary integral inequality $\arcsin u \le u/\sqrt{1-u^2}$ gives

$$q \le \sqrt{\frac{5/4}{1 - (5/4)a^2}} \le \sqrt{100/79} < 7/6.$$

Thus $q \in [5/6, 7/6]$.

In the marked/unmarked two-dimensional subspace, the initial state has coordinates $(\sin\theta, \cos\theta)$, and one Grover iteration sends them to $(\sin 3\theta, \cos 3\theta)$. After m iterations, the probability of measuring a marked input is therefore $\sin^2((2m+1)\theta)$.

For the prescribed rational iteration count there is an e with $|e| \le 1$ such that

$$a(2m+1) = 3/2 + ea.$$

Since $3q/2 \in [5/4, 7/4]$ and $3 < \pi < 22/7$,

$$|(2m+1)\theta - \pi/2| \le 9/28 + 7/60 = 46/105 < \pi/6.$$

Grover success on $E \cap G$ is at least $3/4$. All such marked outputs pass the exact final verification. The unconditional average success over f and measurement is consequently at least

$$\frac{63}{64}\frac{15}{16}\frac{3}{4} = \frac{2835}{4096} > \frac{2}{3}.$$

3 Space complexity

The retained fingerprint table uses rb qubits/bits. Main and selected addresses X, J, the physical address Q if used, and the two full value registers Y, Z use at most $5n$ qubits. The pair-count check and sequential scan use $O(b + \log r)$ scratch/counter bits. Full equality, exact interval-state preparation, classical arithmetic, loop and iteration controls, fixed parameter records and output registers each add $O(n)$. Scratch can be reused, but even summing these allocations gives

$$S \le rb + Cn$$

for an absolute constant C, since $b \le n$ and $\log r \le n$.

The space bound counts classical bits and qubits equally, including the table and the two retained output indices. As usual in the query model, the supplied oracle's internal storage is excluded.

Thus the parameterized costs are

$$T = O(r + \sqrt{N/r}), \qquad S = O(\log N + r \log r).$$

4 Choosing the parameters

Let

$$\ell = \lceil \log_2 n \rceil, \qquad r = \lfloor n/\ell \rfloor$$

for sufficiently large N. Then $r \to \infty$, $r^3 = o(N)$, and $N \ge 64r^2$; hence all the preceding hypotheses eventually hold. Also $r \asymp n/\log n$, and

$$rb \le (n/\ell)(7 + 2\log_2 n) = O(n).$$

All charged workspace is therefore $S = O(n)$. One explicit sufficient threshold is $N \ge 2^{65536}$: then $n \ge 65536$, $\ell \ge 16$, $r \ge 2048$, and $N > 2^{n-1} \ge 64n^2 \ge 64r^2$. Also $rb \le 2n + 7n/\ell < 3n$. Thus the construction applies to every sufficiently large integer N, without a power-of-two restriction. Since preprocessing is negligible,

$$T = O\left(\sqrt{N\ell/n}\right).$$

It follows that

$$T^2 S = O(N\ell) = O(N \log\log N), \qquad \frac{T^2 S}{N \log N} = O(\ell/n) \longrightarrow 0.$$

Consequently, the bound $T^2 S = \Omega(N \log N)$ cannot hold for all algorithms in the stated oracle model.

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Conjugate degradability implies degradability for support-minimal channels with a qubit environment

Abstract

Every finite-dimensional, support-minimal conjugate-degradable channel with a qubit environment is degradable. A bosonic extension lemma converts the conjugate-degrading condition into an ordinary completely positive trace-preserving degrading map. The result includes the regime of input dimension at least four and output dimension larger than the input dimension.

1 Setting and statement

Let A, B, and E be finite complex Hilbert spaces, with $\dim E = 2$, and let $V : A \to B \otimes E$ be an isometry. Define

$$\mathcal{N}(X) = \mathrm{Tr}_E(VXV^*), \qquad \mathcal{N}^c(X) = \mathrm{Tr}_B(VXV^*).$$

Fix the computational basis of E; every transpose below refers to this basis. We use a support-minimal presentation: both $\mathcal{N}(I_A)$ and $\mathcal{N}^c(I_A)$ are positive definite on their displayed output spaces. The construction below uses the full support on B. The condition in the theorem is *conjugate degradability* in the terminology of Brádler, Dutil, Hayden and Muhammad [BDHM10]. On Hermitian matrices, complex conjugation equals transposition; the displayed identity is its complex-linear extension to all input matrices.

Theorem. If a completely positive trace-preserving map $D : \mathcal{L}(B) \to \mathcal{L}(E)$ satisfies

$$D \circ \mathcal{N} = \mathsf{T} \circ \mathcal{N}^c,$$

then a completely positive trace-preserving map $D_0 : \mathcal{L}(B) \to \mathcal{L}(E)$ satisfies $D_0 \circ \mathcal{N} = \mathcal{N}^c$.

The equalities hold on all complex input matrices. In particular, strict conjugate degradability is impossible within this support-minimal environment-qubit class.

Write $d_A = \dim A$ and $d_B = \dim B$. Brádler [B15, Lemma 4] already excludes exclusively conjugate-degradable channels when $d_B \le d_A$. For a qubit environment, the case $d_A \le 3$ also follows from PPT separability in dimensions 2×2 and 2×3 [HHH96]: the complementary Choi matrix is then separable, so the complementary channel is entanglement breaking and its complement is degradable [BDHM10; B15, Section 3.2]. The theorem therefore extends these exclusions to $d_A \ge 4$, $d_B > d_A$, within the stated support-minimal class with a qubit environment.

2 A bosonic extension lemma

Let R be a finite reference space, and let $W \ge 0$ act on $R \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$. Write $W_1 = \mathrm{Tr}_2 W$ and $W_2 = \mathrm{Tr}_1 W$, retaining the reference in both cases.

Lemma. If the 00 reference blocks of W_1 and W_2 are equal, then W_1 has a positive extension $\widetilde{W}$ supported on the symmetric subspace of the two qubits, and both marginals of $\widetilde{W}$ equal W_1.

Proof. Factor $W = ZZ^*$ and split the rows of Z in the order $00, 01, 10, 11$:

$$Z = \begin{pmatrix} A \\ B \\ G \\ D \end{pmatrix}.$$

The equality of the diagonal marginal blocks says $AA^* + BB^* = AA^* + GG^*$, hence $BB^* = GG^*$. There is a unitary U on the column space with $G = BU$. Indeed, the rule carrying each row of B to the corresponding row of G preserves their Gram matrix, hence is an isometry on their spans; extend orthonormal bases of the spans to the common column space. This argument does not require either row factor to be invertible.

Set

$$\widetilde{Z} = \begin{pmatrix} A \\ G \\ G \\ DU \end{pmatrix}, \qquad \widetilde{W} = \widetilde{Z}\widetilde{Z}^*.$$

It is positive and its rows are invariant under swapping the two qubits. Its first marginal has blocks

$$\begin{pmatrix} AA^* + GG^* & AG^* + G(DU)^* \\ GA^* + (DU)G^* & GG^* + (DU)(DU)^* \end{pmatrix} = \begin{pmatrix} AA^* + BB^* & AG^* + BD^* \\ GA^* + DB^* & GG^* + DD^* \end{pmatrix}.$$

This is precisely W_1. Swap invariance gives equality of the two marginals, as required. □

Only diagonal agreement was needed. In particular, the lemma applies when $W_2 = (\mathrm{id}_R \otimes \mathsf{T})(W_1)$.

3 From a purification to a channel

We record the channel reconstruction with its support hypothesis explicit. Let $Z : R \leftarrow B$ have full column rank and let $|Z\rangle = \sum_{r,b} Z_{rb}|r, b\rangle$. If $\sigma \geq 0$ on $R \otimes E$ satisfies $\mathrm{Tr}_E\, \sigma = ZZ^*$, then there is a channel $D_0 : \mathcal{L}(B) \to \mathcal{L}(E)$ such that

$$(\mathrm{id}_R \otimes D_0)(|Z\rangle\langle Z|) = \sigma.$$

To see this, put $L = (Z^*Z)^{-1}Z^*$, so $LZ = I_B$, and define

$$J = (L \otimes I_E)\sigma(L^* \otimes I_E).$$

Then $J \geq 0$ and $\mathrm{Tr}_E\, J = I_B$. Thus J is the unnormalized Choi matrix of a channel: a positive factorization of J, with each column reshaped into an E-by-B matrix, gives Kraus operators whose adjoint products sum to I_B. The marginal identity also implies that σ is supported on $\mathrm{ran}\, Z \otimes E$. Indeed a vector in the complementary subspace has zero quadratic form against the positive matrix σ and therefore lies in its kernel. Since ZL is the orthogonal projection onto $\mathrm{ran}\, Z$, applying $Z \otimes I_E$ to J recovers σ. This proves the displayed identity.

4 Proof of the theorem

Take an unnormalized maximally entangled vector on a reference copy of A and A, and apply V to its second factor. Reorder the resulting purification into reference/environment/output order. Denote it by $|Z\rangle$, where now $Z : (A \otimes E) \leftarrow B$. Its first marginal is the unnormalized Choi matrix $\rho = J(\mathcal{N}^c) = ZZ^*$. Full output support makes Z full column rank: Z^*Z is the transpose of $\mathcal{N}(I_A)$.

Apply the assumed channel D to the output factor. The resulting positive matrix W on $A \otimes E \otimes E$ has first marginal ρ, by trace preservation, and second marginal

$$J(D \circ \mathcal{N}) = (\mathrm{id}_A \otimes \mathsf{T})(\rho).$$

The extension lemma gives a bosonic $\widetilde{W}$ with both marginals equal to ρ. The reconstruction argument gives a channel D_0, defined on the whole output algebra, which maps $|Z\rangle\langle Z|$ to $\widetilde{W}$. Taking its second qubit marginal yields $J(D_0 \circ \mathcal{N}) = \rho = J(\mathcal{N}^c)$. Injectivity of the Choi correspondence proves the theorem. □

5 References

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One ebit is necessary and sufficient for the elegant joint measurement

Abstract

The classical-output elegant joint measurement has exact supplied-entanglement cost one ebit. The same cost holds when the measurement is approached by a sequence of physical LOCC protocols using the same finite pure resource. The protocols may have countably many turns, standard Borel outcomes, separable local workspaces, local mixed ancillary preparations, and a report defined on the entire transcript. The converse combines a separability witness, a determinant identity, and an LOCC concurrence bound. A finite-input approximation argument and compact augmented moments extend it to these protocols and their operational limits.

For one-way LOCC, the resource theorem of Bandyopadhyay, Halder, and Nathanson [6] already gives the one-ebit requirement for the elegant joint measurement. The converse below removes that restriction on communication and applies to the protocol class specified in this paper. Our cost is the entropy of the supplied finite pure resource. It differs from the cost studied by Bandyopadhyay, Brassard, Kimmel, and Wootters [7]: their definition averages the entanglement consumed and credits residual distillable entanglement, with outcome probabilities evaluated on the maximally mixed input. The present one-ebit equality concerns supplied entanglement, not that averaged net-consumption quantity.

1 Measurement, resources, and the protocol class

Let $A = B = \mathbb{C}^2$, let $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ be the Pauli matrices, and set

$$|s\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}, \qquad \mathbf{n}_j \in \frac{1}{\sqrt{3}}\{(1,1,1),(1,-1,-1),(-1,1,-1),(-1,-1,1)\}.$$

Choose one ordering of these four vectors. Define

$$|e_j\rangle = \left[\left(\frac{I}{2} + \frac{\sqrt{3}}{2}\mathbf{n}_j \cdot \boldsymbol{\sigma}\right) \otimes I\right] |s\rangle, \qquad P_j = |e_j\rangle\langle e_j|. \tag{1}$$

These projectors describe the elegant joint measurement [1]. Its shared classical-output channel is

$$\mathcal{M}_{\mathrm{EJM}}(\rho) = \sum_{j=1}^{4} \mathrm{Tr}(P_j\rho)|j\rangle\langle j|_{C_A} \otimes |j\rangle\langle j|_{C_B}. \tag{2}$$

The output registers are four dimensional. The required output in this paper is the channel (2).

A charged resource is any normalized pure state $|\chi\rangle \in A' \otimes B'$ with A', B' finite dimensional. Its cost is the entropy in bits of its entire Alice reduction,

$$E(\chi) = S(\chi_{A'}) = -\,\mathrm{Tr}(\chi_{A'} \log_2 \chi_{A'}), \qquad 0 \log_2 0 = 0.$$

There is no constraint on its finite Schmidt rank. All initial shared entanglement is included in this state. Independent local normal mixed ancillary states and local isometries are free.

We specify the physical LOCC class as follows. There are countably many chronological local turns. At each finite public history, the acting party is a measurable function of the history and applies a normal completely positive instrument whose total operation is trace preserving. The

outcome spaces are standard Borel and may depend on the history. The heterogeneous outcome records are measurably encoded at each turn in a standard Borel space, with a measurable admissible-history graph. The local working spaces are complete separable Hilbert spaces, allowed to change with the history and outcome. Their measurable Hilbert fields have countable dense fundamental sections. Instruments and coordinate wiring are measurable kernels: after measurable isometric embeddings into a fixed separable space, setwise instrument operations on trace-class inputs depend measurably on the history. These conditions include finite, countable, and standard continuous records and impose no bound on local workspace dimension. Normality means that the instruments act on trace-class states and have normal Heisenberg adjoints. This is the usual instrument setting [3, 2].

Each protocol has one input-independent report J taking values in $\{1, 2, 3, 4\}$ on the full common transcript. Stopping can be encoded by absorbing identity turns. A report on an infinite transcript is also allowed. The report may be measurable in the completion of the Born law of the prepared reference

$$\rho_* = V_\chi(I_{AB}/4)V_\chi^\dagger, \qquad V_\chi v = v \otimes |\chi\rangle, \tag{3}$$

with the tensor factors regrouped as $AA' : BB'$. It is reported to both parties. The physical outcomes, instruments, and wiring determine the Born law; the law is not additional protocol data.

Let $\mathcal{M}_{P,\chi}$ be the resulting shared report channel on AB. Define $\mathcal{C}_{\mathrm{ex}}$ to be the set of $E(\chi)$ for which one such protocol satisfies $\mathcal{M}_{P,\chi}(\rho) = \mathcal{M}_{\mathrm{EJM}}(\rho)$ for every density operator ρ on AB. Define $\mathcal{C}_{\mathrm{op}}$ by requiring instead protocols P_k with

$$\mathcal{M}_{P_k,\chi}(\rho) \longrightarrow \mathcal{M}_{\mathrm{EJM}}(\rho) \quad \text{for every density operator } \rho. \tag{4}$$

Here the entire charged resource χ stays fixed. The instruments, outcomes, workspaces, local mixed preparations, and completed reports may vary with k. All limits in (4) concern the fixed finite input and classical output spaces.

Theorem 1. *For this physical LOCC class, every exact or operationally implementing finite pure resource has $E(\chi) \geq 1$. A Bell pair admits a finite-turn exact implementation. Consequently*

$$\inf \mathcal{C}_{\mathrm{ex}} = \inf \mathcal{C}_{\mathrm{op}} = 1,$$

and both infima are attained.

2 Finite effects from physical protocols

The converse will use actual finite complete local Kraus trees. A node in such a tree has finitely many local Kraus operators with Gram sum I; each terminal occurrence has a report. Dimensions are fixed at the original finite local input dimensions.

Lemma 2. *Let the local input spaces be $\mathbb{C}^a : \mathbb{C}^b$, with $a, b > 0$. Every physical protocol with a Borel finite-valued report has report effects F_j satisfying $F_j \geq 0$ and $\sum_j F_j = I$. There are finite complete reported local Kraus trees with effects $G_{N,j}$ such that*

$$\max_j \|G_{N,j} - F_j\|_{\mathrm{op}} \longrightarrow 0.$$

Local normal mixed preparations and separable, history-dependent working spaces are included.

Proof. We give the construction, including the reduction to finite local inputs. Consider a history-dependent normal CP instrument Φ_h with finite domain $M_a(\mathbb{C})$ and output in the trace class of a separable Hilbert space G. For a measurable outcome set B, its positive Choi measure is

$$C_h(B) = \sum_{u,v=1}^{a} \Phi_h(B)(|u\rangle\langle v|) \otimes |u\rangle\langle v|, \qquad \mu_h(B) = \operatorname{Tr} C_h(B)/a.$$

The latter is a probability kernel. In a fixed countable basis of $G \otimes \mathbb{C}^a$, each scalar matrix-entry measure is absolutely continuous with respect to μ_h. Jointly Borel densities can be obtained by taking ratios on the atoms of increasing finite partitions generating the outcome sigma algebra and then taking their martingale limits. There are only countably many matrix entries. Outside a conditional null set, their limits form a positive matrix with diagonal sum a, hence a positive trace-class operator $D(h,x)$ with trace a. Set $D = 0$ on the exceptional set. Finite compressions show that D is trace-class-valued Borel, and uniform polynomial approximation to the square root on $[0,a]$ makes $D^{1/2}$ operator-norm Borel.

Unvectorize the columns $D^{1/2}e_\ell$ into maps $K_\ell(h,x) : \mathbb{C}^a \to G$. Equality of the Choi measures gives, for every complex matrix Z,

$$\Phi_h(B)(Z) = \int_B \sum_\ell K_\ell(h,x) Z K_\ell(h,x)^\dagger \, \mu_h(dx), \qquad \int \sum_\ell K_\ell^\dagger K_\ell \, d\mu_h = I. \tag{5}$$

The integrals converge in trace norm because $\sum_\ell \|K_\ell\|_{\mathrm{HS}}^2 = \operatorname{Tr} D \le a$. Choose $p_\ell > 0$ with $\sum_\ell p_\ell = 1$, replace the record by (x,ℓ), and use the density $K = K_\ell/\sqrt{p_\ell}$ against the probability kernel $\lambda_h(dx,\ell) = \mu_h(dx)p_\ell$. Then $\int K^\dagger K \, d\lambda_h = I$ and $\int \|K\|_{\mathrm{HS}}^2 \, d\lambda_h = a$. Revealing ℓ and allowing the continuation to ignore it recovers the original operation.

For varying Hilbert fibers, measurable Gram–Schmidt applied to their fundamental sections gives measurable embeddings into ℓ^2. The recorded output projection supports the Choi measure. Positivity therefore supports its density and every square-root column in the actual output fiber, almost everywhere. The preceding construction thus preserves the correct output space. Null branches admit an arbitrary complete continuation.

For every rectangular K, put $A = (K^\dagger K)^{1/2}$ and let V be its polar partial isometry. With $Q = I - V^\dagger V$ and a unit vector ϕ in the output space, the map

$$R_K(Z) = VZV^\dagger + \operatorname{Tr}(QZ)|\phi\rangle\langle\phi| \tag{6}$$

is normal CPTP and satisfies $R_K(AZA) = KZK^\dagger$. Its Kraus operators are V and $|\phi\rangle\langle i|Q$, whose Gram sum is I. Both V and its support projection are Borel functions of K: use the limits $K(K^\dagger K + t^{-1}I)^{-1/2}$ and $K^\dagger K(K^\dagger K + t^{-1}I)^{-1}$. This handles rank changes and $K = 0$. Zero-dimensional fibers carry only zero branches and can be replaced there by an absorbing one-dimensional fiber.

Carry two pending local CPTP maps from the fixed original finite input spaces into the actual working spaces. Initially these are the actual local preparation maps, including mixed preparations. When Alice acts, compose her instrument with her pending map and apply (5). In the compressed process she uses the square density A on $\mathbb{C}^a$ and replaces her pending map by R_K. Bob's map stays pending; the roles are reversed when he acts. The identity (6), including its reference extensions, proves branch equality after the pending maps. Local pending maps commute with remote operations. Induction proves equality of every finite-cylinder record probability. Consistency and the unique countable standard-Borel probability extension give the same full transcript law after the additional

Kraus labels are forgotten. Thus a measurable process with square local Kraus densities on the fixed finite spaces has exactly the original report probabilities.

Fix a finite horizon and a Borel report of its history. Backward recursion gives remaining report effects $E_j(h)$ with $0 \le E_j(h) \le I$ and $\sum_j E_j(h) = I$. At an Alice node, let $L_h(z)$ be the square density and $S_h(z) = L_h(z) \otimes I_b$. The recursion reads

$$E_j(h) = \int S_h(z)^\dagger E_j(h,z) S_h(z)\, \lambda_h(dz).$$

Sample the entire integrable finite-dimensional Hermitian tuple

$$(L_h(z)^\dagger L_h(z),\ S_h(z)^\dagger E_1(h,z) S_h(z), \dots, S_h(z)^\dagger E_q(h,z) S_h(z)).$$

An integrable finite-dimensional barycenter is in the closed convex hull of the values on every full-measure set: otherwise a separating linear functional contradicts its integral. Thus finitely many actual valid points z_i and positive weights w_i summing to one simultaneously approximate the tuple. For any $0 < \delta < 1$ they may be chosen with

$$\|Q - I\|_{\mathrm{op}} < \delta, \quad \|C_j - E_j(h)\|_{\mathrm{op}} < \delta, \quad Q = \sum_i w_i L_i^\dagger L_i, \quad C_j = \sum_i w_i S_i^\dagger E_j(h, z_i) S_i.$$

Approximate the finitely many sampled continuation effects by induction with error at most η. Use the local operators $M_i = \sqrt{w_i} L_i Q^{-1/2}$ and attach those continuation trees. The Gram sum of (M_i) is exactly I. If $R = Q^{-1/2}$ and $u_\delta = (1-\delta)^{-1/2} - 1$, the resulting effect error is at most

$$\eta + \frac{\delta}{1-\delta} + 2u_\delta + u_\delta^2.$$

Indeed the continuation error, before conjugation, lies between $-\eta(Q \otimes I)$ and $\eta(Q \otimes I)$, so after conjugation it is at most η; the remaining terms bound $(R \otimes I) C_j (R \otimes I) - E_j(h)$. Bob's case is identical. Small enough η, δ at each finite induction step give a finite complete tree approximating the finite-horizon report as accurately as desired. Sampling the whole tuple retains outcome-dependent continuation policies.

It remains to approximate an infinite-transcript report. Let $d = ab$. Trace preservation gives consistent probability laws $\mathbb{P}_\rho$ on finite prefixes and hence a unique full law. Its affine dependence on ρ, extended homogeneously to positive inputs, gives event effects by finite-dimensional trace duality. Use the faithful reference I/d with law μ. Since $\rho \le d(I/d)$, positivity gives $\mathbb{P}_\rho(B) \le d\mu(B)$ for every event B. Let $\mathcal{F}_N$ be the first-N-record sigma algebra and set $q_{N,j} = \mathbb{E}_\mu[\mathbf{1}_{J=j} \mid \mathcal{F}_N]$. Choose Borel finite-history versions and report a smallest maximizing index J_N. Upward martingale convergence implies $\mu(J_N \ne J) \to 0$. The effect difference is bounded uniformly by

$$\|F_{N,j} - F_j\|_{\mathrm{op}} \le d\mu(J_N \ne J).$$

Combine this with a finite complete tree approximation of error $1/N$. Appending local orthonormal-basis measurements at all terminal nodes gives literal product-bra leaves without changing the report effects. □

A report completed only at (3) causes no restriction in this lemma's application. Choose one Borel representative under its prepared-reference law. For every density ρ on AB, $V_\chi \rho V_\chi^\dagger \le 4\rho_*$. Positivity of the event operations implies domination of the corresponding full Born laws. The same Borel representative therefore agrees with the completed report for every charged density input. Apply Lemma 2 to that representative on the whole augmented finite input. This supplies effects on the whole input while retaining the required physical probabilities on the resource code.

3 Exact effects and determinant costs

Use local Schmidt coordinates for the charged resource, retaining all its nonzero coefficients:

$$|\chi\rangle = \sum_{k=1}^{r} \sqrt{\lambda_k}|k,k\rangle, \qquad \lambda_1 \geq \cdots \geq \lambda_r > 0, \qquad \sum_k \lambda_k = 1.$$

Set $x = \lambda_1$ and $D = \mathrm{diag}(\sqrt{\lambda_k})$. Passing to Schmidt coordinates is a local isometry on the support and preserves the entire reduced entropy. Rectangular original coefficient matrices and zero Schmidt coefficients are included: their local support embeddings, or their zero padding into a common square space, preserve normalization and all nonzero reduced eigenvalues.

Write $E_j \in M_2(\mathbb{C})$ for the coefficient matrix of e_j. The tetrahedral identities give

$$\langle e_i|e_j\rangle = \frac{1 + 3\mathbf{n}_i \cdot \mathbf{n}_j}{4} = \delta_{ij}.$$

The matrix of s has both singular values $1/\sqrt{2}$; multiplying it by $(I + \sqrt{3}\mathbf{n}_j \cdot \boldsymbol{\sigma})/2$ gives squared singular values $(2 \pm \sqrt{3})/4$. Consequently

$$|\det E_j| = \frac{1}{4}, \quad |e_j\rangle = \tfrac{1}{2}|s\rangle + \tfrac{\sqrt{3}}{2}|t_j\rangle, \quad |t_j\rangle = (\mathbf{n}_j \cdot \boldsymbol{\sigma} \otimes I)|s\rangle, \quad |\langle t_j|e_j\rangle|^2 = \frac{3}{4}. \tag{7}$$

Each t_j is normalized and maximally entangled because $\mathbf{n}_j \cdot \boldsymbol{\sigma}$ is unitary.

Suppose effects F_j on $AA' : BB'$ satisfy

$$F_j \geq 0, \qquad \sum_j F_j = I, \qquad V_\chi^\dagger F_j V_\chi = P_j. \tag{8}$$

For $u_j = V_\chi e_j$, positivity of $I - F_j$ gives $F_j u_j = u_j$. The line spanned by u_j reduces the Hermitian F_j, so $F_j \geq |u_j\rangle\langle u_j|$.

If the F_j are separable, compressing their resource factors onto $|1,1\rangle$ yields separable two-qubit effects $G_j \geq xP_j$ with $\sum_j G_j = I$. For a positive separable two-qubit operator G and a maximally entangled unit vector t, $2\langle t|G|t\rangle \leq \mathrm{Tr}\, G$: it holds for a product projector by its Schmidt overlap and hence for every positive sum. Equation (7) therefore gives

$$4 = \sum_j \mathrm{Tr}\, G_j \geq \sum_j 2x|\langle t_j|e_j\rangle|^2 = 6x, \qquad x \leq \frac{2}{3}. \tag{9}$$

For any finite product decomposition $F_j = \sum_v |a_{jv}, b_{jv}\rangle\langle a_{jv}, b_{jv}|$, absorb positive weights into the vectors. Let $A_{jv}, B_{jv} \in M_{2,r}(\mathbb{C})$ be their coefficient matrices with columns $a_{jv,k}, b_{jv,k}$. Compressing a positive sum to the rank-one P_j forces

$$A_{jv} D B_{jv}^T = c_{jv} E_j, \qquad c_{jv} = \langle u_j|a_{jv}, b_{jv}\rangle, \qquad \sum_v |c_{jv}|^2 = 1. \tag{10}$$

The identity $F_j u_j = u_j$ also gives

$$\sum_v \overline{c_{jv}} a_{jv,1} b_{jv,1}^T = \sqrt{x} E_j. \tag{11}$$

Define

$$T_{jv} = \sum_{k \geq 2} \sqrt{\lambda_k} a_{jv,k} b_{jv,k}^T = c_{jv} E_j - \sqrt{x} a_{jv,1} b_{jv,1}^T, \quad L_{jv} = \sqrt{x} b_{jv,1}^T E_j^{-1} a_{jv,1}.$$

Taking the trace of E_j^{-1} times (11) yields $\sum_v \overline{c_{jv}} L_{jv} = 2x$. The two-dimensional rank-one determinant formula implies

$$\det T_{jv} = (\det E_j) c_{jv}(c_{jv} - L_{jv}).$$

If $c_{jv} = 0$, the tail matrix has rank at most one and determinant zero. Otherwise multiply by $\overline{c_{jv}}/c_{jv}$, sum over v, and use (10). The triangle inequality gives

$$\sum_v |\det T_{jv}| \geq \frac{1}{4}|1 - 2x|, \qquad \sum_{j,v} |\det T_{jv}| \geq |1 - 2x|. \tag{12}$$

This algebraic conclusion holds for every product decomposition of exact effects.

4 The LOCC bound and passage to limits

For a finite complete tree, refine all terminal local outputs to product bras. Put $t = 1 - x$ and suppose $t > 0$. Run that same complete tree on

$$|\tau\rangle = \frac{1}{\sqrt{t}} \sum_{k \geq 2} \sqrt{\lambda_k} |k, k\rangle, \qquad |\Phi_2\rangle_{AR_A} \otimes |\Phi_2\rangle_{BR_B},$$

where R_A, R_B are untouched references local to their respective parties. The only initial interparty entanglement is in τ. The reference coefficient matrix at leaf (j, v) is $\overline{T_{jv}}/(2\sqrt{t})$. Its probability is $\|T_{jv}\|_F^2/(4t)$ and its normalized two-qubit concurrence is $2|\det T_{jv}|/\|T_{jv}\|_F^2$. Zero-probability leaves contribute zero. Thus its probability-weighted concurrence is $|\det T_{jv}|/(2t)$.

For a pure bipartite state define the I-concurrence [4]

$$C_I(\psi) = \sqrt{2(1 - \operatorname{Tr} \psi_A^2)}.$$

This is an average LOCC monotone. The map $\rho \mapsto 1 - \operatorname{Tr} \rho^2$ is concave, and composing with the increasing concave square root preserves concavity. At each local measurement, the opposite party's average reduced state is unchanged, so Jensen's inequality bounds the average new concurrence by the previous one. Its two reduced states have the same nonzero spectrum, so either party may act. Local pure ancillas leave it unchanged. This is the pure-state concavity criterion [5], proved here for the quantity used. On two-qubit pure states it equals the determinant concurrence above.

With $\mu_k = \lambda_k/t$ for $k \geq 2$, monotonicity gives the bound

$$\sum_{j,v} |\det T_{jv}| \leq 2t \sqrt{2\left(1 - \sum_{k \geq 2} \mu_k^2\right)} = 4 \sqrt{\sum_{2 \leq k < \ell} \lambda_k \lambda_\ell}. \tag{13}$$

The tree need not implement the EJM on τ; only its completeness is used in this reference experiment.

We now preserve (13) when effects become exact only in a limit. On unit product vectors $\alpha = a \otimes b$ in the fixed augmented input, form the compact set of augmented moments

$$\mathcal{K} = \{(|\alpha\rangle\langle\alpha|, |\det T(\alpha)|) : \|a\| = \|b\| = 1\},$$

where $T(\alpha)$ contracts the resource tail. If a product projector has weight w, its effect and determinant cost are both multiplied by w. The weights in a decomposition of outcome j sum to $\operatorname{Tr} F_j$, which is bounded by the augmented input dimension. Its augmented moment lies in $\operatorname{Tr} F_j \operatorname{conv} \mathcal{K}$. This

set of possible effect/cost pairs is compact, including trace zero. Along a common subsequence for the four outcomes, convergent effects therefore have convergent determinant costs. Carathéodory's theorem represents each limiting moment by a finite product decomposition with the same limiting cost. The total cost still satisfies (13). If the limiting compressed effects are P_j, apply (12) to these particular decompositions. Separability of the effects is also preserved by this compactness. Hence both bounds hold together and, when $x > 1/2$,

$$(2x-1)^2 \le 16 \sum_{2\le k<\ell} \lambda_k \lambda_\ell. \tag{14}$$

For an exact physical protocol, Lemma 2 supplies the finite trees needed in this argument. For a sequence in (4), apply that lemma to each Borel report representative, choosing a tree with effect error at most $1/k$. Every effect is bounded between 0 and I on the same finite augmented input, so take a convergent subsequence. Its compressed effects equal P_j: convergence on every system density state in (4) identifies the compressed Hermitian matrices by trace duality. Apply the same augmented-moment argument. This proves (9) and (14) in the exact and operational cases. In particular rank one is excluded by (9), justifying $t > 0$ whenever the tail reference test is needed.

5 From the spectrum bounds to one bit

We complete the entropy estimate explicitly. Write h_2 for binary entropy. If $x \le 1/2$, then $H(\lambda) \ge -\log_2 x \ge 1$. Otherwise (9) gives $1/2 < x \le 2/3$. Put

$$z = 2x-1, \qquad t = 1-x, \qquad c = \frac{z}{2t} \in (0,1/2], \qquad F(c) = h_2\left(1-\sqrt{1-c^2/2}\right).$$

Equation (14) bounds the normalized tail purity by $P = \sum_{k\ge 2} \mu_k^2 \le 1 - c^2/2$. If its largest entry m exceeds $1/2$, then $m^2 \le P$ and grouping that entry against the rest gives $H(\mu) \ge h_2(m) \ge h_2(\sqrt{1-c^2/2})$. If $m \le 1/2$, then $H(\mu) \ge 1$, which implies the same bound. Thus

$$H(\mu) \ge F(c). \tag{15}$$

The following elementary inequalities suffice:

$$F(c) \ge \tfrac{3}{2}c^2 \qquad (0 \le c \le 1/3), \tag{16}$$

$$F(c) \ge \tfrac{4}{3}c^2 \qquad (0 \le c \le 1/2), \tag{17}$$

$$1 - h_2(x) \le \tfrac{3}{4}(2x-1)^2 \qquad (1/2 \le x \le 2/3). \tag{18}$$

For $c > 0$ set $q = 1 - \sqrt{1-c^2/2}$. Then $c^2 = 4q(1-q/2)$ and $-\ln(1-q) \ge q$ gives

$$h_2(q) \ge \frac{q[\ln(1/q) + 1 - q]}{\ln 2}.$$

For $c \le 1/3$, one has $q < 1/32$, so $\log_2(1/q) \ge 5$ and $(1-q)/\ln 2 \ge 1$; hence $h_2(q) \ge 6q \ge (3/2)c^2$. For $c \le 1/2$, one has $q < 1/15$. The ratio

$$R(q) = \frac{\ln(1/q) + 1 - q}{4 \ln 2 (1 - q/2)}$$

decreases on $(0,1)$: its derivative has the sign of $-1/q + \frac{1}{2}\ln(1/q) < 0$. Consequently

$$\frac{F(c)}{c^2} \ge R(1/15) = \frac{\log_2 15 + (14/15)/\ln 2}{58/15} > \frac{31/8 + 4/3}{58/15} = \frac{625}{464} > \frac{4}{3}.$$

Here $\log_2 15 > 31/8$ follows from $15^8 > 2^{31}$, and $\ln 2 < 7/10$ implies $(14/15)/\ln 2 > 4/3$. The inequalities at $c = 0$ follow directly.

For $0 \le z \le 1/3$, expansion of the logarithms yields

$$1 - h_2((1+z)/2) = \frac{1}{\ln 2} \sum_{n \ge 1} \frac{z^{2n}}{2n(2n-1)} \le \frac{z^2}{\ln 2} \left(\frac{1}{2} + \frac{z^2}{12(1-z^2)} \right) \le \frac{49 z^2}{96 \ln 2} \le \frac{3}{4} z^2.$$

For these constants, the first two terms and a geometric tail bound in $\ln 2 = 2 \sum_{n \ge 0} [(2n+1) 3^{2n+1}]^{-1}$ give $56/81 < \ln 2 < 56/81 + 1/540 < 7/10$ and $56/81 > 49/72$. This proves (16)–(18) without numerical estimates.

Finally $H(\lambda) = h_2(x) + tH(\mu)$. If $x \le 5/8$, then $c \le 1/3$ and $t \le 1/2$, so (15) and (16) give

$$tH(\mu) \ge \frac{3z^2}{8t} \ge \frac{3}{4} z^2.$$

If $5/8 \le x \le 2/3$, then $t \le 3/8$, and (17) gives

$$tH(\mu) \ge \frac{z^2}{3t} \ge \frac{8}{9} z^2 \ge \frac{3}{4} z^2.$$

Combining either bound with (18) proves $H(\lambda) \ge 1$.

6 Attainment

Supply $|\Phi_2\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, whose reduced entropy is one bit. Alice measures her input qubit and her resource half in a Bell basis and sends the two-bit outcome to Bob. Bob's corresponding Pauli correction transfers Alice's input state, including all its correlations with his input, to his resource half. He measures the two qubits locally in the orthonormal basis (e_j) and sends j to Alice. For every density input the resulting shared label probabilities are $\mathrm{Tr}(P_j \rho)$, so the protocol implements (2) exactly. It is a finite complete physical LOCC protocol. A constant sequence gives an operational implementation with the same resource. Both cost sets therefore contain 1, and the established lower bound on every member proves Theorem 1.

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Third-level Clifford families with unbounded controlled levels

Abstract

We construct explicit quantum gates of exact Clifford-hierarchy level three and fixed dyadic Pauli period two whose controlled gates attain every exact level at least five. The number of qubits grows with the controlled level. A two-parameter extension gives exact hierarchy levels and periods, with all Pauli phases retained.

Xu and Wang ask how high the controlled level can be when the number of qubits, the level of the original gate, and its dyadic Pauli period are prescribed [1, Open Problem 1]. Our family gives unbounded controlled levels with original level three and fixed period two, as the number of qubits grows. This is a partial answer: it does not determine the maximum at a fixed qubit count. We use the block-extraction principle of [1, Lemma 2] and the necessary period condition of Anderson and Weippert [2], also stated in [1, Theorem 1]. The proofs below keep the specified Pauli phases explicit.

1 Definitions and the explicit family

On n qubits use the actual Pauli group

$$\mathcal{P}_n = \{aX_xZ_z : a \in \{1,-1,i,-i\},\ x,z \in \mathbb{F}_2^n\}, \quad X_x = \prod_j X_j^{x_j}, \quad Z_z = \prod_j Z_j^{z_j}.$$

The Clifford hierarchy is defined recursively by

$$\mathcal{C}_1(n) = \mathcal{P}_n, \qquad \mathcal{C}_{r+1}(n) = \{U \text{ unitary} : UPU^\dagger \in \mathcal{C}_r(n) \text{ for every } P \in \mathcal{P}_n\}.$$

The exact level $\mathrm{lev}(U)$ is the least positive r containing U, when such a level exists. Define the actual dyadic Pauli period by

$$\mathrm{per}(U) = \min\{a \ge 0 : U^{2^a} \in \mathcal{P}_n\},$$

when this set is nonempty. All phases of U are retained. Its controlled matrix is $\mathrm{C}(U) = \mathrm{diag}_c(I,U)$, on a genuine additional input bit c.

For $s \ge 1$ define two layers on an s-qubit block, numbered from 1:

$$C_1 = H_1 \prod_{j\ge1,\ 2j+1\le s} \mathrm{CNOT}_{2j+1\to 2j}, \tag{1.1}$$

$$C_2 = \mathrm{CZ}_{1,2} \prod_{j\ge1,\ 2j+2\le s} \mathrm{CNOT}_{2j+2\to 2j+1} \quad (s \ge 2), \tag{1.2}$$

and $C_2 = I$ when $s = 1$. Gates in each displayed layer have disjoint supports. Let S swap the two complete s-qubit blocks, and put

$$R = C_2C_1, \quad B = (C_1 \otimes I)S, \quad D = C_2 \otimes C_1, \quad A = BD = (I \otimes C_2)S. \tag{1.3}$$

The selector target, on $2s+1$ qubits, is

$$U_s = \mathrm{diag}_b(B,A), \qquad W_s = \mathrm{C}(U_s). \tag{1.4}$$

Both block registers, the selector b, and the outer control c range over their entire input spaces.

For integers $h \geq 1$ and $m \geq 2$ set

$$s_h = \begin{cases} 1, & h = 1, \\ 2^{h-2} + 1, & h \geq 2, \end{cases} \qquad \lambda_m = \exp(i\pi/2^{m+1}), \qquad U_{h,m} = \lambda_m U_{s_h}. \tag{1.5}$$

Theorem 1.1. *For every $h \geq 1$ and $m \geq 2$, the unitary $U_{h,m}$ acts on*

$$n_h = \begin{cases} 3, & h = 1, \\ 2^{h-1} + 3, & h \geq 2, \end{cases}$$

qubits and satisfies

$$\operatorname{lev}(U_{h,m}) = 3, \qquad \operatorname{per}(U_{h,m}) = m, \qquad \operatorname{lev}(\mathrm{C}(U_{h,m})) = \max\{h+4, m+2\}. \tag{1.6}$$

Corollary 1.2. *For every integer $r \geq 5$ there is a finite target with exact level three, actual period two, and exact controlled level r. It is $U_{r-4,2}$ in (1.5).*

2 Elementary hierarchy and period facts

We use the correction $[V, P] = VPV^\dagger P^\dagger$. Membership in $\mathcal{C}_r$, for $r \geq 2$, is equivalent to all its Pauli corrections belonging to $\mathcal{C}_{r-1}$, since multiplication by an actual Pauli preserves every hierarchy level. Induction also proves that the hierarchy is nested, that Clifford conjugation preserves every level, and that multiplication on either side by a Clifford preserves $\mathcal{C}_r$ for $r \geq 2$. Unit scalar multiplication preserves these levels as well. Exact levels at least three are therefore preserved by Clifford factors and unit scalars. These are the usual recursive hierarchy properties; controlled-Clifford jump criteria are treated in [1].

If $\operatorname{diag}(V_0, V_1) \in \mathcal{C}_r$, then both blocks belong to $\mathcal{C}_r$. At level one this follows from the block form of an actual Pauli. The induction step applies the previous-level assertion to the conjugations of $I \otimes P$. Tensoring an identity spectator preserves membership in both directions by the same argument.

Lemma 2.1 (Necessary actual period)**.** *If $\mathrm{C}(V) \in \mathcal{C}_r$ with $r \geq 2$, then*

$$V^{2^{r-2}} \in \mathcal{P}_n. \tag{2.1}$$

Also $V, V^\dagger \in \mathcal{C}_{r-1}$.

Proof. The first outer-control correction is

$$[\mathrm{C}(V), X_c] = \operatorname{diag}(V^\dagger, V) \in \mathcal{C}_{r-1}.$$

Block extraction gives the second assertion. Repeated correction by X_c gives $\operatorname{diag}(V^{-2^j}, V^{2^j}) \in \mathcal{C}_{r-1-j}$ for $0 \leq j \leq r-2$. At the last step its blocks are actual Paulis, including their fourth-root phases. This proves (2.1). □

Lemma 2.2. *If D is a Clifford, $D^2 = I$, and $D \notin \mathcal{P}_n$, then $\mathrm{C}(D)$ has exact level three.*

Proof. For a target Pauli P, its image is $\operatorname{diag}(P, DPD)$, a Pauli times a controlled Pauli, and hence Clifford. The image of X_b is $(I \otimes D)X_b$, also Clifford; Z_b is fixed. Since the destination $\mathcal{C}_2$ is a group, this proves the assertion for every full Pauli string and puts $\mathrm{C}(D)$ in $\mathcal{C}_3$. If it were in $\mathcal{C}_2$, Lemma 2.1 would force D to be an actual Pauli. □

Here a controlled actual Pauli is Clifford: a Hermitian Pauli can be Clifford-conjugated to a product of Z operators, and its controlled version is a product of CZ gates. The allowed factor i^k contributes S_b^k on the control. This verifies the phase convention used in Lemma 2.2.

3 The physical Clifford cycle and its signed endpoint

Both C_1 and C_2 are real Hermitian Clifford involutions. Their product satisfies the actual identity

$$C_1 R C_1 = R^{-1}. \tag{3.1}$$

We compute its action on the complete binary Pauli-label space $V = \mathbb{F}_2^{2s}$. In this computation Y_1 denotes the label $X_1 + Z_1$. Let $F_R = I + N$. Direct propagation through (1.1)–(1.2) gives, for $s \geq 2$,

$$\begin{aligned}
&NX_2 = Z_1, && NX_3 = X_2 + Z_1, \\
&NX_i = X_{i-1} && \text{for even } i \geq 4, \\
&NX_i = X_{i-1} + X_{i-2} && \text{for odd } i \geq 5, \\
&NZ_1 = Y_1 + Z_2, && NY_1 = Z_2, \\
&NZ_i = Z_{i+1} && \text{for odd } i \geq 3, \\
&NZ_i = Z_{i+1} + Z_{i+2} && \text{for even } i \geq 2.
\end{aligned} \tag{3.2}$$

Only existing input labels occur; labels past s in these formulas are zero. The X_3 formula is used only if $s \geq 3$. In the ordered basis

$$X_s, \ldots, X_2, Z_1, Y_1, Z_2, \ldots, Z_s$$

every basis vector is sent to the next one with coefficient one, plus a combination of later vectors. Therefore

$$N^{2s} = 0, \qquad N^{2s-1} X_s = Z_s \neq 0, \qquad \ker N = \langle Z_s \rangle. \tag{3.3}$$

The nonzero power has one surviving path of length $2s-1$, using all adjacent steps. For $s = 1$, use the ordered basis Z_1, Y_1; then $NZ_1 = Y_1$, $NY_1 = 0$, and $R = H_1$.

Let $h = \lceil \log_2(2s) \rceil$ and $q = 2^h$. In characteristic two,

$$(I + N)^{2^a} = I + N^{2^a}.$$

Thus the exact dyadic order of the label action is q: its earlier dyadic powers are nonidentity. In particular no earlier dyadic power of R is even a scalar multiple of a Pauli.

We require an actual matrix endpoint as well. An identity label action means the Clifford is proportional to a Pauli: match its X/Z conjugation signs by a Pauli, and the remaining operator commutes with all Paulis. Commutation with all Z strings makes it diagonal, and commutation with all X strings makes all diagonal entries equal. Hence the remaining operator is scalar. Apply this to R^q. Its Pauli label p is fixed by F_R, since R commutes with R^q. For $s \geq 2$, (3.3) gives $p = 0$ or Z_s. Both corresponding ordered Paulis are real Hermitian. Reality and unitarity consequently force

$$R^q \in \{I, -I, Z_s, -Z_s\}, \qquad R^{2q} = I. \tag{3.4}$$

For $s = 1$, $R^2 = I$ directly. The endpoint label also satisfies

$$p \in \operatorname{im} N^{q/2}. \tag{3.5}$$

Indeed $2s > q/2$, and the last vector of the regular chain lies in this image; at $s = 1$ the actual endpoint label is zero. This derives both the endpoint sign and the image condition from the physical circuit. The regular-unipotent Clifford period mechanism and its resource scale are also discussed in [1, Theorem 4, Proposition 9, Appendix A].

4 Exact target level, period, and controlled lower bounds

From the actual involution identities,

$$B^2 = C_1 \otimes C_1, \quad A^2 = C_2 \otimes C_2, \quad B^4 = A^4 = D^2 = I. \tag{4.1}$$

The second factor of D contains the Hadamard H_1. Its computational columns have two nonzero entries, since the other gates in that layer are disjoint permutations. Tensoring with the unitary C_2 preserves a two-entry column witness. Every scalar multiple of a Pauli has at most one nonzero entry in each column. Thus D is not even projectively Pauli, and in particular is not an actual Pauli. Lemma 2.2 and

$$U_s = (I_b \otimes B)\, \mathrm{C}_b(D)$$

show that U_s has exact level three. Its fourth power is I. Its square is not even projectively Pauli, because its $b = 0$ block is $C_1 \otimes C_1$, which has the same multi-entry obstruction. Therefore $\mathrm{per}(U_s) = 2$.

The scalar λ_m has primitive order 2^{m+2}, satisfies $\lambda_m^{2^m} = i$, and obeys

$$\lambda_m^{2^a} \in \{1, -1, i, -i\} \quad \Longleftrightarrow \quad a \geq m. \tag{4.2}$$

The early powers of U_s are not scalar Paulis; for $a \geq 2$ its dyadic power is I. Equation (4.2) consequently proves $\mathrm{per}(\lambda_m U_s) = m$. Unit scalar multiplication preserves the exact target level three.

Put $E = BDB^\dagger$. Direct use of the block swap gives

$$E = C_1 \otimes C_2, \qquad DE = R \otimes R^{-1} =: R_{\mathrm{big}}. \tag{4.3}$$

The qth power of R_{big} is an actual Pauli by (3.4); earlier dyadic powers have nonidentity label action on a block. Thus $\mathrm{per}(R_{\mathrm{big}}) = h$.

Suppose $W = \mathrm{C}(\lambda_m U_s)$ has a finite level r. Block extraction gives $r \geq 3$; applying it to the first control correction gives $r \geq 4$. The two literal corrections by X_c and then X_b give

$$\begin{aligned} H_0 &= \mathrm{diag}_c(\overline{\lambda_m} U_s^\dagger, \lambda_m U_s), \\ [H_0, X_b] &= \mathrm{diag}_c(D, E) \otimes I_b \in \mathcal{C}_{r-2}. \end{aligned} \tag{4.4}$$

The scalars cancel within the selector correction. Removing D, a constant Clifford, is valid because $r - 2 \geq 2$ and gives $\mathrm{C}(R_{\mathrm{big}}) \otimes I_b \in \mathcal{C}_{r-2}$. Lemma 2.1 implies

$$r \geq h + 4. \tag{4.5}$$

Applied directly to $\mathrm{C}(\lambda_m U_s)$, the same lemma and its actual target period imply

$$r \geq m + 2. \tag{4.6}$$

These inequalities exclude every smaller level once a finite upper bound is supplied.

5 Signed corrections for all physical Pauli strings

We now prove the upper bound first for $W_s = \mathrm{C}(U_s)$, with $q = 2^h$ as above. On the two-block state space set

$$T_0 = R \otimes I, \qquad T_1 = I \otimes R, \qquad V_a = \operatorname{im} N^a \oplus \operatorname{im} N^a.$$

The shifts commute, have actual order dividing $2q$, and satisfy

$$BT_0B^{-1} = T_1, \qquad BT_1B^{-1} = T_0^{-1}, \qquad D = T_0B^2. \tag{5.1}$$

The four (c, b) branches of W_s are therefore the actual word

$$F(c, b) = T_1^{cb}B^{c+2cb}, \qquad c, b \in \{0, 1\}. \tag{5.2}$$

All powers here and below are powers of actual unitary matrices, including negative integer powers.

For a state label $v = (x, z)$ let $P_v = X_xZ_z$. The full signed action is computable by the following elementary rules, together with composition:

$$\begin{aligned} H_jP_{x,z}H_j &= (-1)^{x_jz_j}P_{x',z'}, & (x'_j, z'_j) &= (z_j, x_j), \\ \mathrm{CNOT}_{a\to b}P_{x,z}\mathrm{CNOT}_{a\to b} &= P_{x',z'}, & x'_b &= x_b + x_a, \quad z'_a = z_a + z_b, \\ \mathrm{CZ}_{a,b}P_{x,z}\mathrm{CZ}_{a,b} &= (-1)^{x_ax_b}P_{x,z'}, & z'_a &= z_a + x_b, \quad z'_b = z_b + x_a. \end{aligned}$$

All unlisted coordinates are unchanged. Block swap simply swaps the two full labels. Hence every real Clifford occurring in the proof has an actual action

$$FP_vF^\dagger = (-1)^{\kappa_F(v)}P_{L_Fv} \tag{5.3}$$

on every state Pauli. The signs are retained, including those produced by the Hadamard gates. Ordered Paulis obey

$$P_vP_w = (-1)^{z_v\cdot x_w}P_{v+w}, \qquad P_v^\dagger = (-1)^{z_v\cdot x_v}P_v. \tag{5.4}$$

Consider an intermediate diagonal multiplexer whose branches are

$$(-1)^{f(u)}P_{y(u)}F(u), \qquad u = (c, b) \in \mathbb{F}_2^2. \tag{5.5}$$

The functions f and y may be arbitrary. A full input Pauli has arbitrary control translation $a \in \mathbb{F}_2^2$, arbitrary control Z label, arbitrary state label w, and a scalar in $\{1, -1, i, -i\}$. The input scalar and control Z phases cancel in its normalized correction. Put

$$G(u) = F(u)F(u + a)^\dagger.$$

Direct matrix multiplication and (5.4) give its state label

$$y'(u) = y(u) + L_G(u)y(u + a) + L_F(u)w + L_G(u)w. \tag{5.6}$$

For completeness, its real branch sign is explicitly determined as follows. Put

$$v_1 = y(u), \quad v_2 = L_F(u)w, \quad v_3 = L_G(u)y(u + a), \quad v_4 = L_G(u)w.$$

Then the new sign exponent, in $\mathbb{F}_2$, is

$$f'(u) = f(u) + f(u+a) + \kappa_{F(u)}(w) + \kappa_{G(u)}(y(u+a)) + \kappa_{G(u)}(w) + z_{v_3} \cdot x_{v_3} + z_{v_4} \cdot x_{v_4} + \sum_{k<l} z_{v_k} \cdot x_{v_l}. \tag{5.7}$$

Thus arbitrary mixed labels and all allowed input phases are accounted for by actual matrix identities. Subsequent steps remain in the real signed class (5.5).

6 A filtration closed under every Pauli correction

For $0 \le j \le h-1$, consider profiles with

$$\begin{aligned} F(u) &= T_0^{t_0(u)} T_1^{t_1(u)}, \\ t_i(c,b) &= t_{i,c}c + t_{i,b}b + t_{i,12}cb, \qquad (6.1) \\ 2^j &\mid t_{i,c}, t_{i,b}, \qquad 2^{j+1} \mid t_{i,12}, \\ y_{12} &\in V_{2^j}. \qquad (6.2) \end{aligned}$$

Here $y_{12} = y(0,0) + y(1,0) + y(0,1) + y(1,1)$ is the quadratic Boolean coefficient. Integer coefficients are kept unreduced; if a modulus is used it is the actual period $2q$, not the label period q.

Lemma 6.1. *If (6.2) holds at depth $j < h-1$, every full Pauli correction, after removing a constant right Clifford, has the same form at depth $j+1$.*

Proof. For a control translation $a = (a_c, a_b)$, normalizing the constant branch replaces each exponent polynomial $t(u)$ by

$$t(u) - t(u+a) + t(a).$$

Write Boolean addition as $c \oplus a_c = c + a_c - 2ca_c$ over the integers. Expansion gives the new coefficients

$$\begin{aligned} t'_c &= 2a_c t_c - a_b(1-2a_c)t_{12}, \\ t'_b &= 2a_b t_b - a_c(1-2a_b)t_{12}, \qquad (6.3) \\ t'_{12} &= 2(a_c + a_b - 2a_c a_b)t_{12}. \end{aligned}$$

The two linear divisibilities and the deeper quadratic divisibility advance by one power of two.

We verify the label condition as well. Put $k = 2^j$. A shift exponent divisible by k has label action I plus a polynomial in N divisible by N^k on its block, and thus raises the filtration by k. This holds for negative exponents too, since $I + N$ is invertible and its inverse is a finite polynomial in N. For a single block, the quadratic coefficient of the label profile with exponents $\alpha c + \beta b + \gamma cb$ is

$$((I+N)^\alpha - I)((I+N)^\beta - I) + (I+N)^{\alpha+\beta}((I+N)^\gamma - I). \tag{6.4}$$

When $k \mid \alpha, \beta$ and $2k \mid \gamma$, both terms raise the filtration by $2k$. Products between differences supported on separate blocks are zero.

Before constant normalization, the difference profile G has a constant action G_0 with $G_0 - I$ raising by k. Its nonconstant coefficients raise by $2k$, by (6.3) and (6.4). In (5.6), the quadratic coefficient of $y + G_0 y(\cdot + a)$ is $(I + G_0)y_{12} \in V_{2k}$. The other terms involving the nonconstant part of G or the inserted state label w are already in V_{2k}. Thus $y'_{12} \in V_{2k}$. Removing the constant right

Clifford changes the Clifford profile but leaves the left Pauli label unchanged. The sign (5.7) is still an arbitrary actual real sign function. □

Lemma 6.2 (Terminal level)**.** *Every signed profile satisfying* (6.2) *at* $j = h - 1$ *belongs to* $\mathcal{C}_3$.

Proof. Let $k = q/2$. The quadratic exponents are multiples of q, and the linear exponents are multiples of k. By (3.4), each T_i^q is a real Hermitian actual Pauli with label in V_k, and $T_i^{2q} = I$. Absorb the qth-power factors into the left Pauli dressing. This preserves $y_{12} \in V_k$. The remaining profile is

$$F_*(c, b) = S_c^c S_b^b, \tag{6.5}$$

where S_c, S_b are commuting products of T_0^k, T_1^k. Their squares are actual signed Paulis and their fourth powers are I.

Their label differences raise into V_k, annihilate V_k, and have pairwise product zero because $N^q = 0$. Their label matrices are involutions and the label profile (6.5) is affine in c, b. For an X_c correction, the exact Clifford difference is

$$F_*(c, b)F_*(c + 1, b)^{-1} = S_c^{-1}(S_c^2)^c.$$

It is a constant Clifford times a linear controlled actual Pauli; the X_b formula is identical with S_b. These corrections have constant label action, which fixes V_k. Equation (5.6) therefore has zero quadratic coefficient, since $(I + L_G)y_{12} = 0$ and $L_F w$ is affine. For a state X or Z generator, $a = 0$; then (5.6) reduces to $L_F w + w$, again affine. Control Z generators commute with the multiplexer.

An affine Pauli-label dressing is Clifford: factor it into a constant Pauli and two controlled Paulis. The resulting product signs are functions of only c, b. Every real sign function on two bits is a product of a constant sign, Z_c, Z_b, and $\mathrm{CZ}_{c,b}$, so it too is Clifford. Equations (5.7) and (6.5) now show that each generator correction is Clifford. The group property of $\mathcal{C}_2$ extends this to every full Pauli. This generator argument is confined to this $\mathcal{C}_3$ terminal step. □

Backward induction using Lemmas 6.1 and 6.2 proves

$$\text{every depth-}j\text{ profile is in } \mathcal{C}_{h-j+2}. \tag{6.6}$$

At each backward step the removed right factor is Clifford and the destination level is at least three, so the hierarchy factor rule applies.

7 Entry of the circuit after two corrections

We show that two arbitrary full Pauli corrections of (5.2), allowing zero control translations and arbitrary state labels, enter depth zero. Define the normalized profile difference

$$\mathcal{D}_a F(u) = F(u)F(u + a)^{-1}F(a).$$

The last factor is a constant Clifford, so this normalization preserves membership at all levels used below. Write $a = (a_c, a_b)$ and treat its bits as integers in the exponent formulas. Direct substitution of the four branches of (5.2) into (5.1) gives

$$\mathcal{D}_a F(c, b) = T_0^{t_0(c,b)} T_1^{t_1(c,b)} K^{(a_c+a_b)c+a_c b}, \qquad K = B^2, \tag{7.1}$$

where the exponent of the actual involution K is binary, and

$$t_0(c,b) = -a_c a_b c + a_c b + a_c(2a_b - 1)cb,$$
$$t_1(c,b) = (1 - a_c)a_b c + a_c cb. \tag{7.2}$$

Also $K^2 = I$ and $KT_iK^{-1} = T_i^{-1}$ for $i = 0, 1$.

For any second translation $d = (d_c, d_b)$ put

$$\delta = (-1)^{(a_c+a_b)d_c + a_c d_b} \in \{1, -1\}.$$

A second normalized difference of (7.1) cancels K and has shift exponents

$$t_i(u) - \delta t_i(u + d) + \delta t_i(d). \tag{7.3}$$

For a polynomial $t = t_c c + t_b b + t_{12} cb$ the new quadratic coefficient is

$$\big[1 - \delta(1 - 2d_c)(1 - 2d_b)\big]t_{12}, \tag{7.4}$$

which is even for either value of δ. The new linear coefficients are integers. Thus (6.2)'s exponent condition at depth zero holds without a phase quotient.

It remains to include the Pauli dressings generated by those same two arbitrary corrections. From (3.1), C_1 preserves every $\operatorname{im} N^a$: conjugating N by C_1 gives $(I + N)^{-1} - I = N(I + N)^{-1}$. Its action on the one-dimensional quotient $V/\operatorname{im} N$ is the identity. On the two-block head quotient V_0/V_1, B consequently swaps the heads, with action J satisfying $J^2 = I$, while D and both shifts act trivially. The original profile (5.2) has head action J^c.

Starting from zero dressing, (5.6) gives on this quotient, for the first state label w,

$$y_1(c,b) = J^c w + J^{a_c} w.$$

This is affine in c and independent of b. The first normalized Clifford profile (7.1) has identity head action: the shifts and $K = B^2$ all act trivially there. Its difference profile also has identity head action. Hence for any second state label w', (5.6) reduces on the head quotient to

$$y_2(u) = y_1(u) + y_1(u + d) + w' + w' = y_1(u) + y_1(u + d),$$

which is constant. Its quadratic coefficient therefore lies in V_1. Constants removed during normalization only conjugate subsequent arbitrary state Paulis by fixed Cliffords, and those Paulis still range over the entire group. All sign factors remain those of (5.7).

We have proved entry into depth zero for every pair of full Pauli corrections. By (6.6), the second corrections belong to $\mathcal{C}_{h+2}$. Recursing upward twice proves

$$W_s \in \mathcal{C}_{h+4}. \tag{7.5}$$

The proof also covers $h = 1, s = 1$: then $q = 2$, depth zero is already terminal, and the actual $R^2 = I$ supplies its endpoint. No initialized register or restricted input subspace is used.

8 The actual scalar extension and exactness

Lemma 8.1 (Branch phase)**.** *Suppose* $V = \operatorname{diag}(V_0, V_1) \in \mathcal{C}_r$, $r \geq 2$, *and* $|z| = 1$. *If* $z^{2^a} \in \{1, -1, i, -i\}$, *then*

$$\operatorname{diag}(V_0, zV_1) \in \mathcal{C}_{\max\{r, a+2\}}. \tag{8.1}$$

Proof. Induct on a. For $a = 0$ the phase factor is a controlled actual scalar Pauli, hence Clifford. If $r = 2$, factor out the one-bit phase $\mathrm{diag}(1, z)$. Its repeated X corrections square its relative phase, so $z^{2^a} \in \{1, -1, i, -i\}$ puts it in $\mathcal{C}_{a+2}$. Multiplication by the original Clifford V preserves that level.

For $r \geq 3$, take any full input Pauli and its correction. If its control X bit is zero, the branch scalar cancels and the correction is unchanged. If that bit is one, direct block multiplication gives, up to the unit scalar $\bar{z}$, a correction of the form $\mathrm{diag}(M, z^2 N)$, where $\mathrm{diag}(M, N)$ is the original correction in $\mathcal{C}_{r-1}$. Apply the induction hypothesis with $a - 1$, $r - 1$, and z^2. This gives level at most $\max\{r - 1, a + 1\}$ for that full correction. The common unit scalar is immaterial at this level, which is at least two. Control Z components and actual input scalar phases cancel in the same matrix identity. Recursion proves (8.1). □

For the chosen s_h, the inequality $2^{h-1} < 2s_h \leq 2^h$ shows that (7.5) uses precisely the prescribed h in (1.5). Lemma 8.1, with $z = \lambda_m$ and $a = m$, yields

$$\mathrm{C}(U_{h,m}) \in \mathcal{C}_{\max\{h+4, m+2\}}.$$

The actual lower bounds (4.5) and (4.6) prove exactness. The target and period assertions were proved in Section 4; the qubit count is $2s_h + 1$. This establishes Theorem 1.1 and, by $h = r - 4, m = 2$, Corollary 1.2.

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